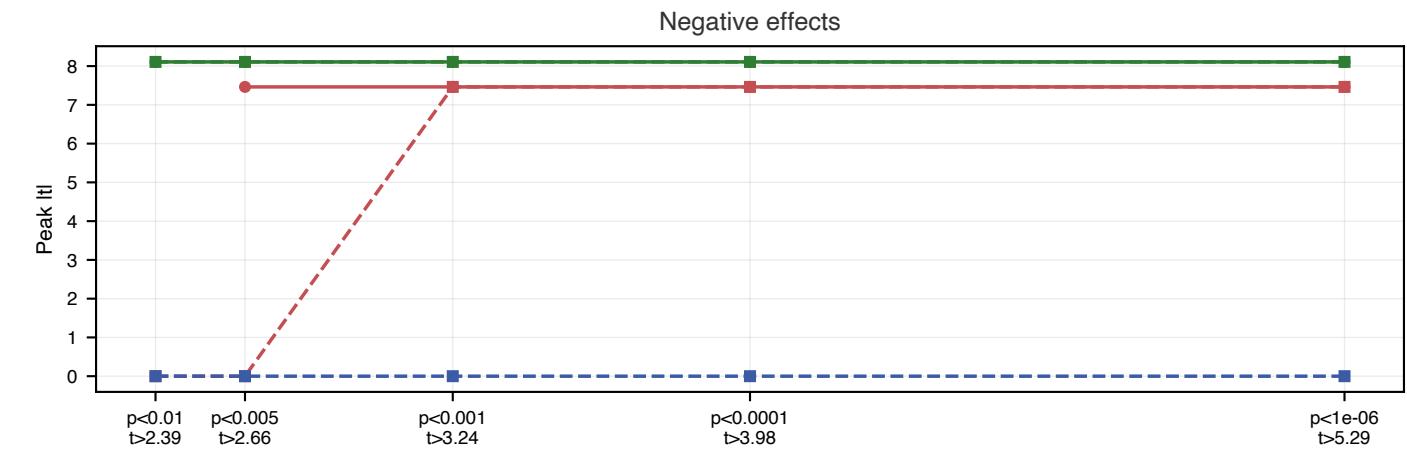
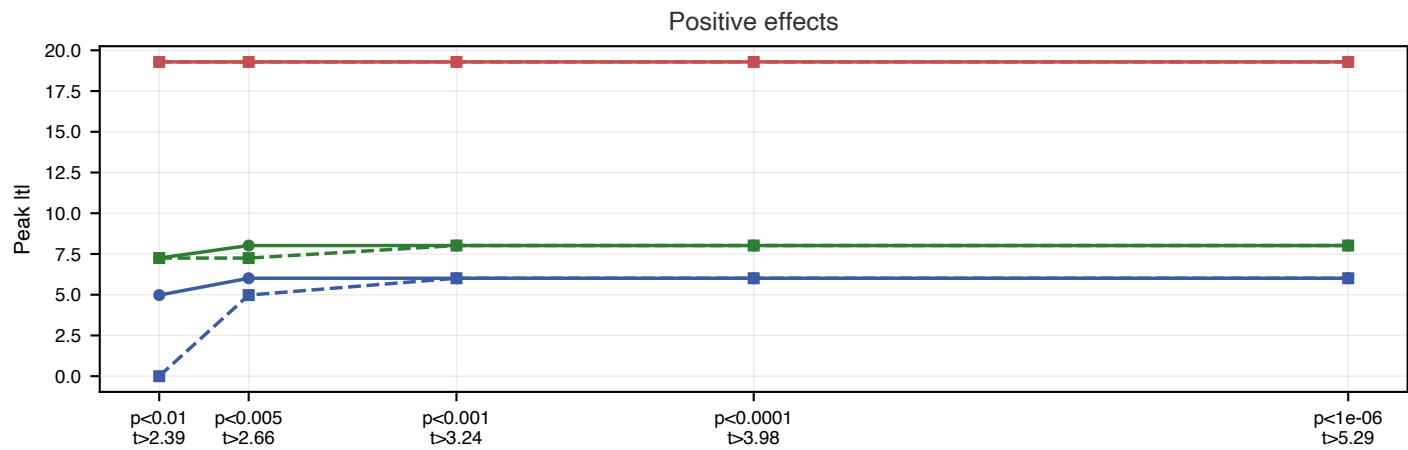
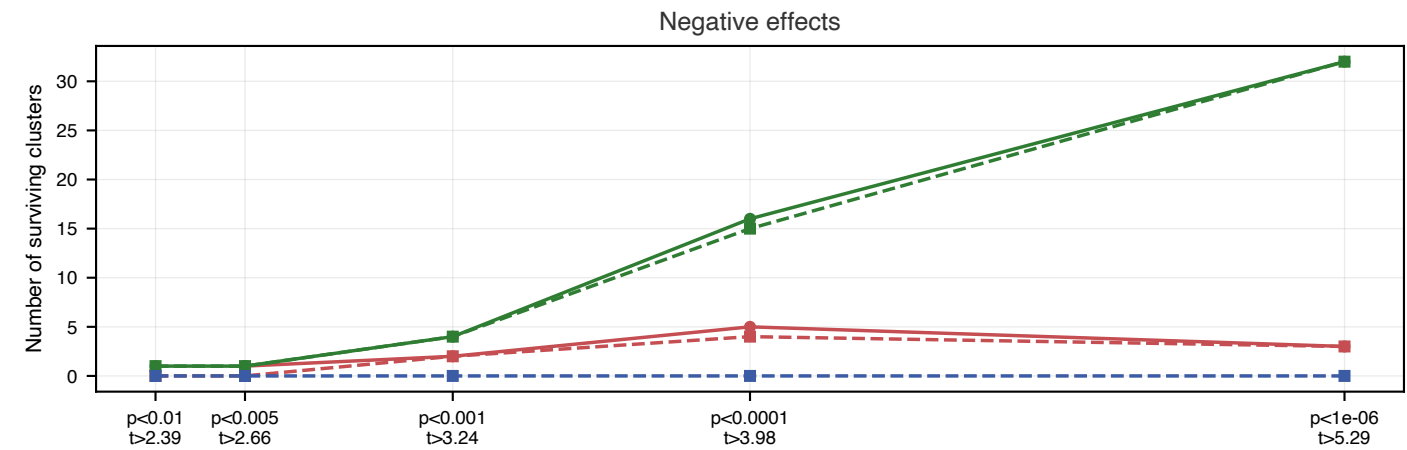
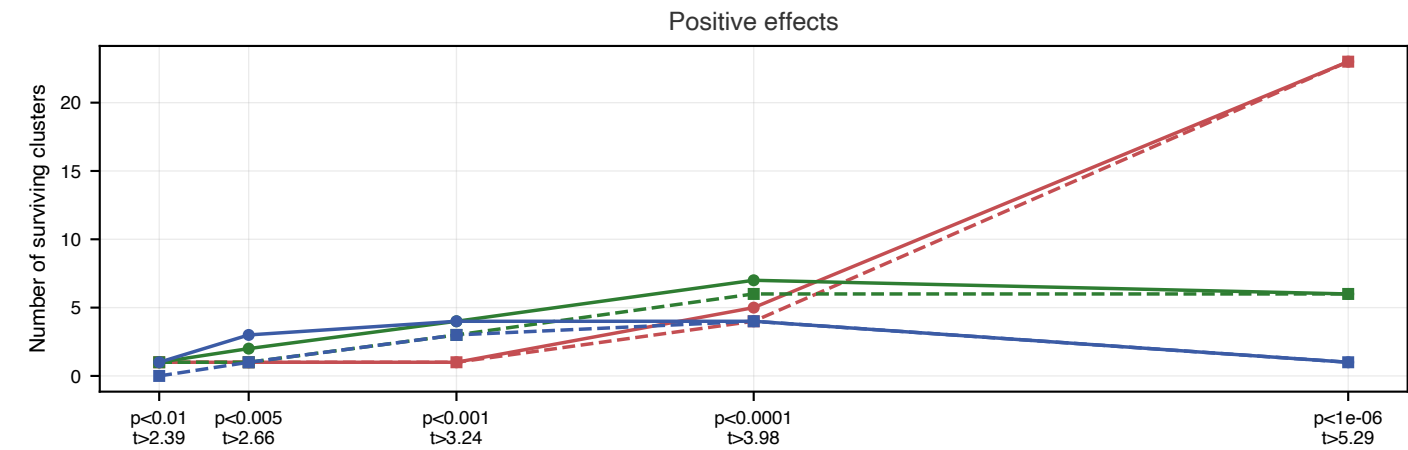
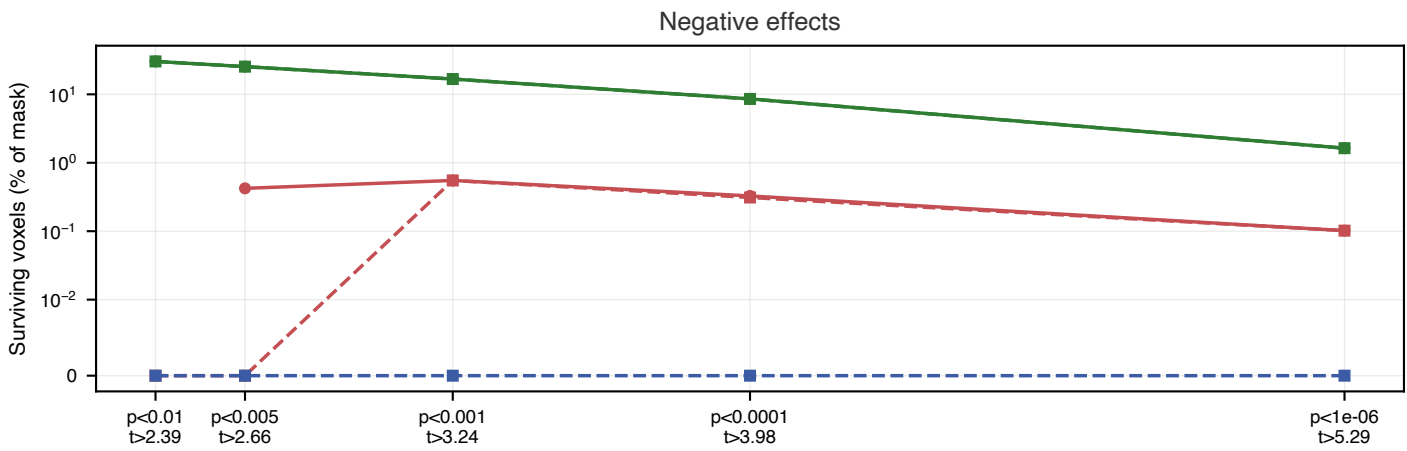
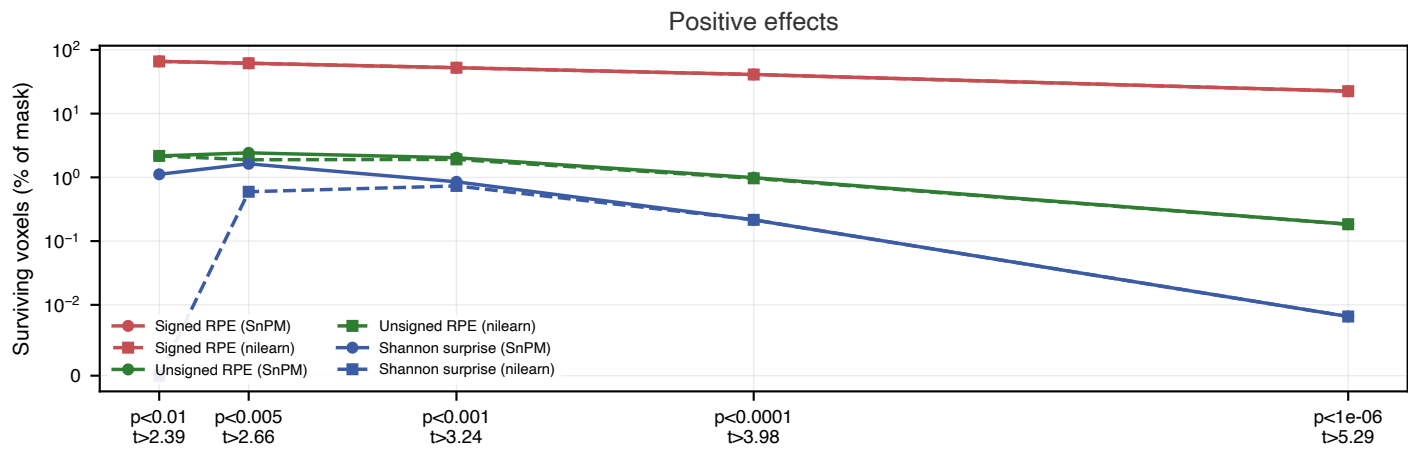


How much of the brain survives — model2

Extent of the surviving map as a function of the cluster-forming threshold. If a signal's advantage is sensitivity rather than anatomy, it shows up here as a curve that sits above the others at every threshold.



Engine comparison and caveats

SnPM and Nilearn run the same sign-flipping permutation test on the same 58 contrast images and the same analysis mask (SnPM's own), but they are not identical, and three differences matter for reading the pages that follow:

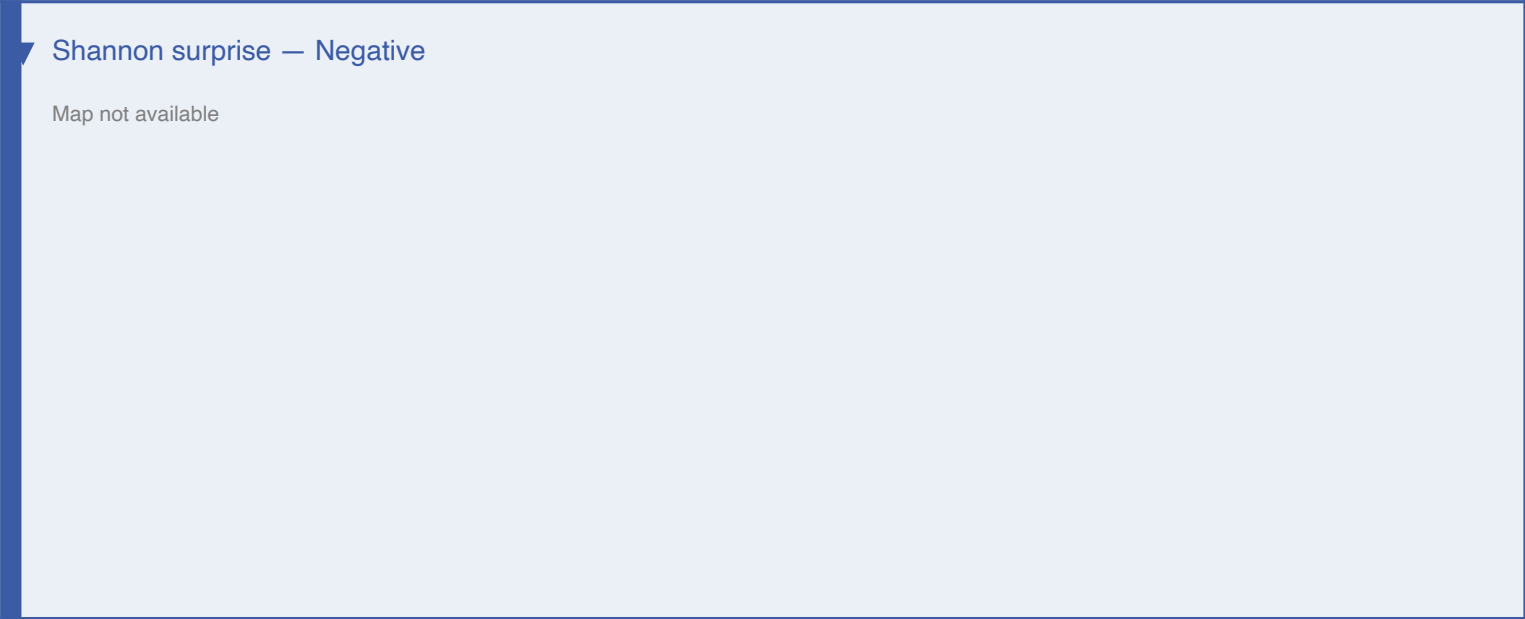
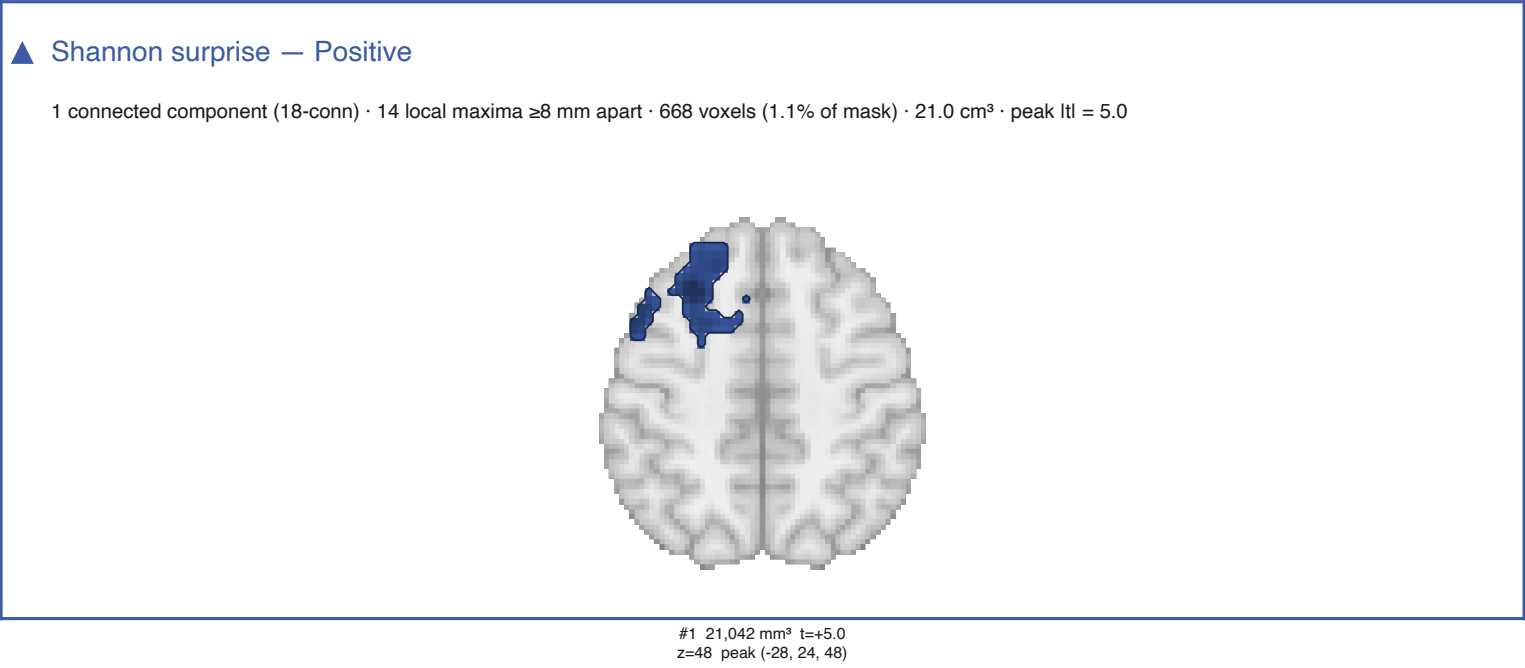
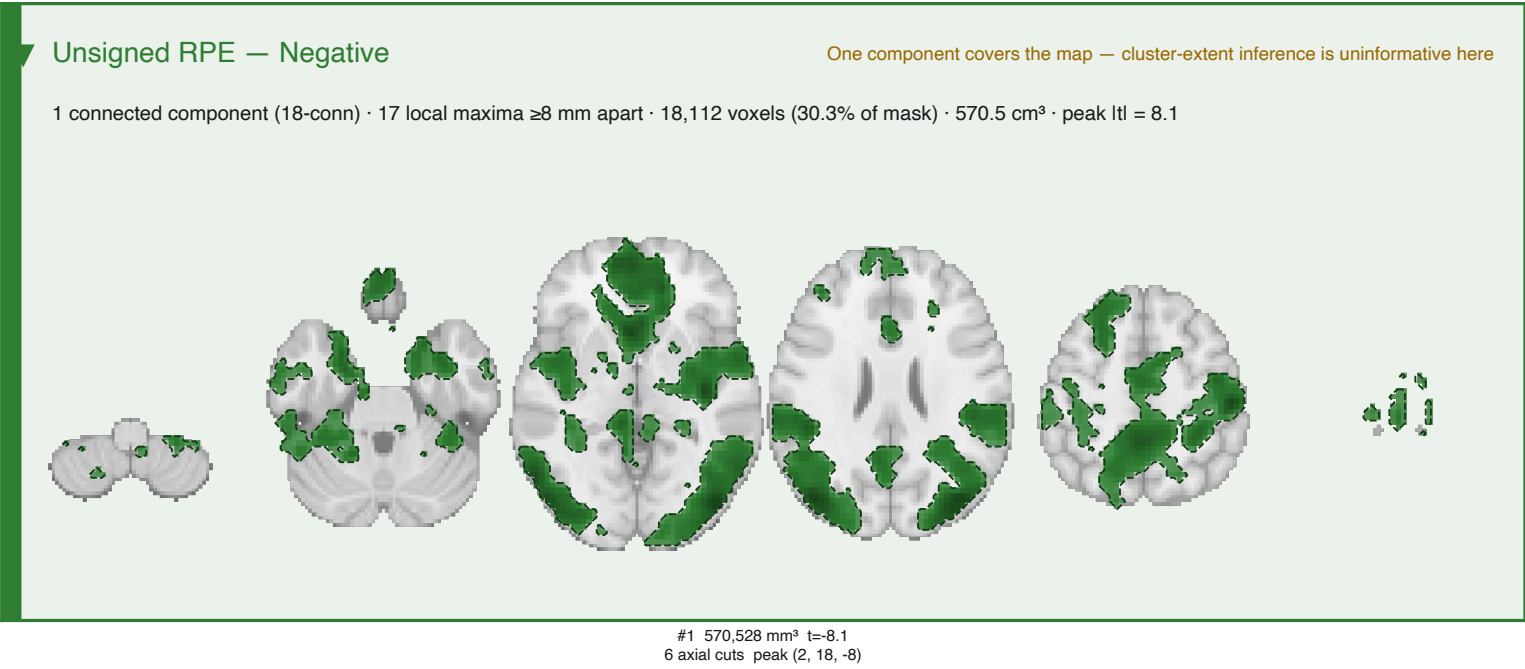
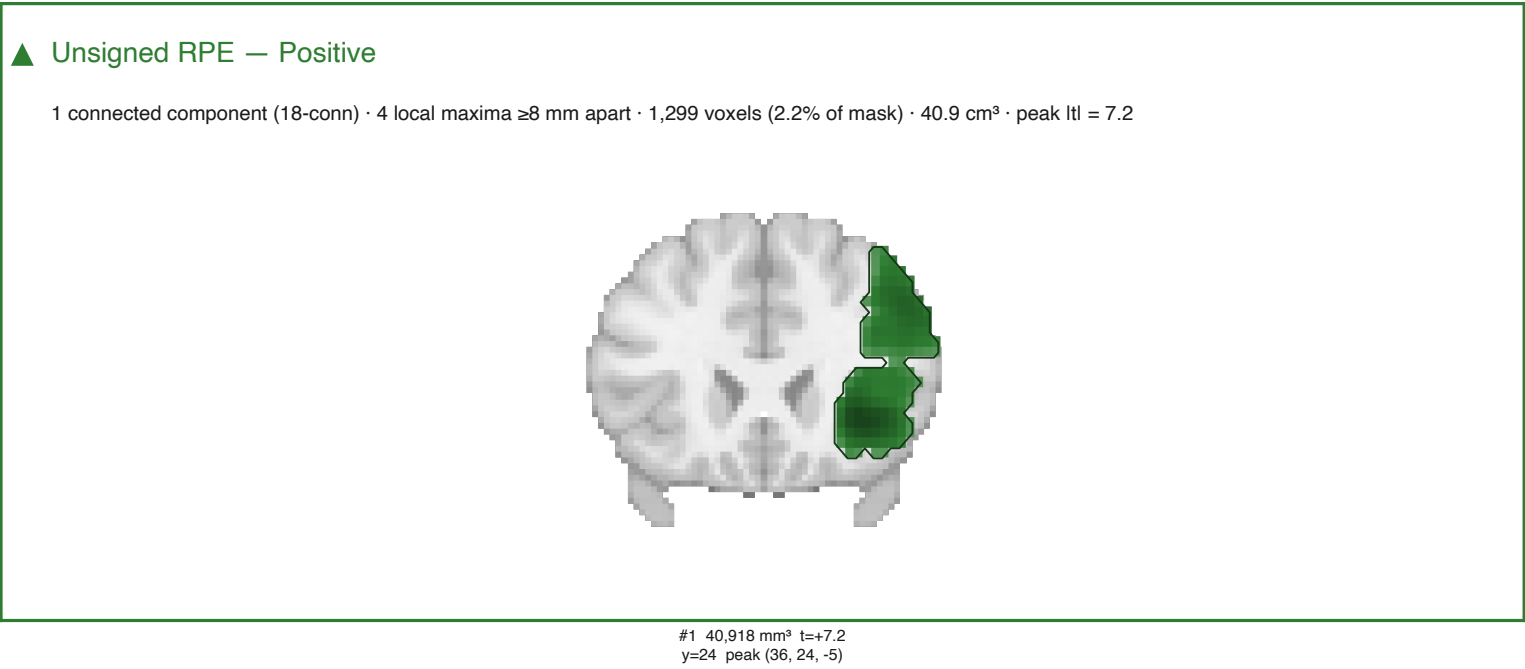
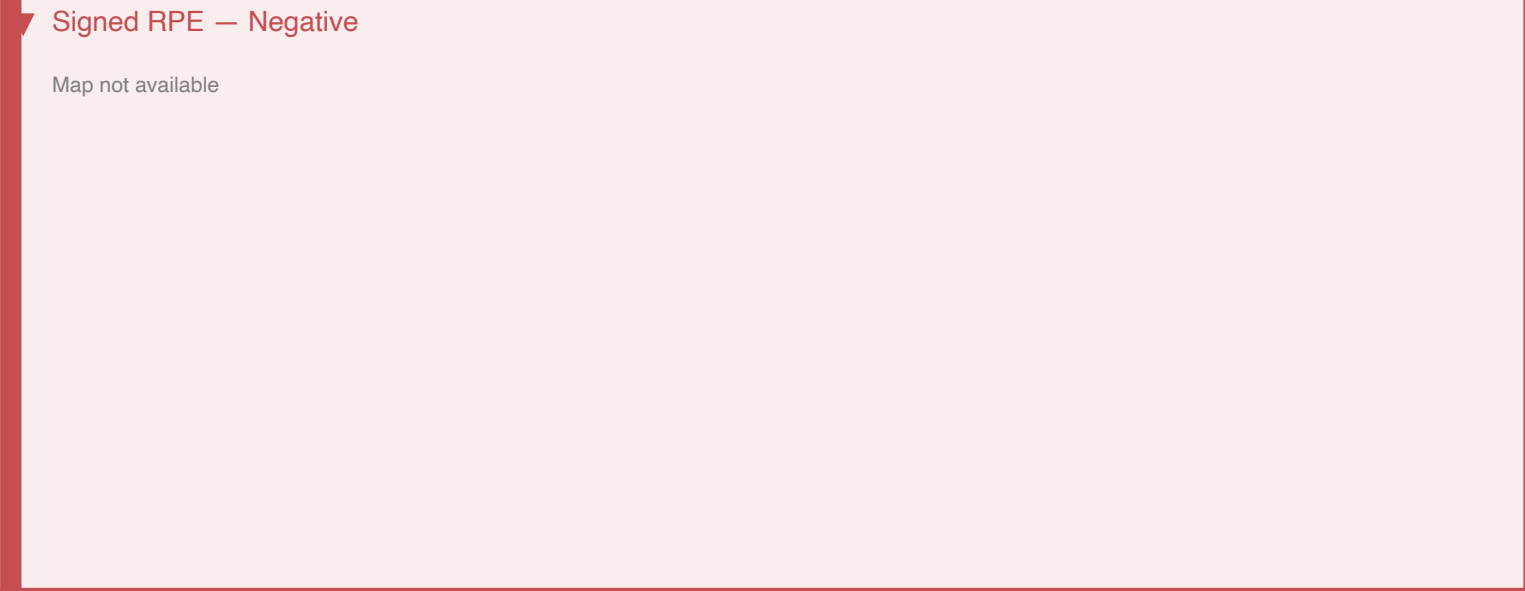
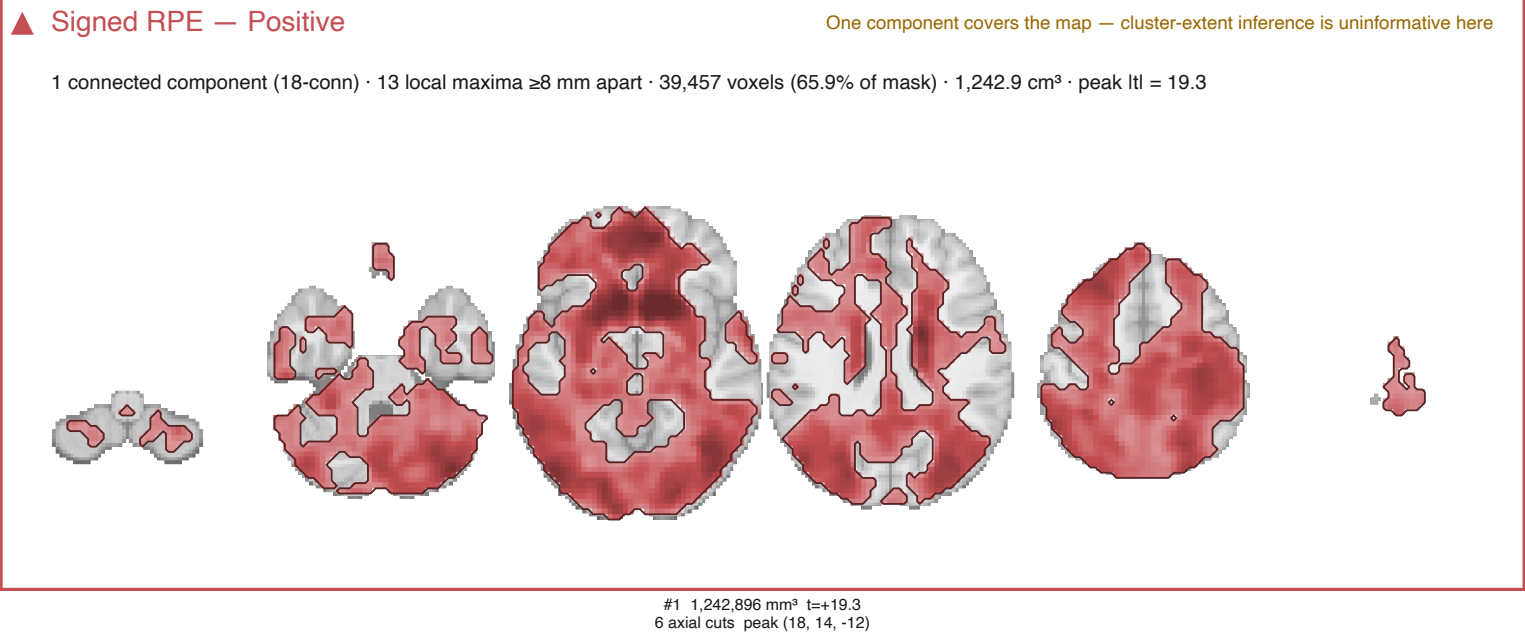
1. Family. SnPM's `Tsign` treats each tail as its own one-tailed family, which is what the paper reports; Nilearn's `two\_sided\_test=True` controls the family-wise error over both tails at once. Nilearn is therefore the stricter test by roughly a factor of two in the tail, and its maps should be slightly smaller wherever a threshold is near the edge of significance.
2. Connectivity. SPM labels clusters with 18-connectivity (`spm\_clusters`); Nilearn hardcodes 6-connectivity. These runs monkeypatch Nilearn to 18 so the two engines parcellate identically -- otherwise Nilearn splits a single SPM cluster into several smaller ones, each of which is then easier to kill on extent.
3. Statistic. SnPM's cluster-extent test uses cluster size only. Nilearn additionally provides cluster *mass* (the summed excess of t over the threshold) and TFCE, both of which reward tall-and-narrow clusters that pure extent penalises.

TFCE and voxel-wise FWE are threshold-free and appear once, not once per cluster-forming threshold. Where the two engines disagree, the Dice overlap on the summary panels is the place to look first; large disagreement almost always means the map is sitting on the edge of the extent criterion, which is itself the point of this document.

Across all 24 signal × sign × threshold panels in this document, the SnPM and Nilearn survivor maps overlap with a median Dice of 1.00 (range 0.00–1.00).

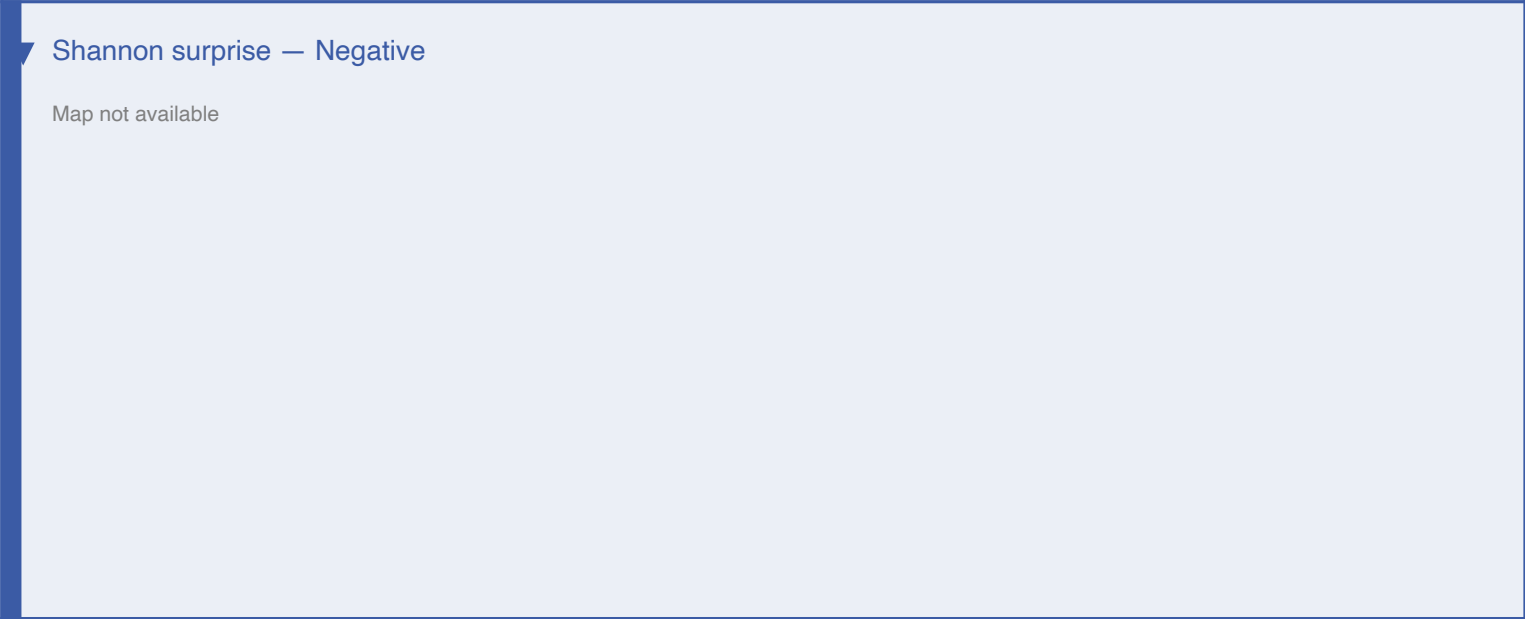
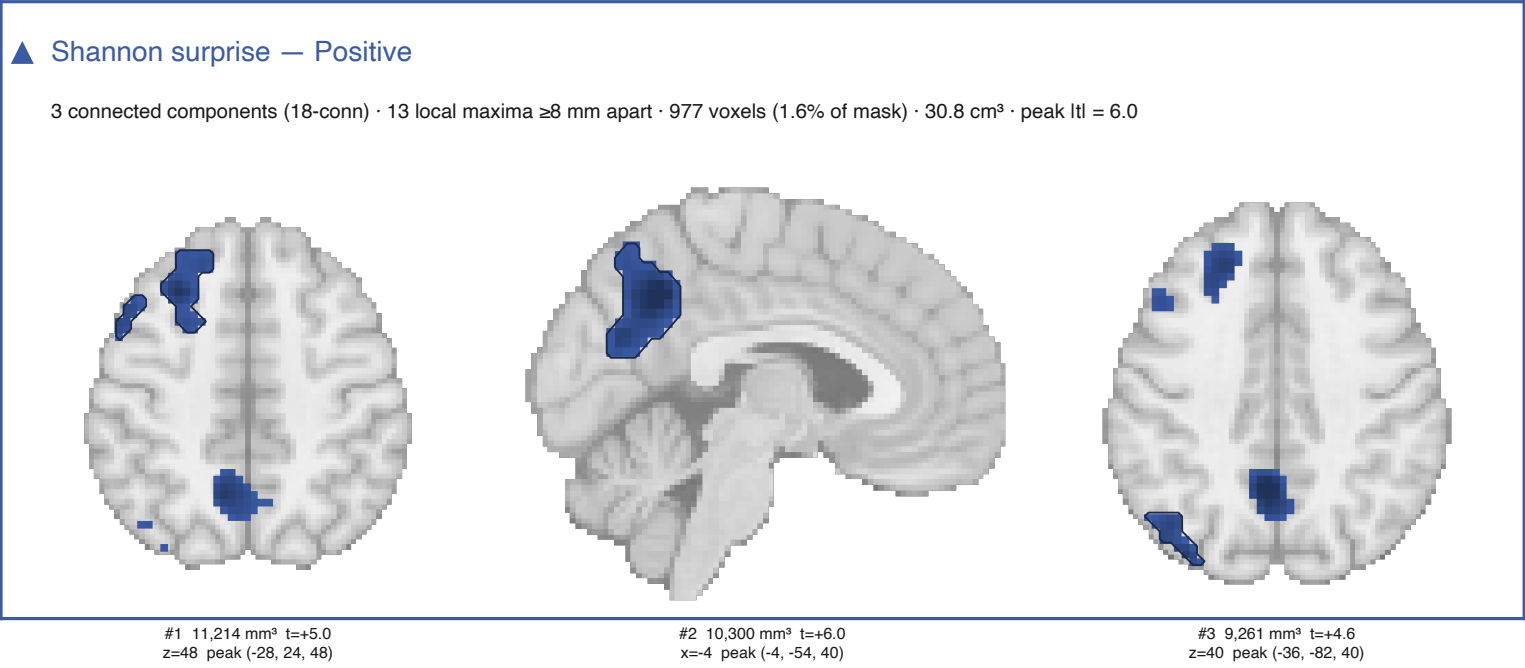
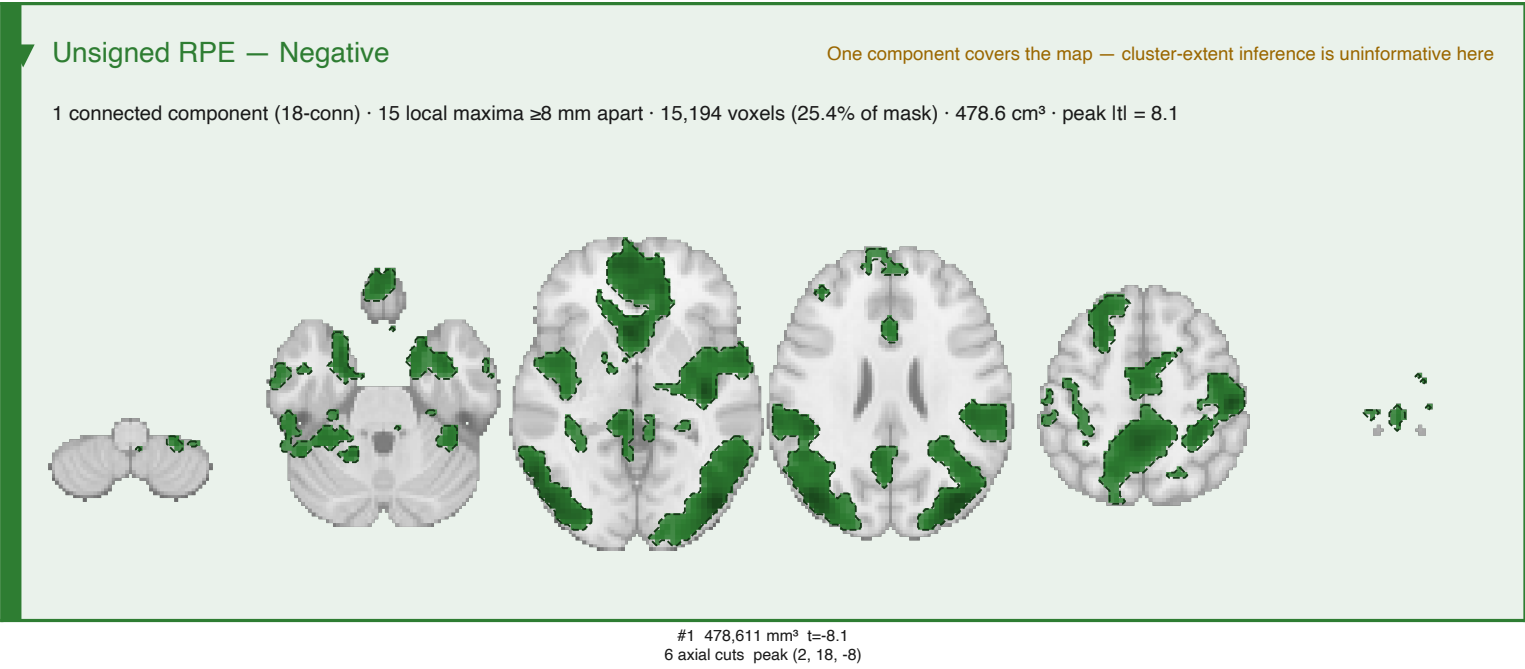
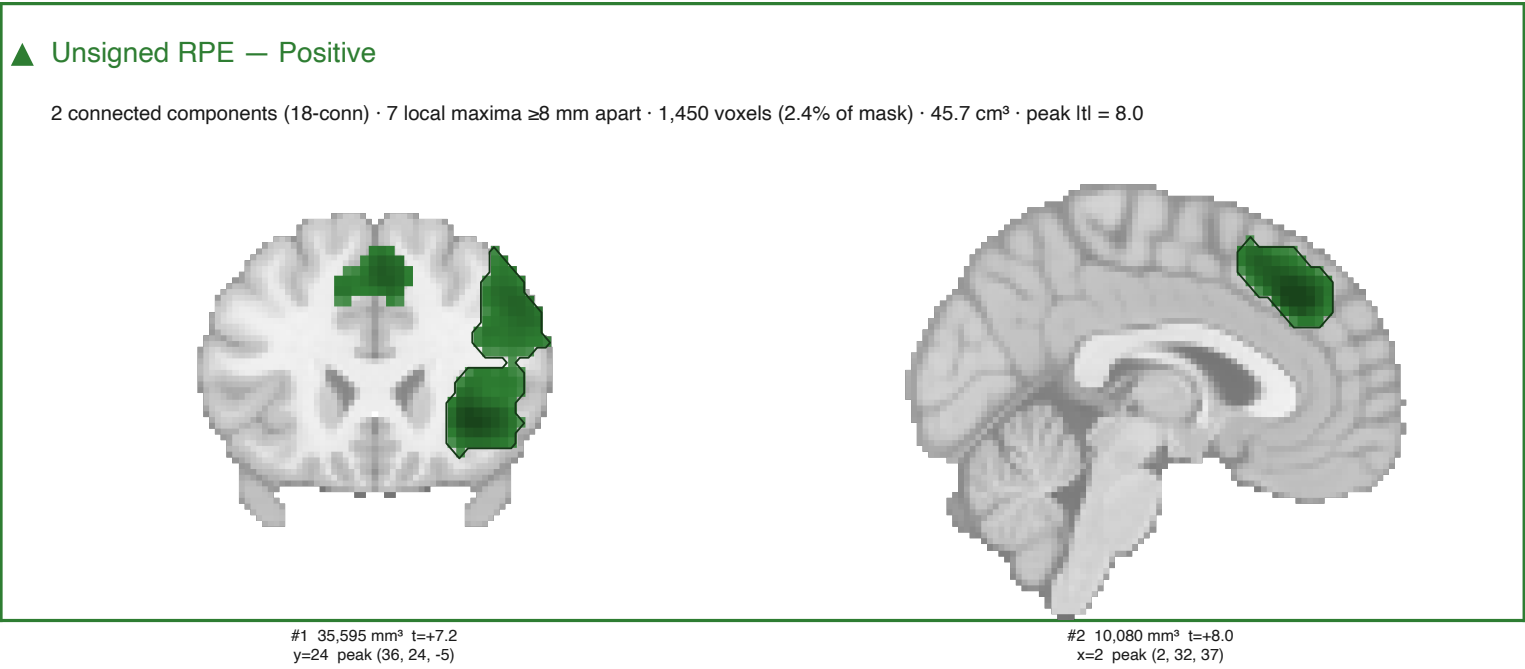
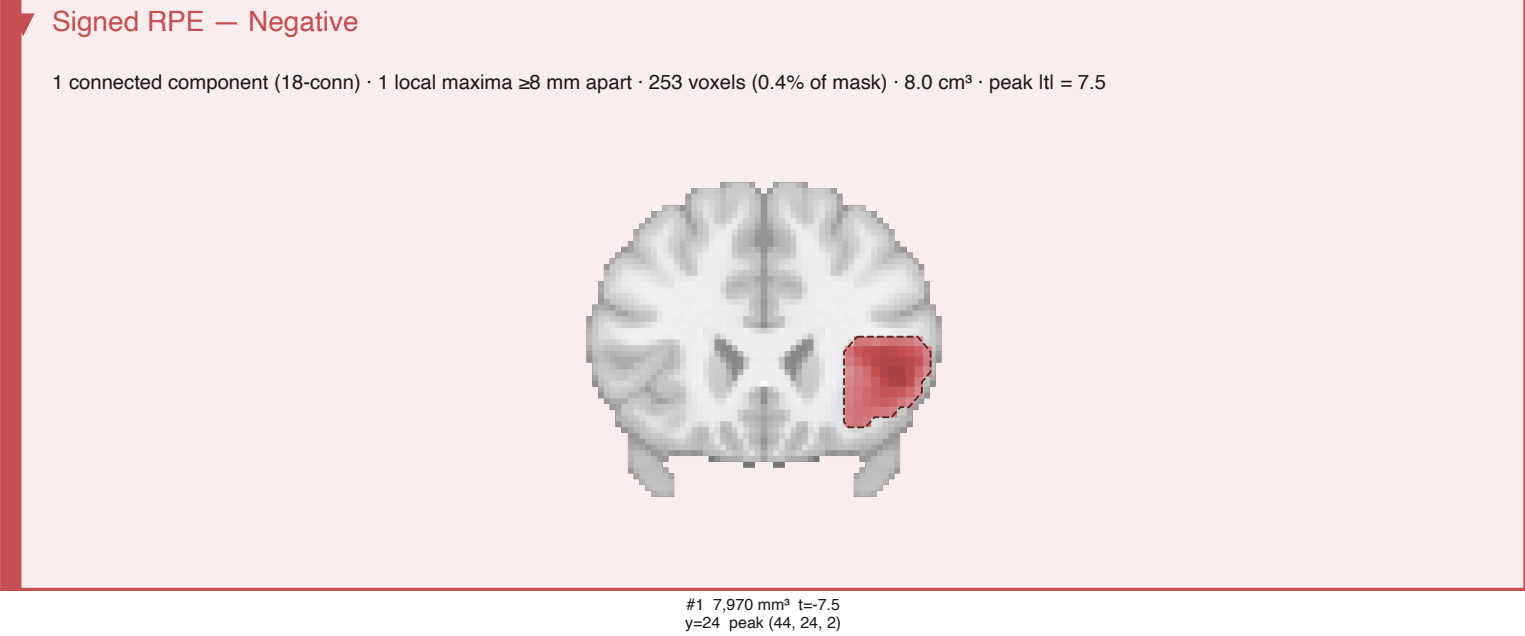
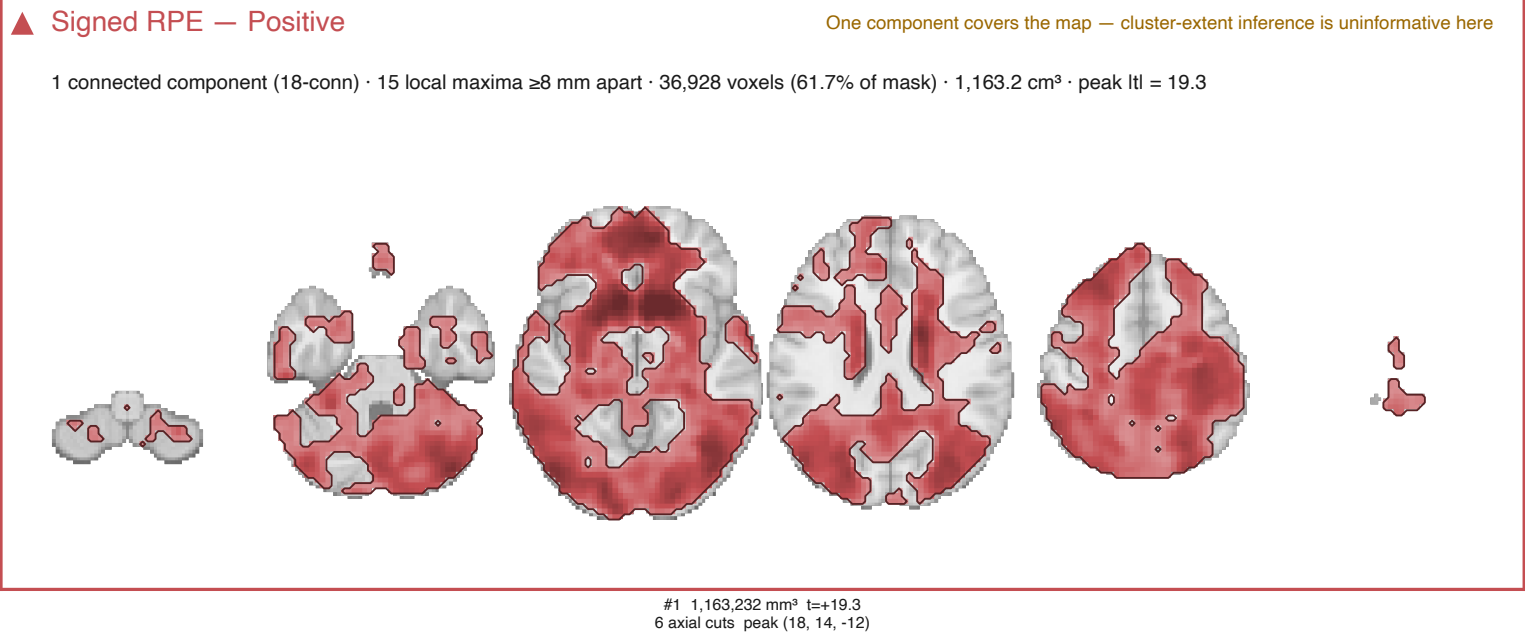
SnPM cluster-extent FWE — model2

Cluster-forming $t > 2.3936$ (one-tailed $p < 0.01$, $df = 57$); clusters with $p_FWE < .05$, 5000 permutations, one-tailed family per tail. Each slice is cut through one cluster's peak.



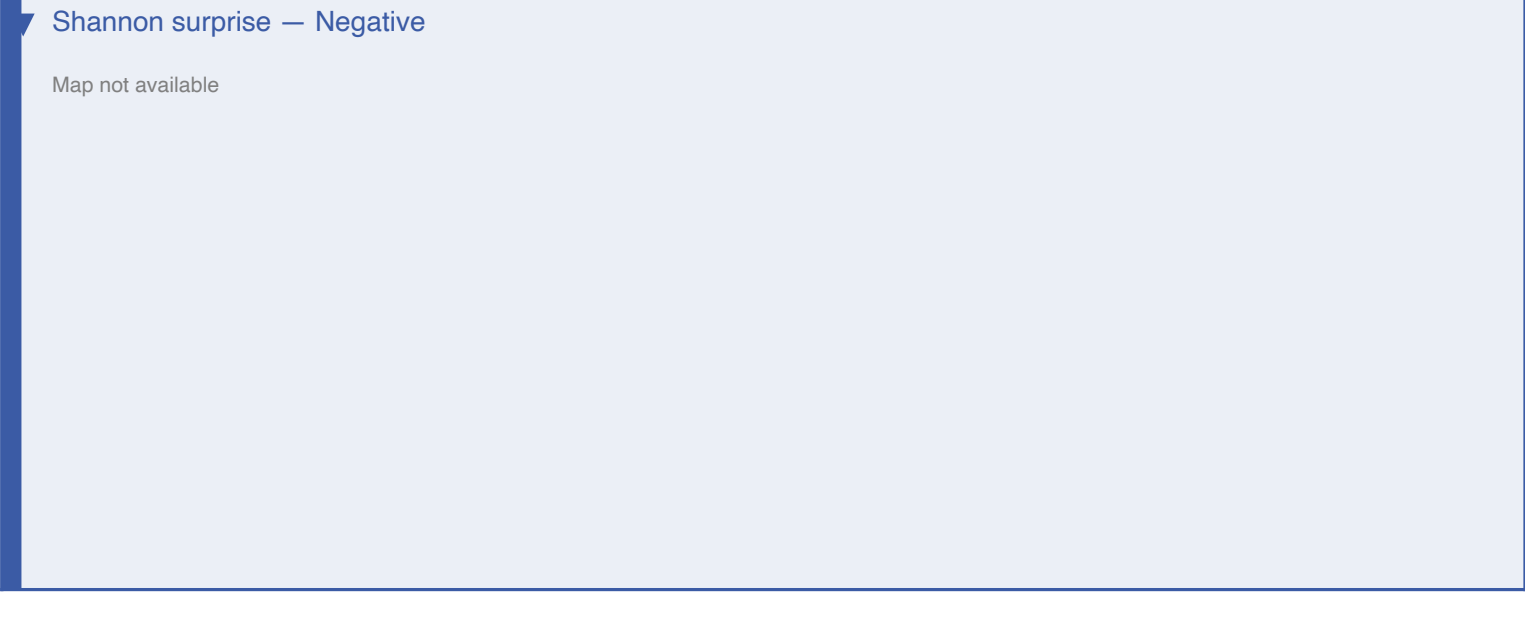
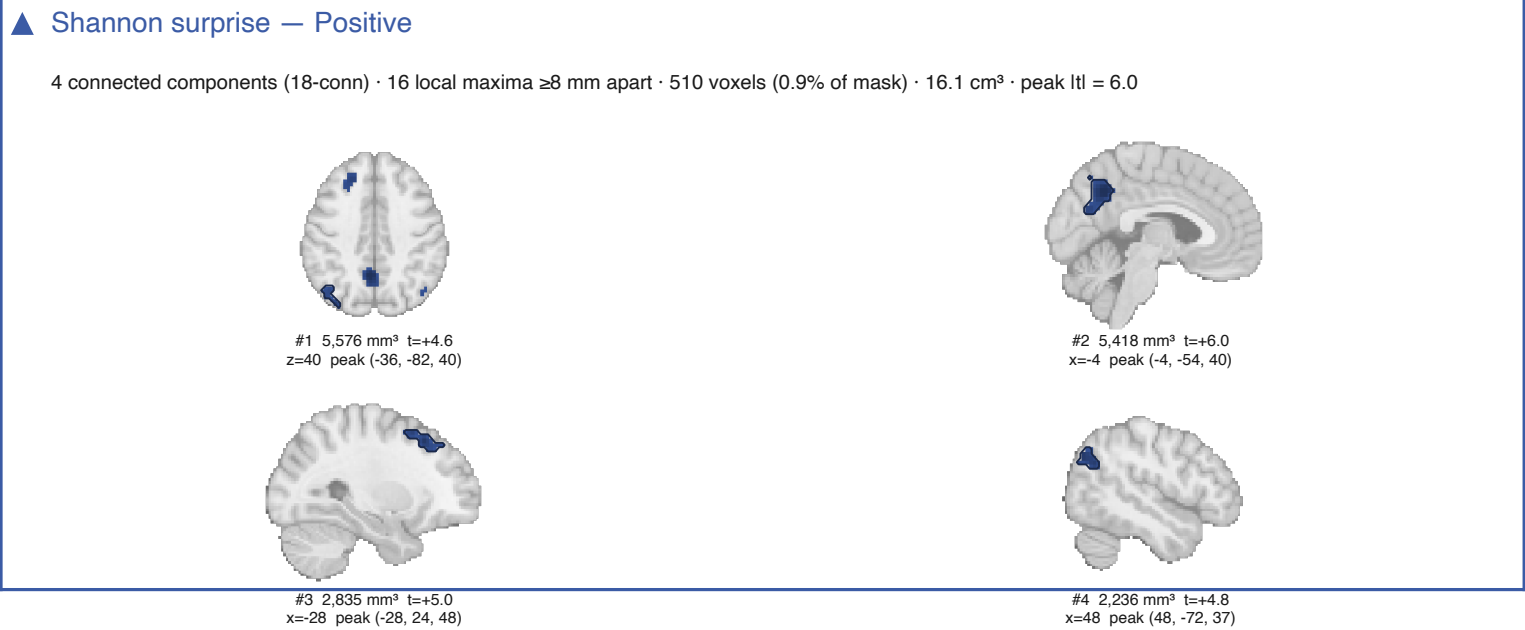
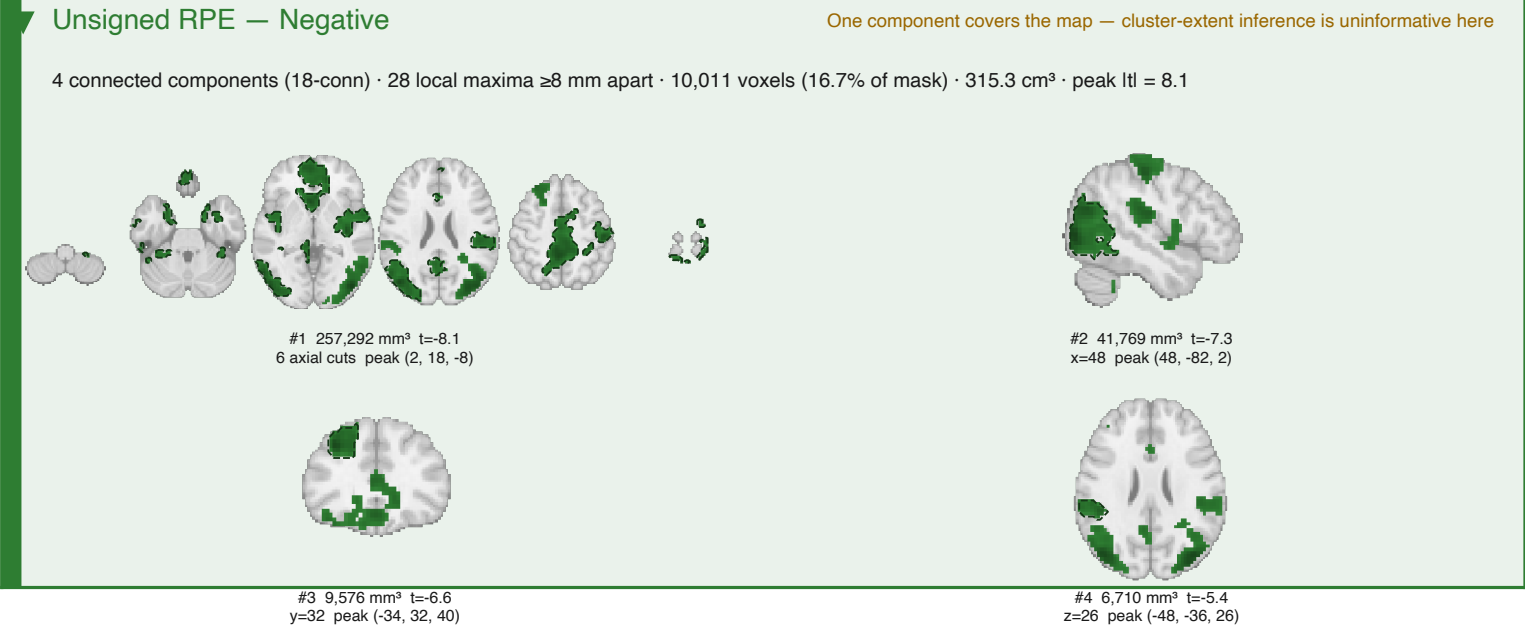
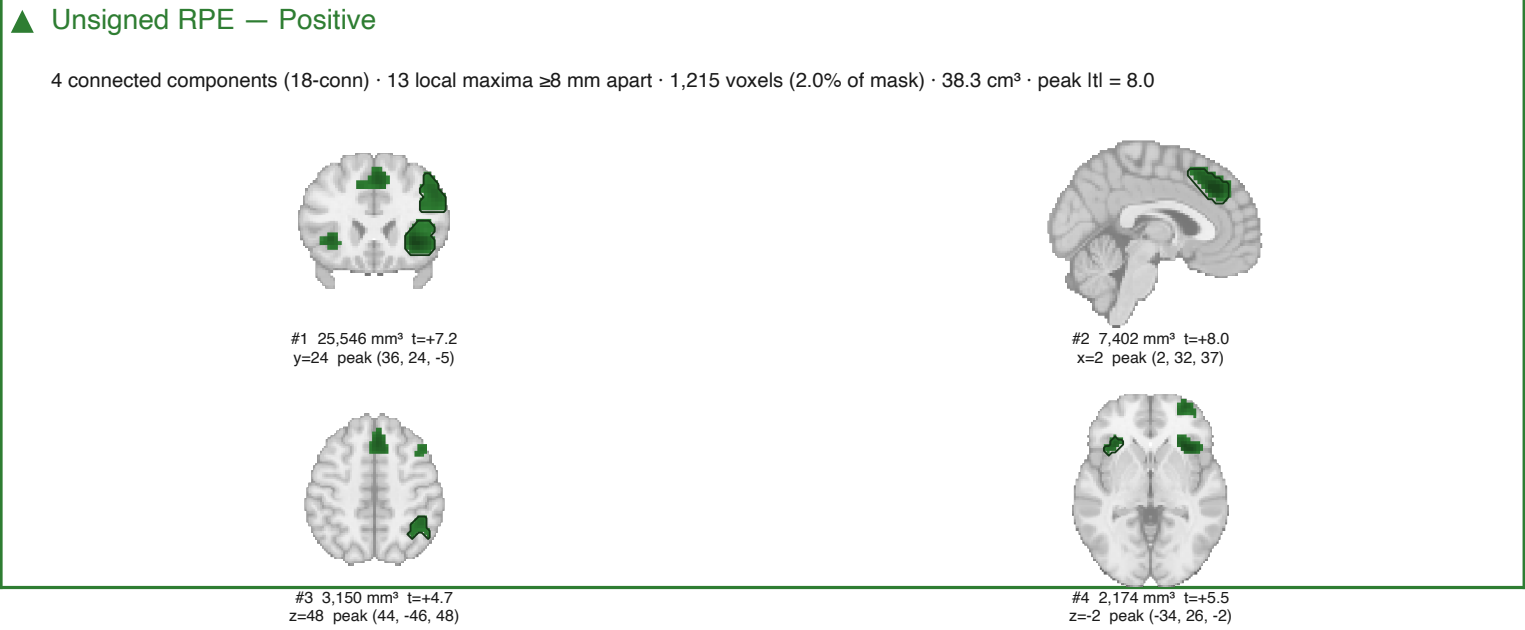
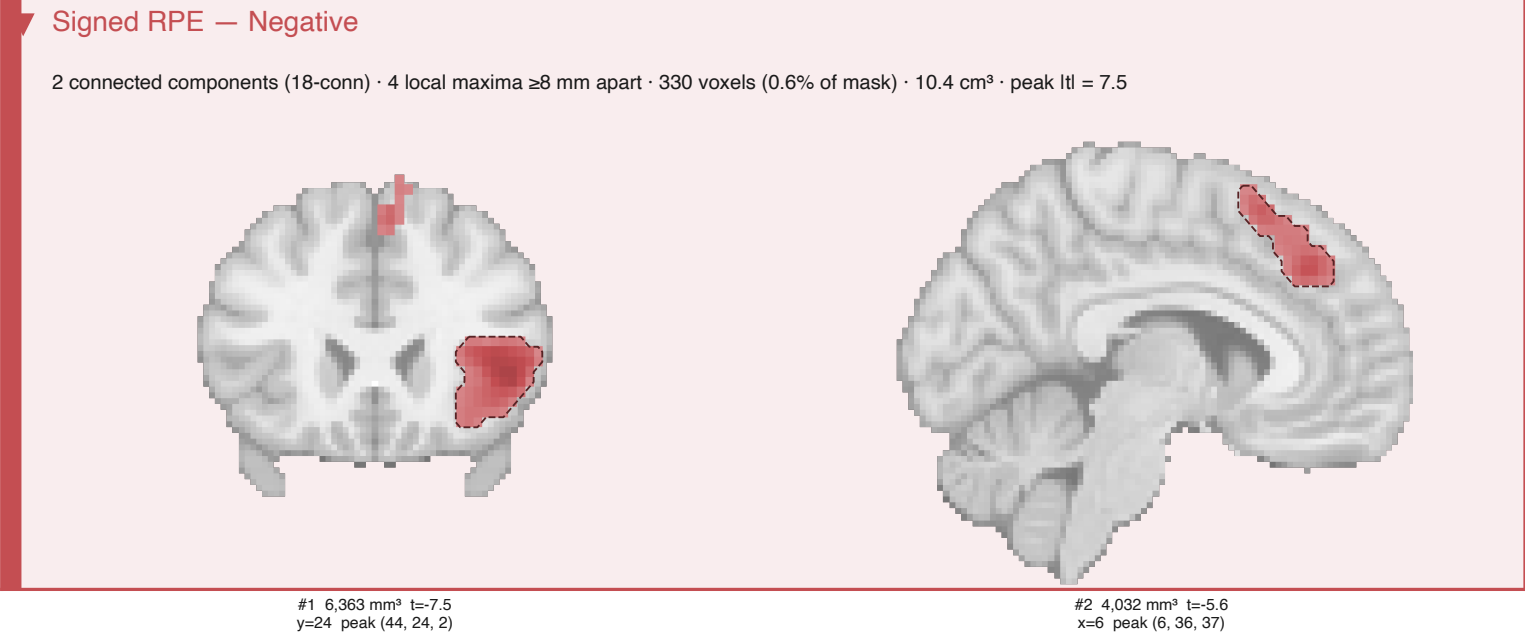
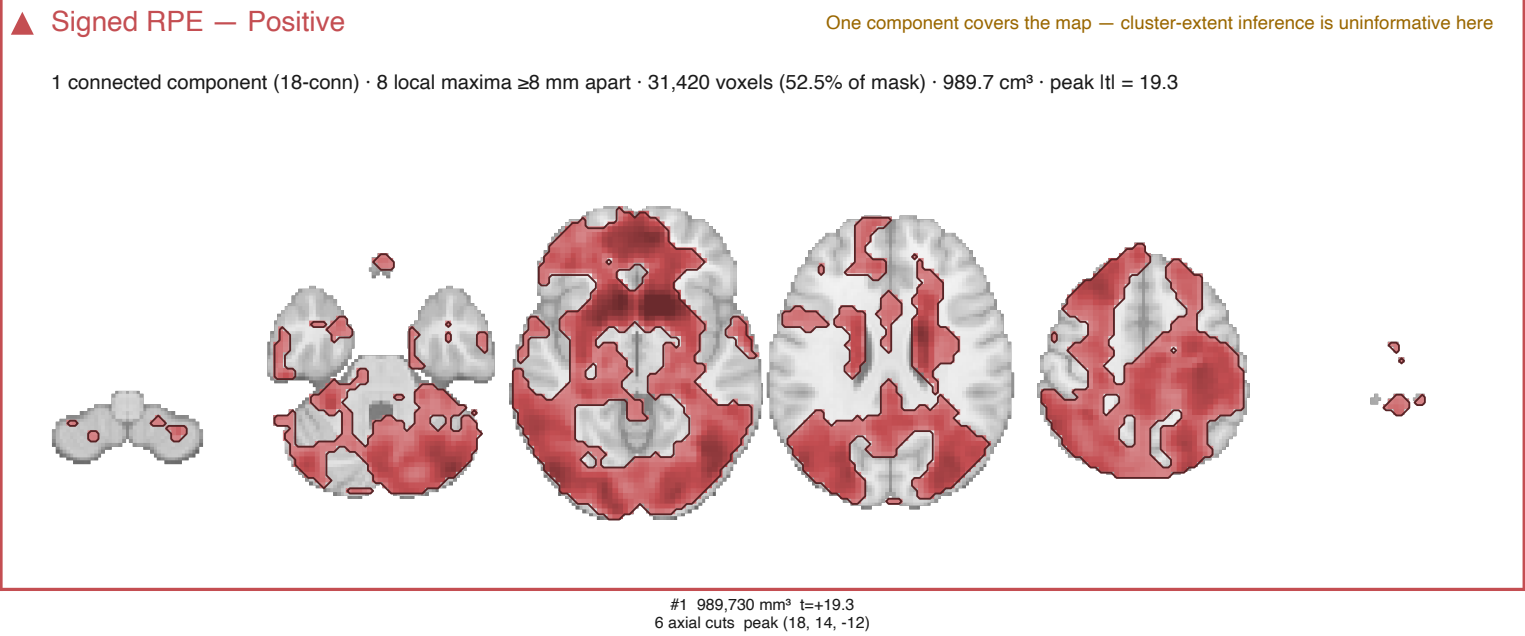
SnPM cluster-extent FWE — model2

Cluster-forming $t > 2.6649$ (one-tailed $p < 0.005$, $df = 57$); clusters with $p_FWE < .05$, 5000 permutations, one-tailed family per tail. Each slice is cut through one cluster's peak.



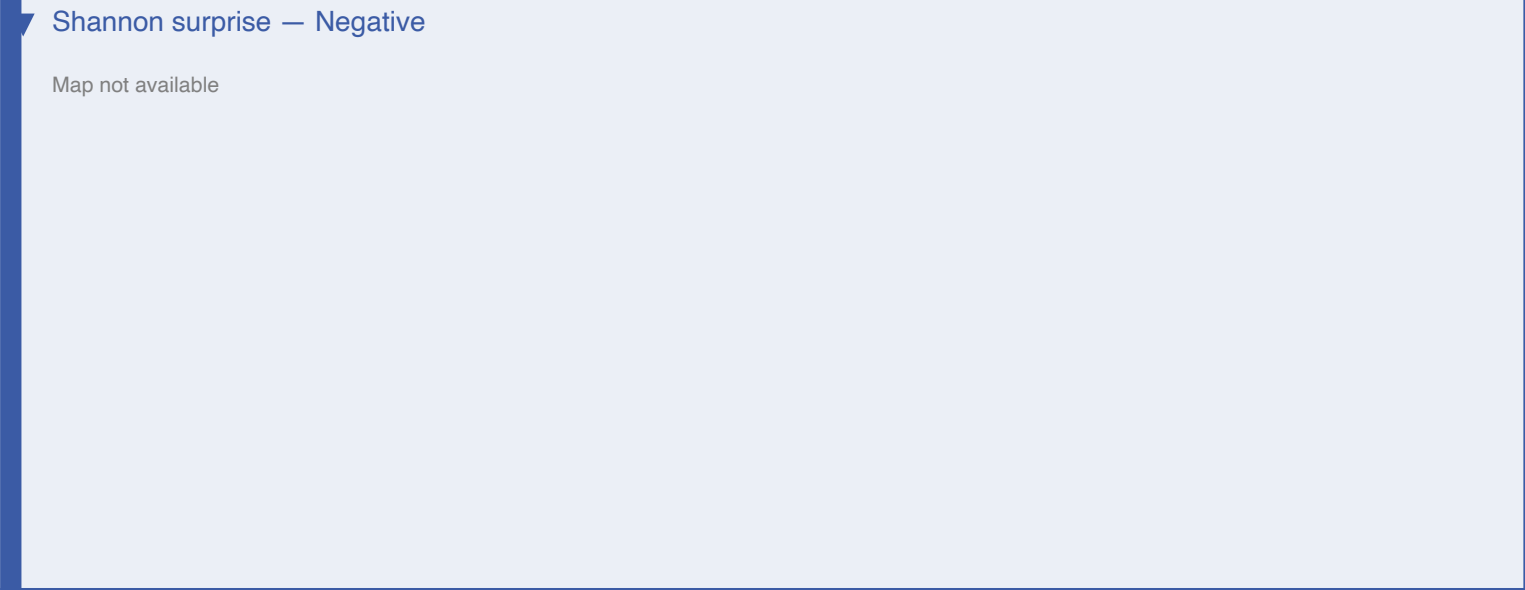
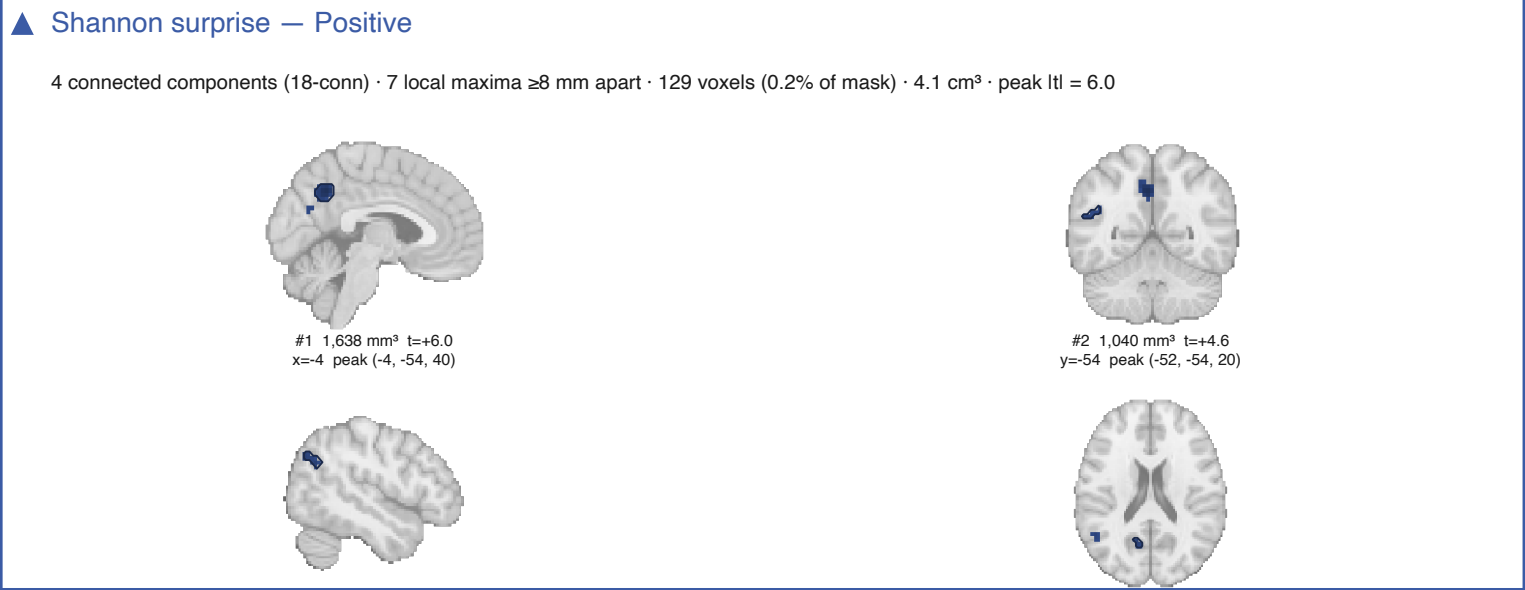
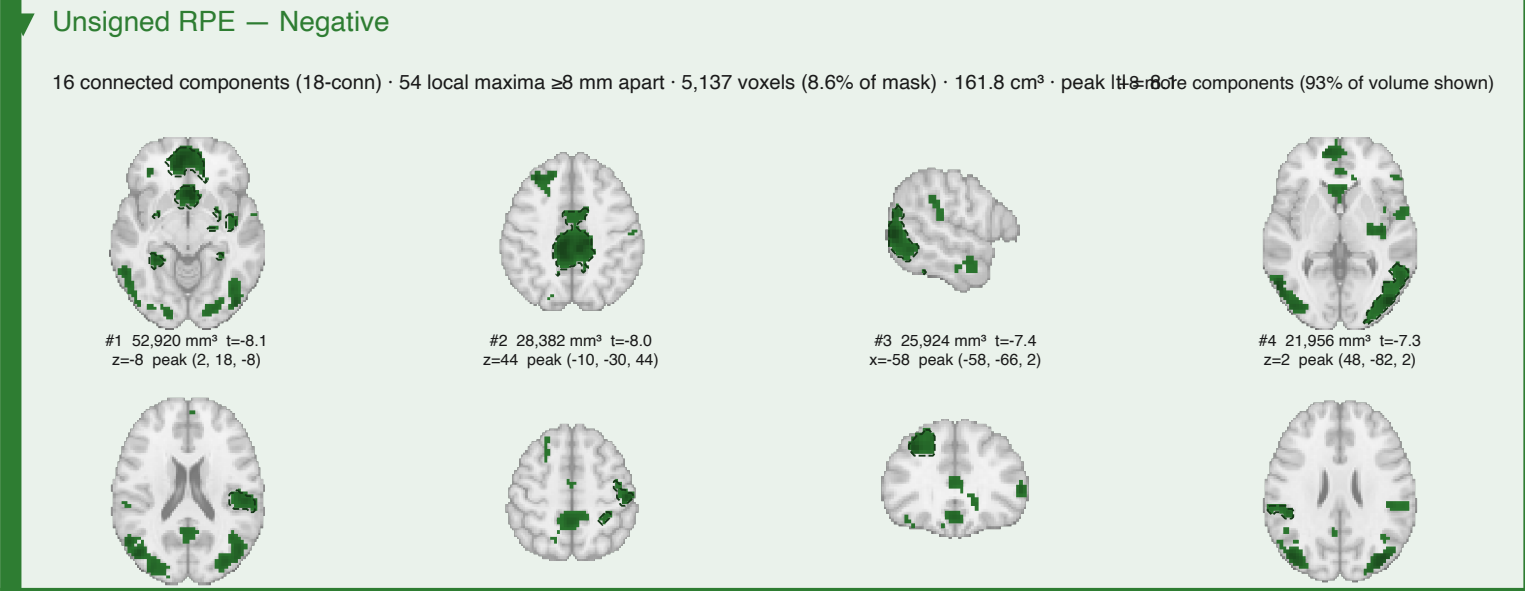
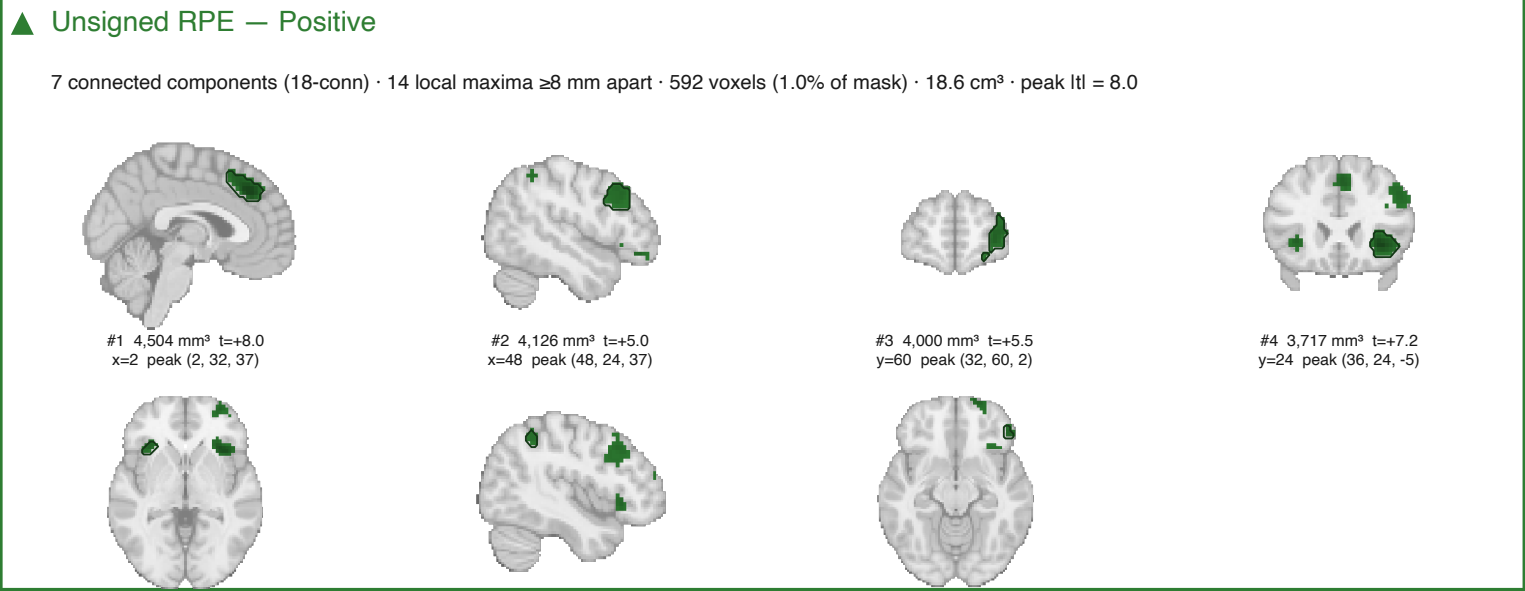
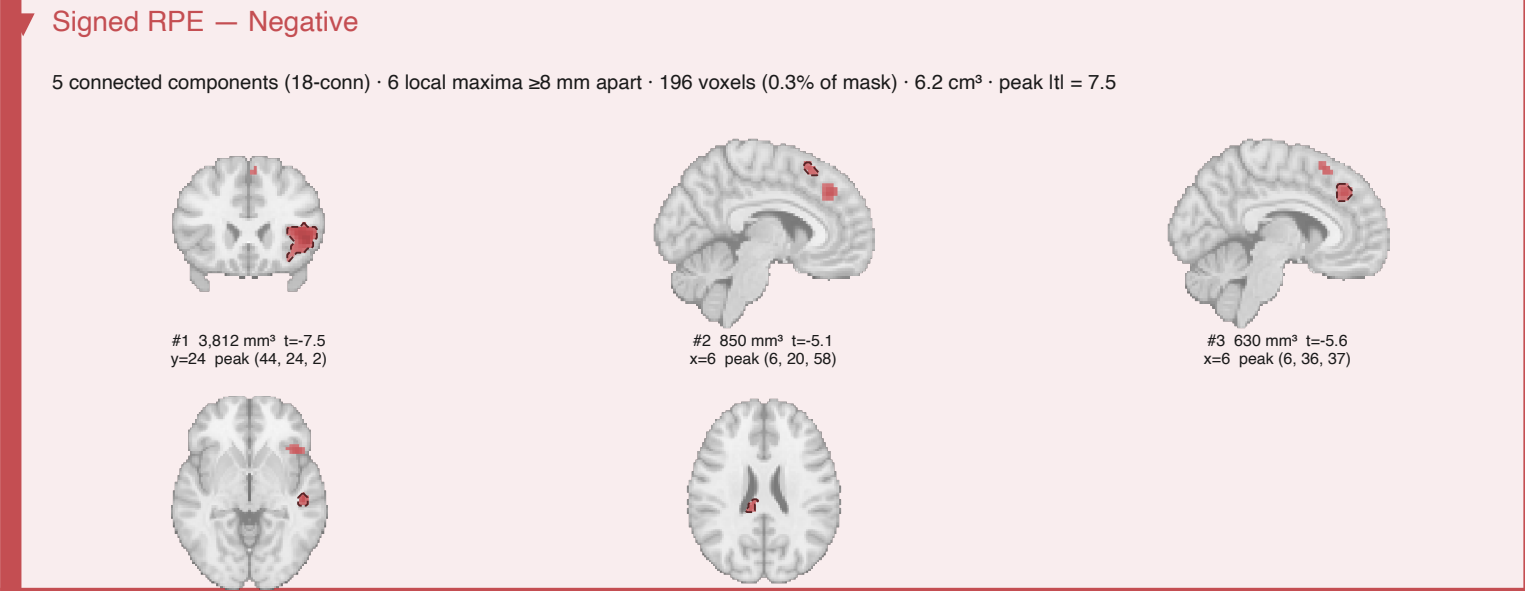
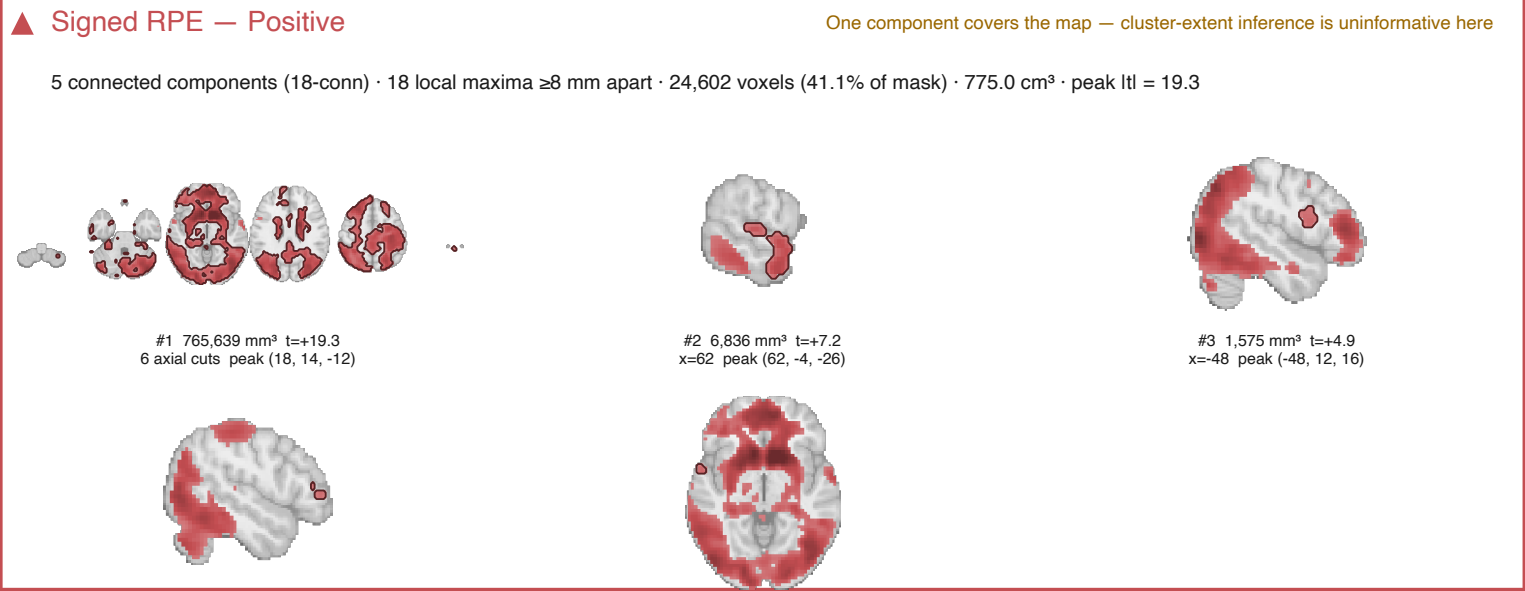
SnPM cluster-extent FWE — model2

Cluster-forming $t > 3.2395$ (one-tailed $p < 0.001$, $df = 57$); clusters with $p_FWE < .05$, 5000 permutations, one-tailed family per tail. Each slice is cut through one cluster's peak.



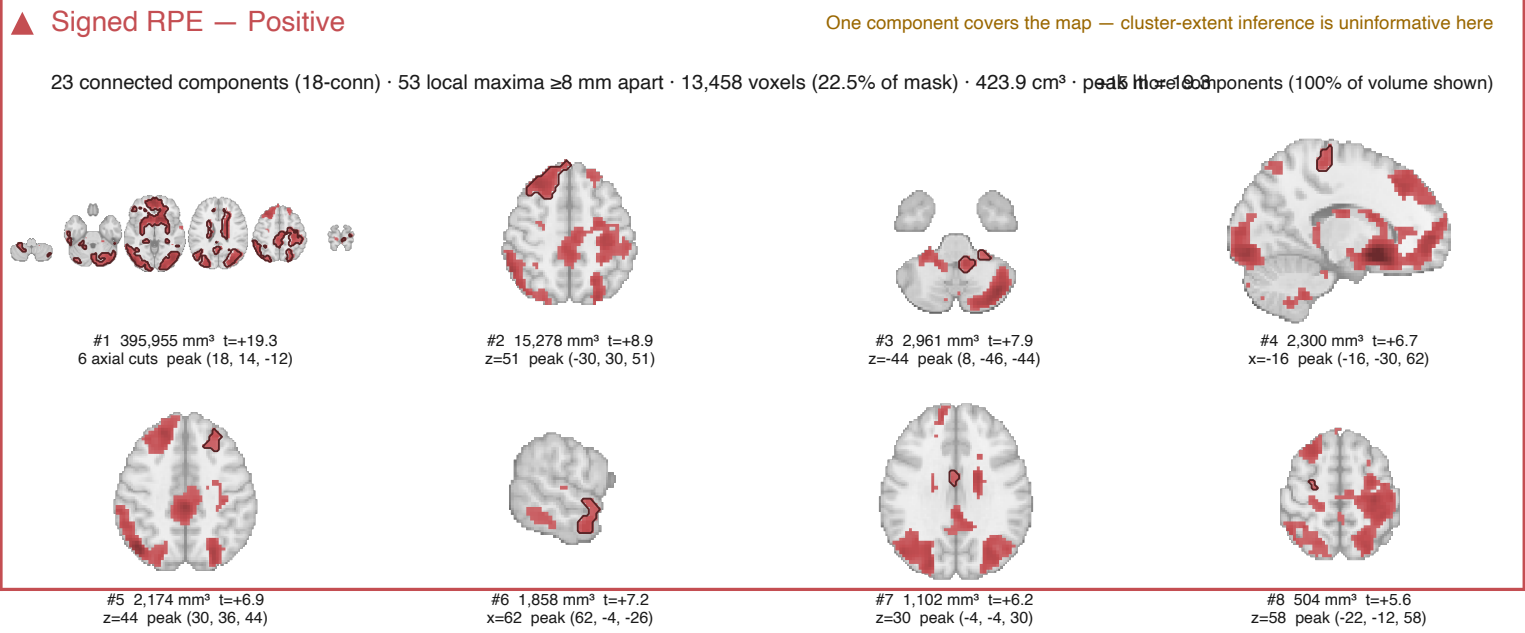
SnPM cluster-extent FWE — model2

Cluster-forming $t > 3.9756$ (one-tailed $p < 0.0001$, $df = 57$); clusters with $p_FWE < .05$, 5000 permutations, one-tailed family per tail. Each slice is cut through one cluster's peak.



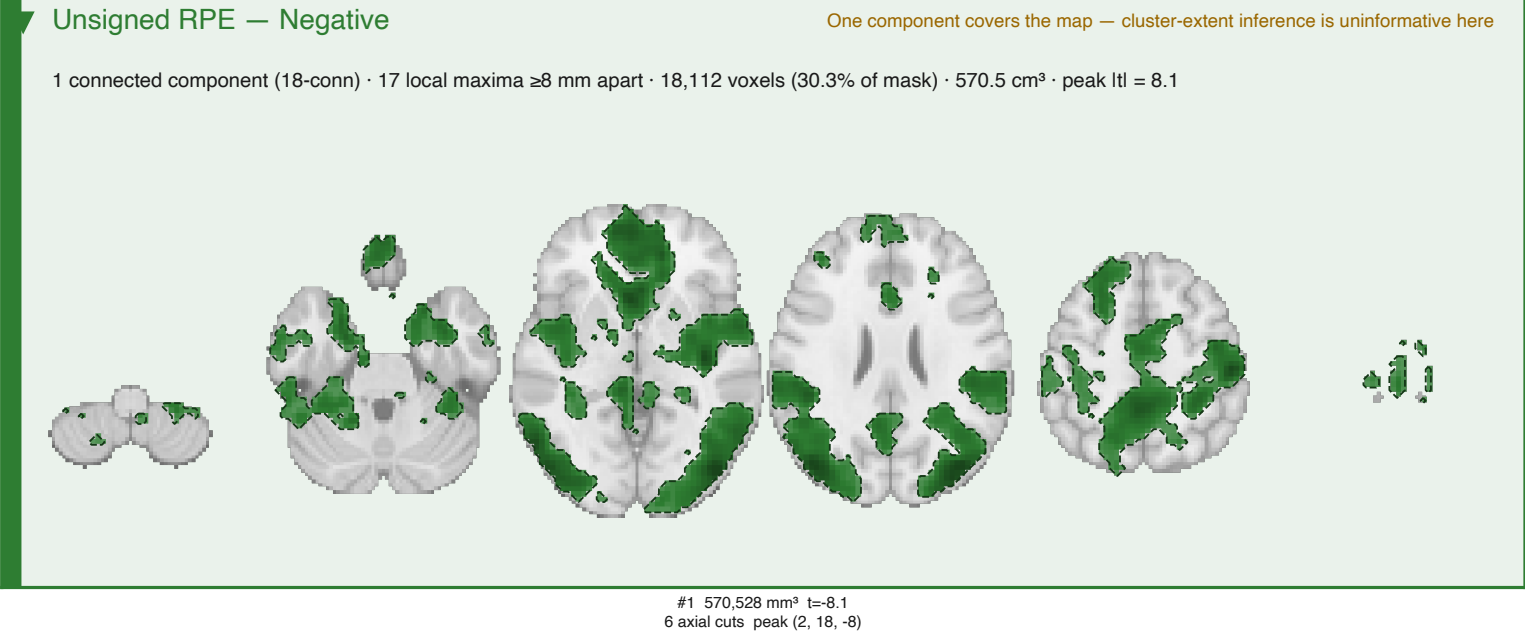
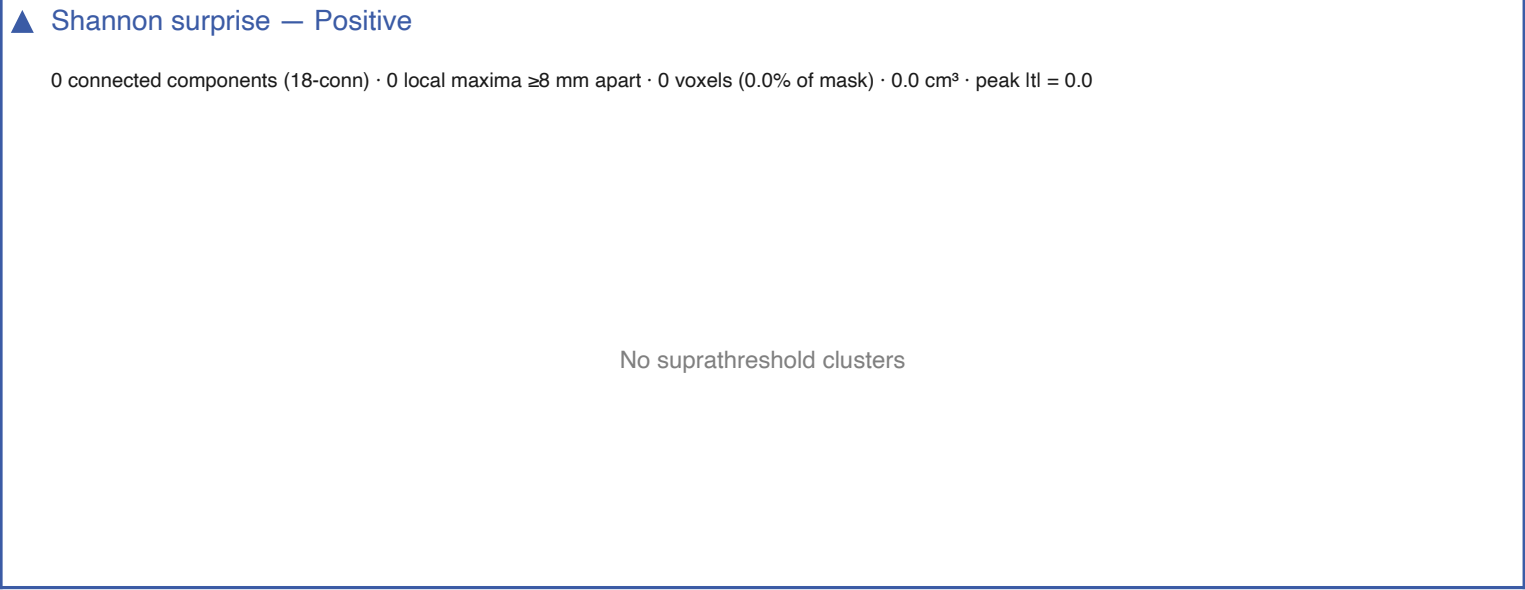
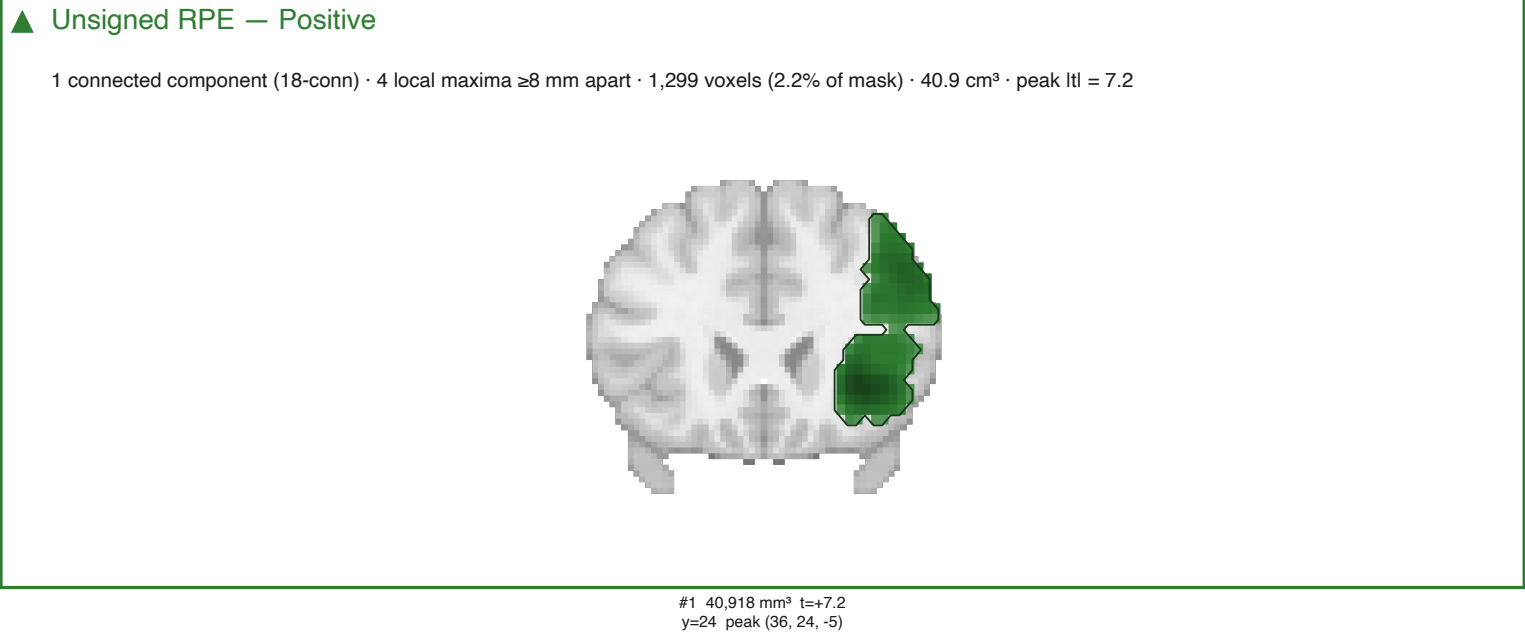
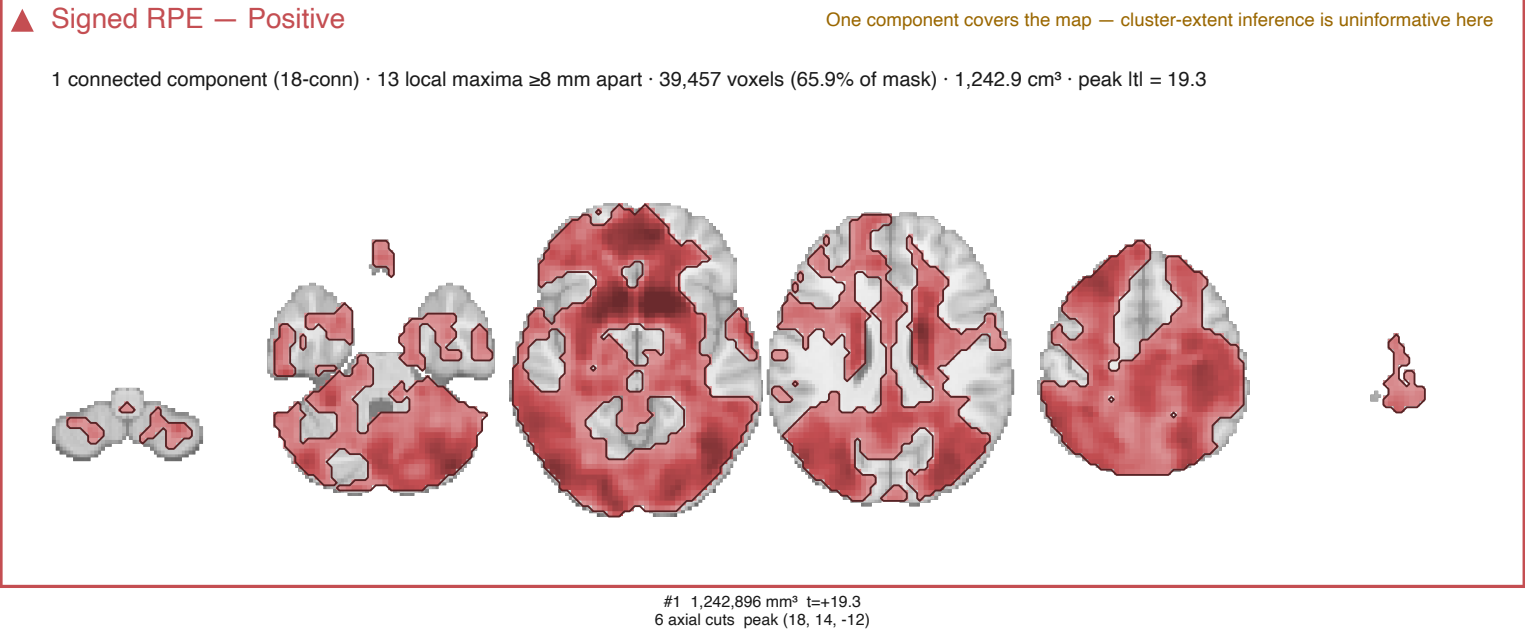
SnPM cluster-extent FWE — model2

Cluster-forming $t > 5.2929$ (one-tailed $p < 1e-06$, $df = 57$); clusters with $p_FWE < .05$, 5000 permutations, one-tailed family per tail. Each slice is cut through one cluster's peak.



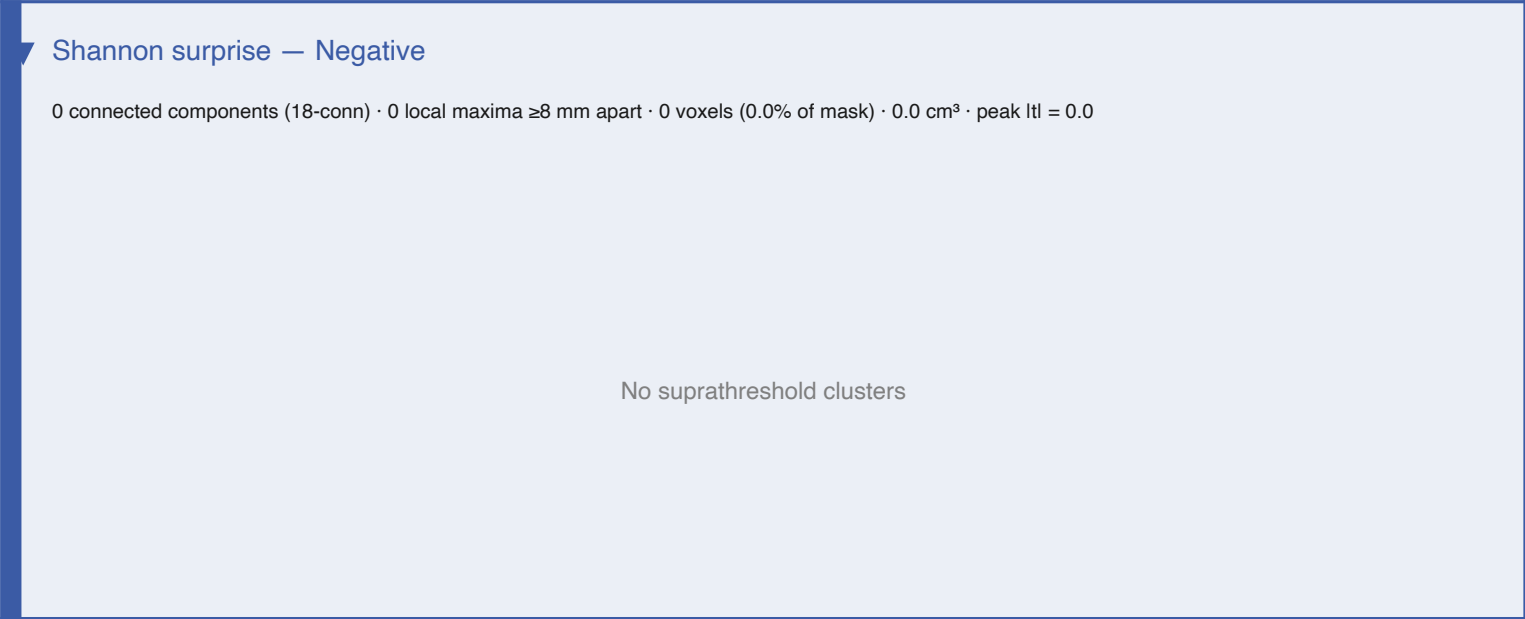
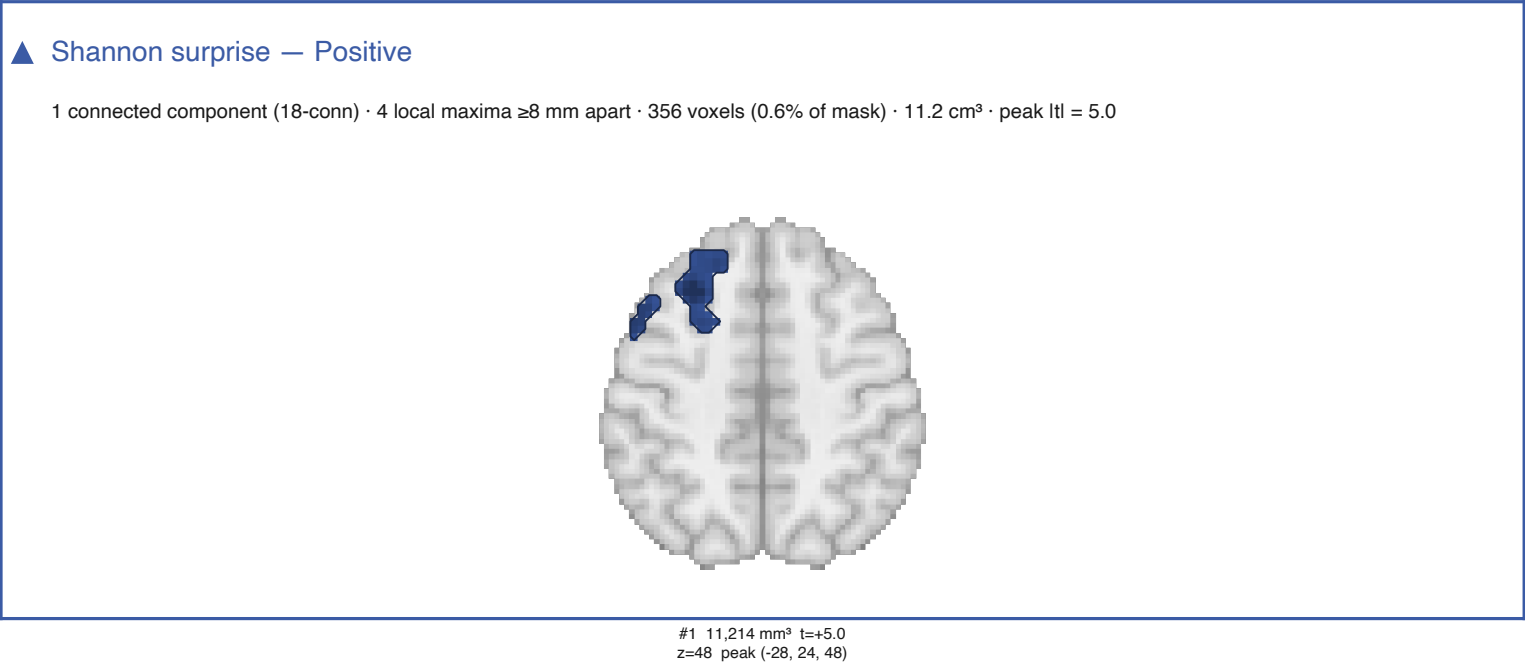
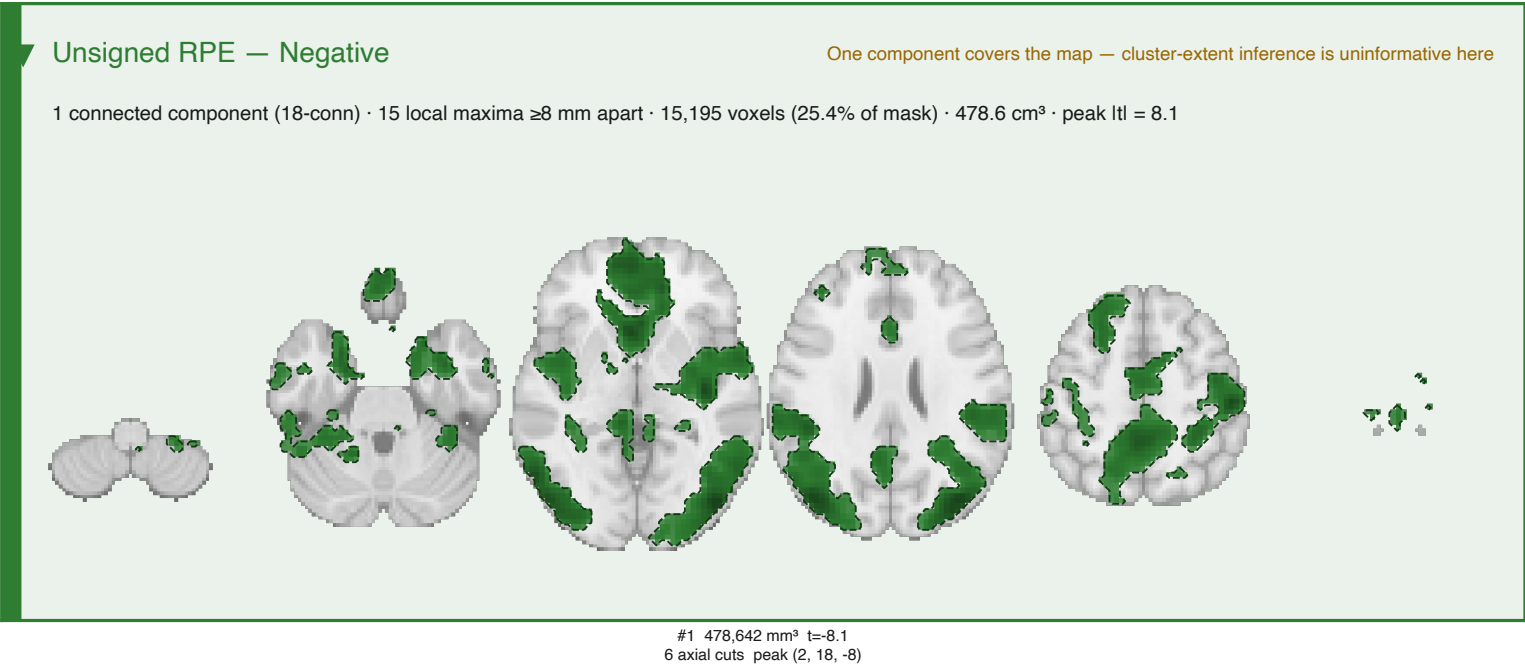
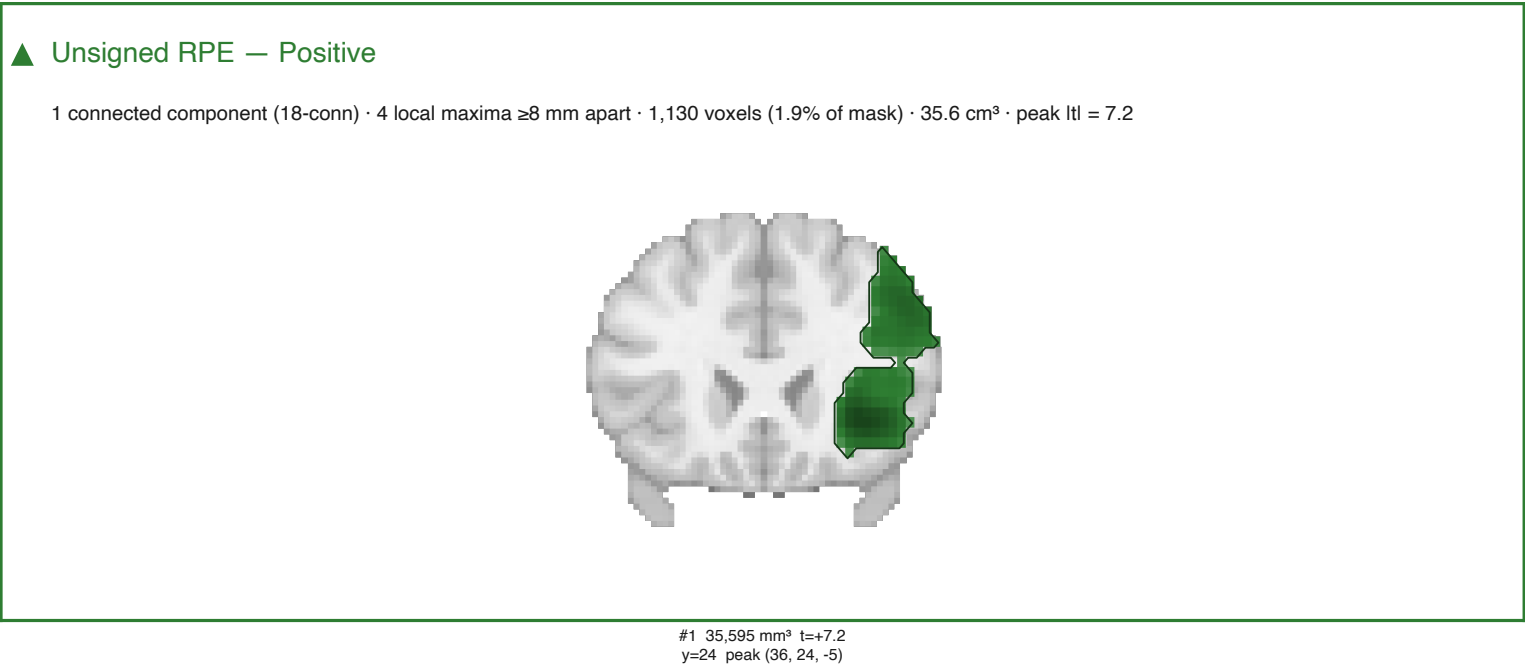
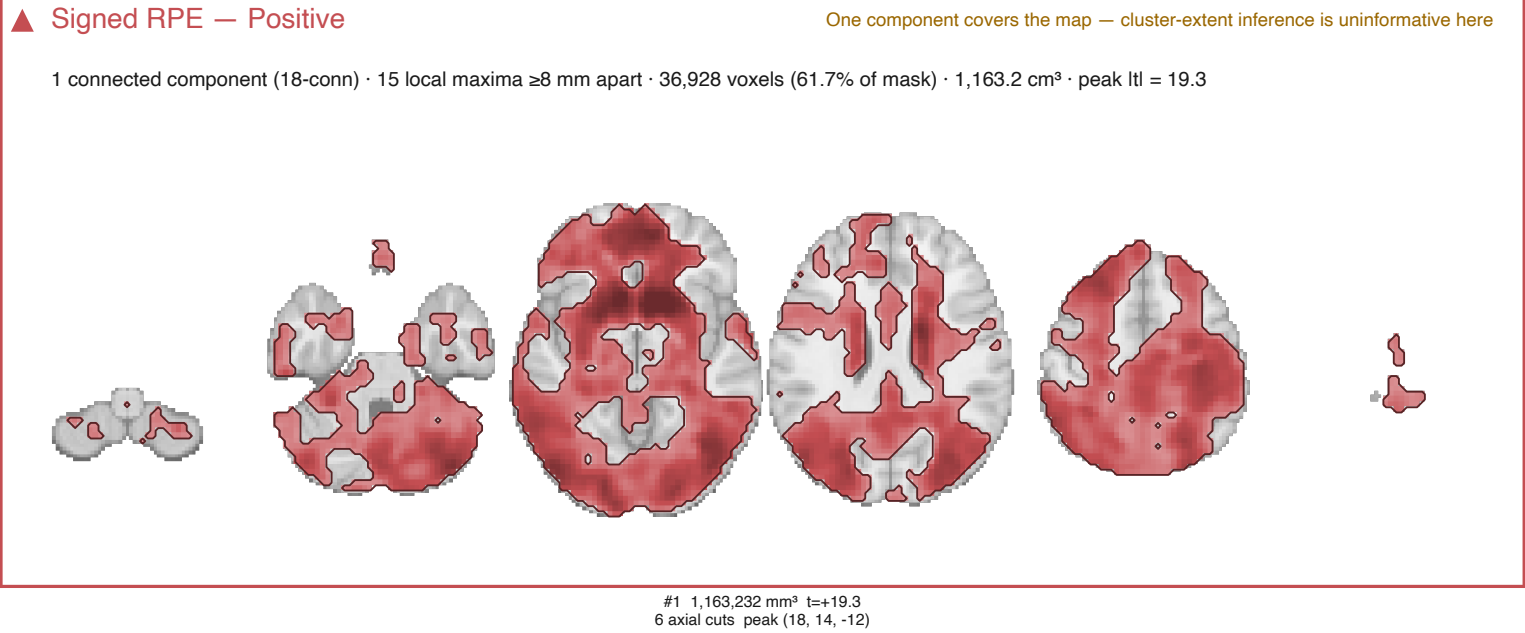
nilearn cluster-extent FWE — model2

Cluster-forming $t > 2.3936$ (one-tailed $p < 0.01$); clusters with two-tailed $p_{\text{FWE}} < .05$, 5000 sign flips, 18-connectivity.



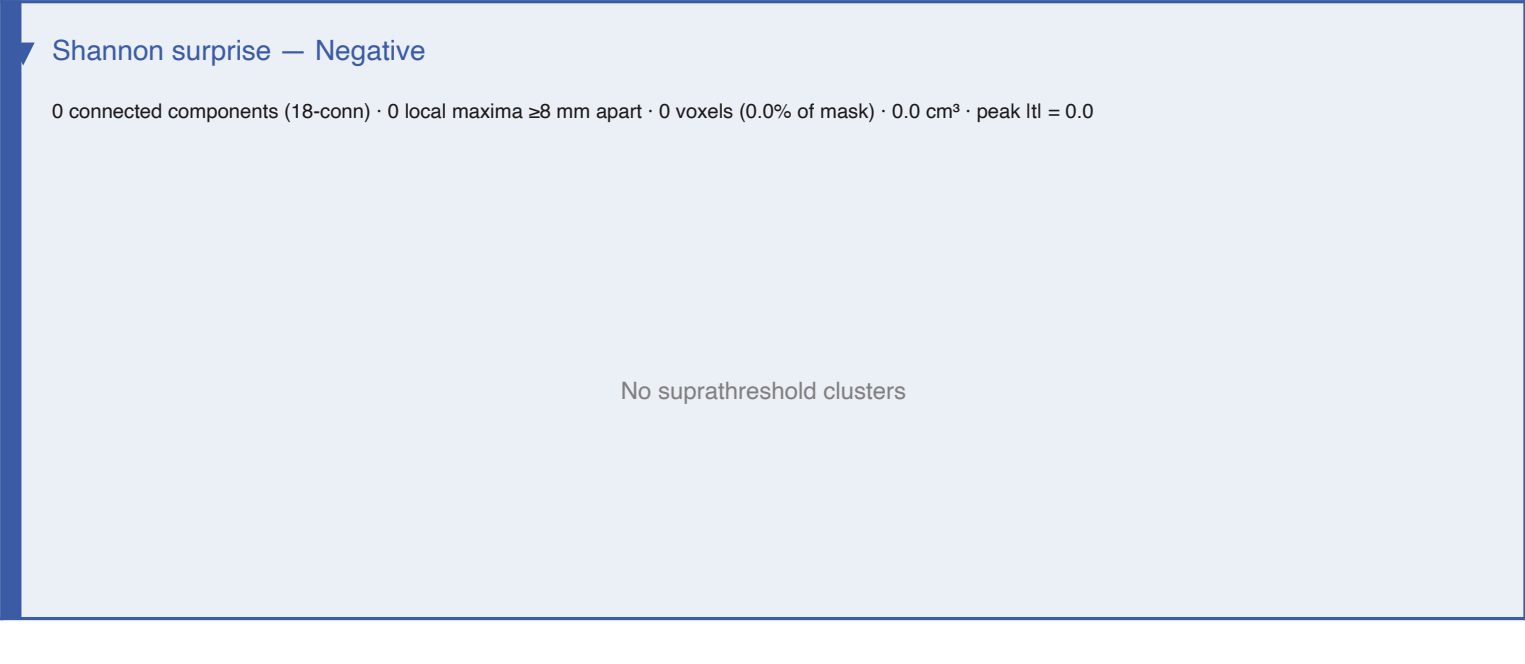
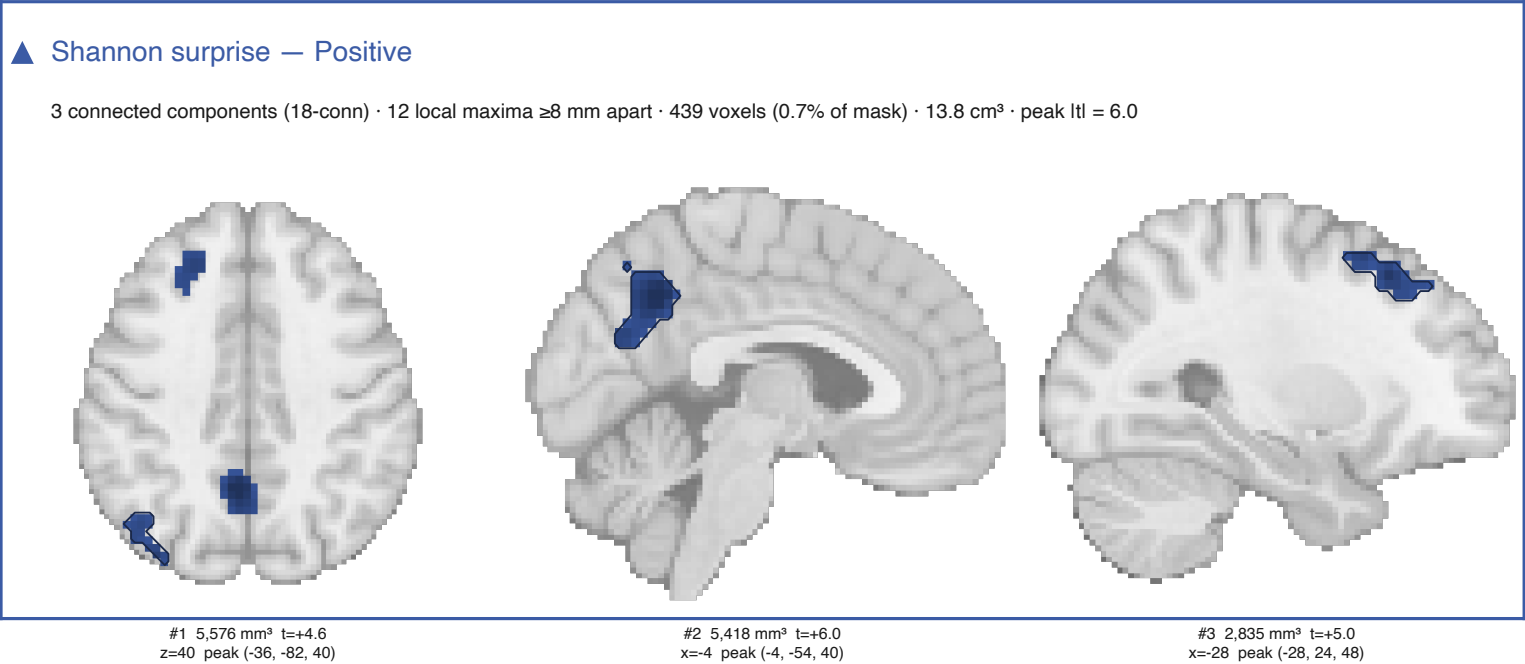
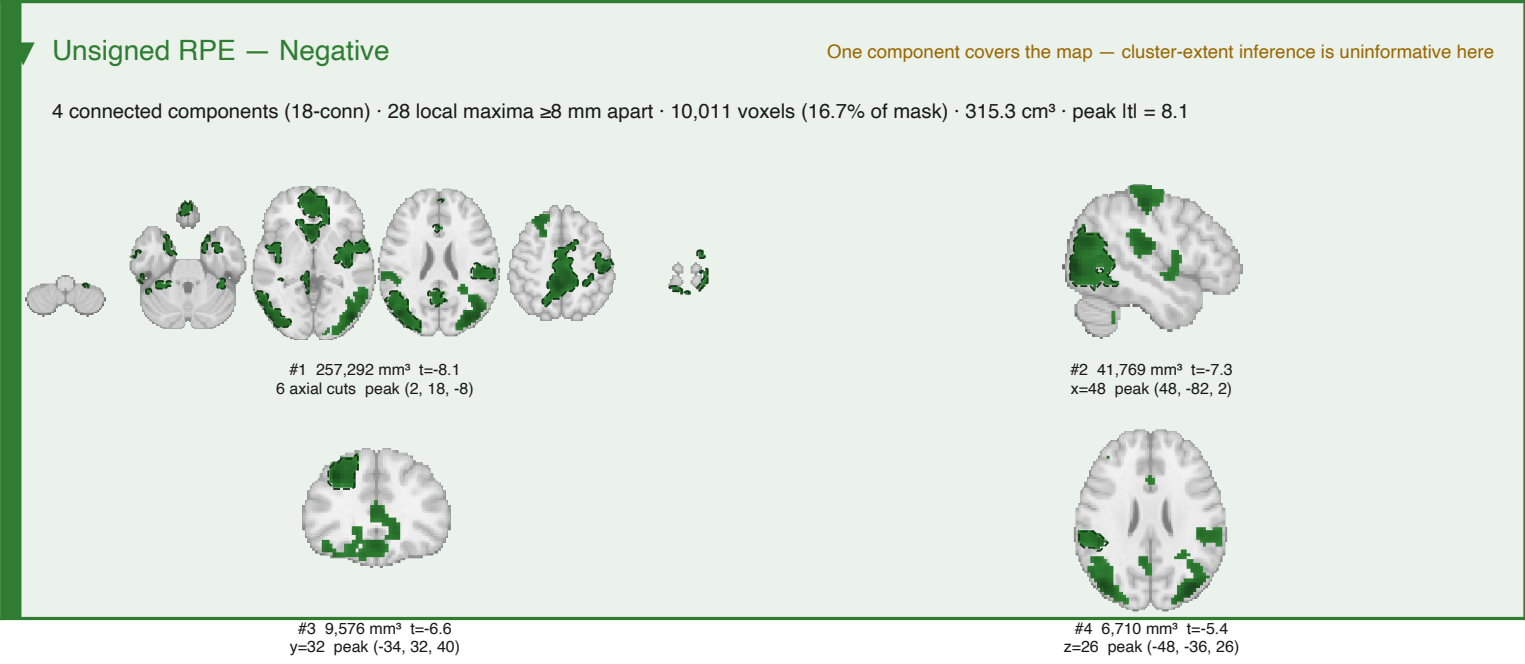
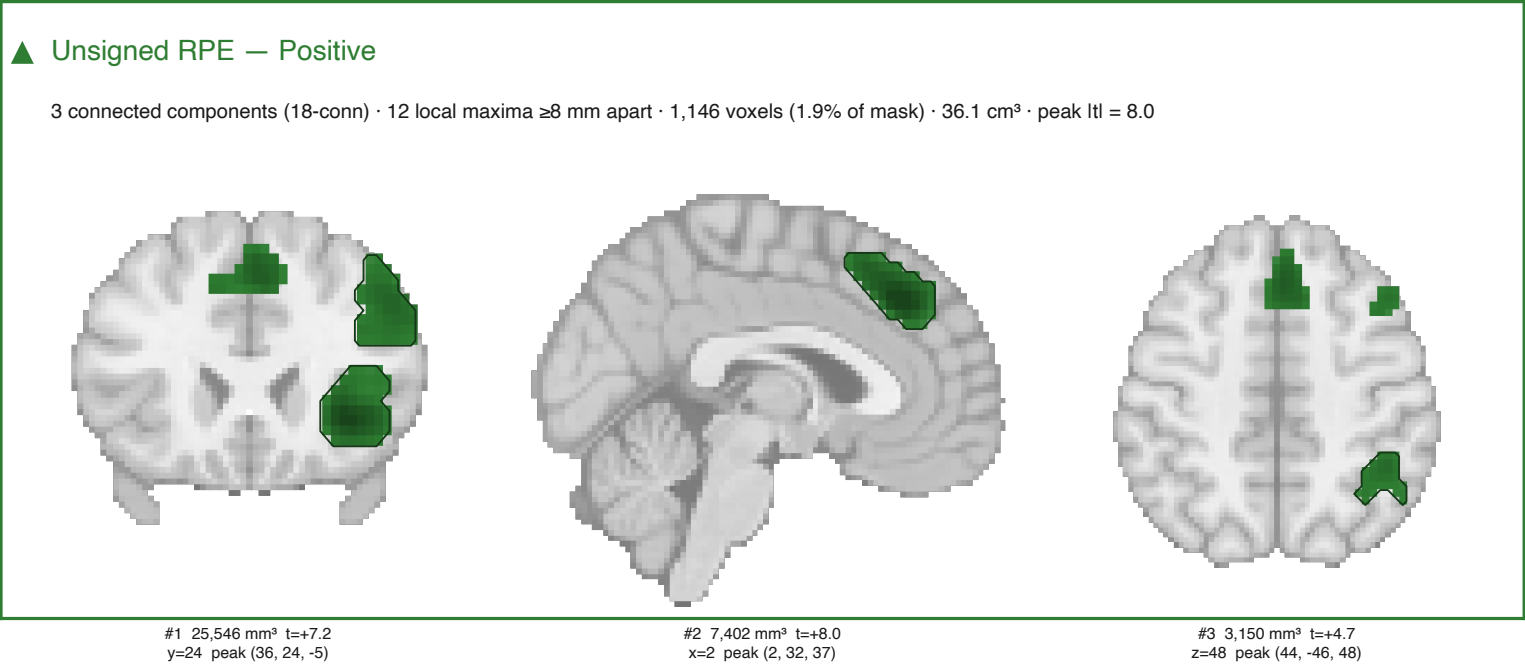
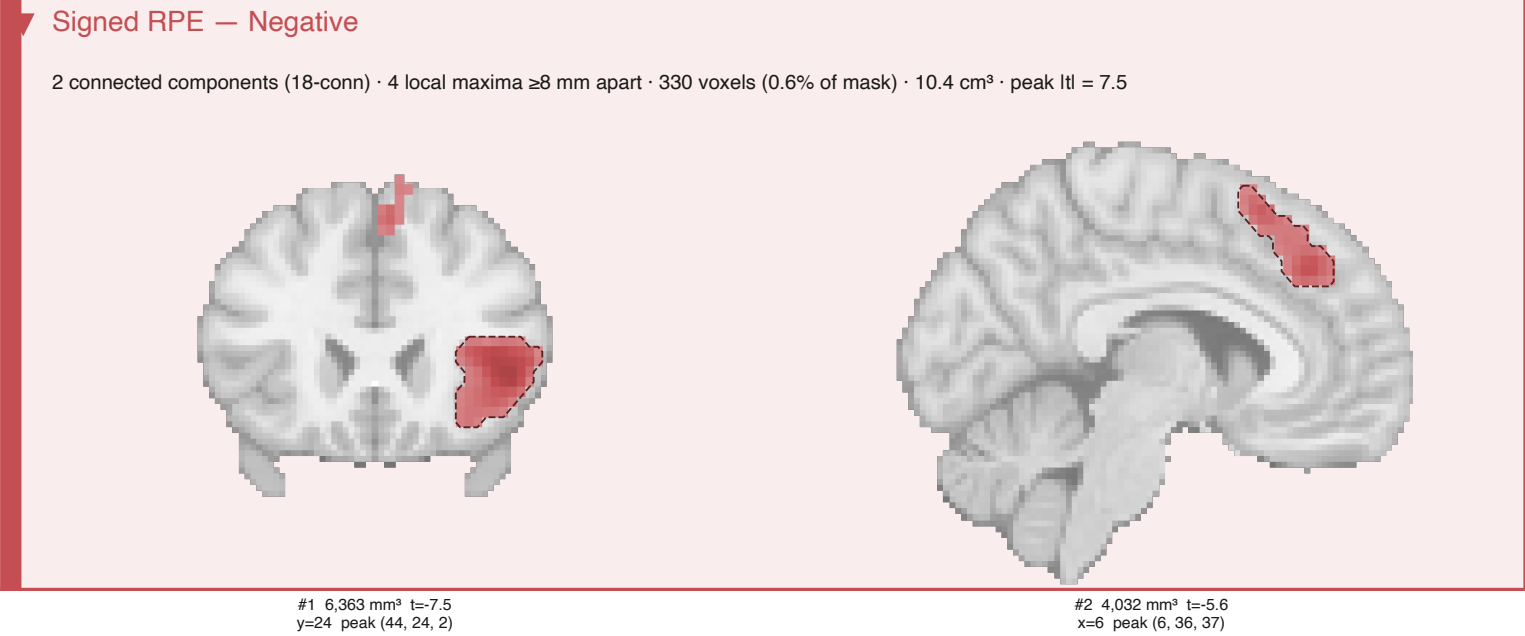
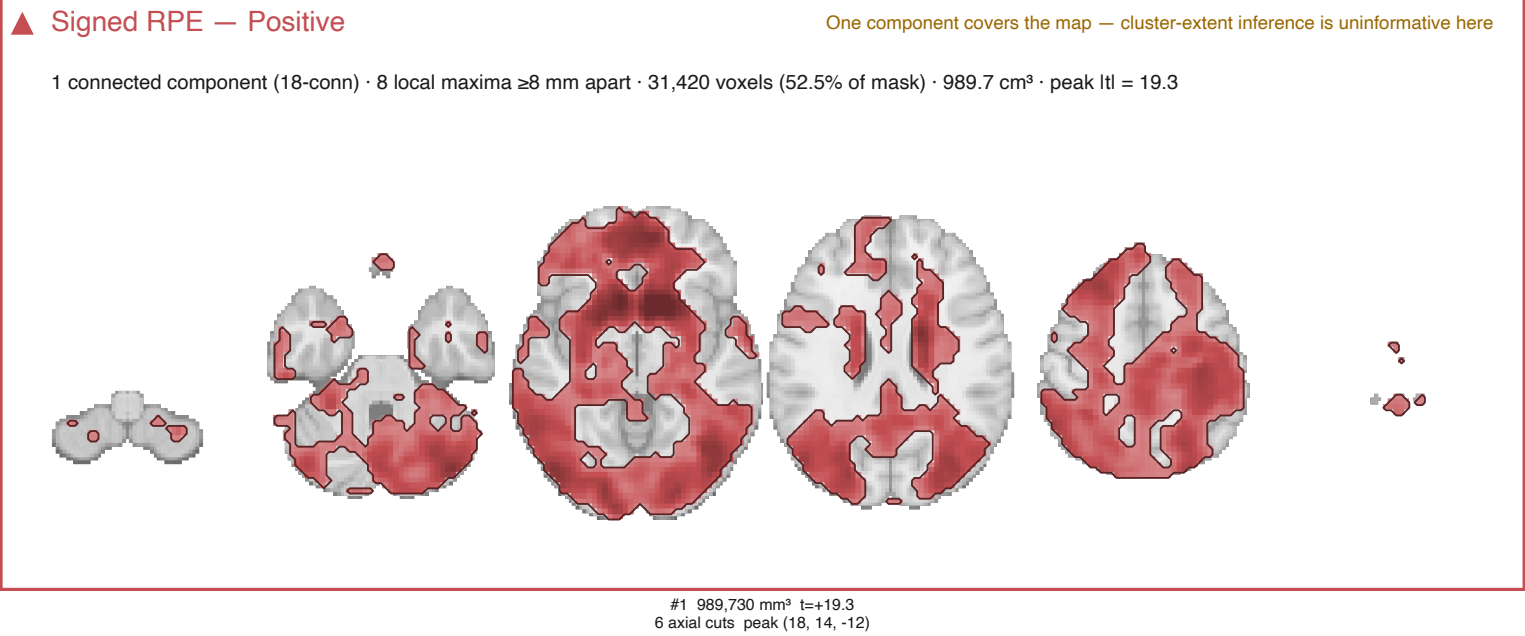
nilearn cluster-extent FWE — model2

Cluster-forming $t > 2.6649$ (one-tailed $p < 0.005$); clusters with two-tailed $p_{\text{FWE}} < .05$, 5000 sign flips, 18-connectivity.



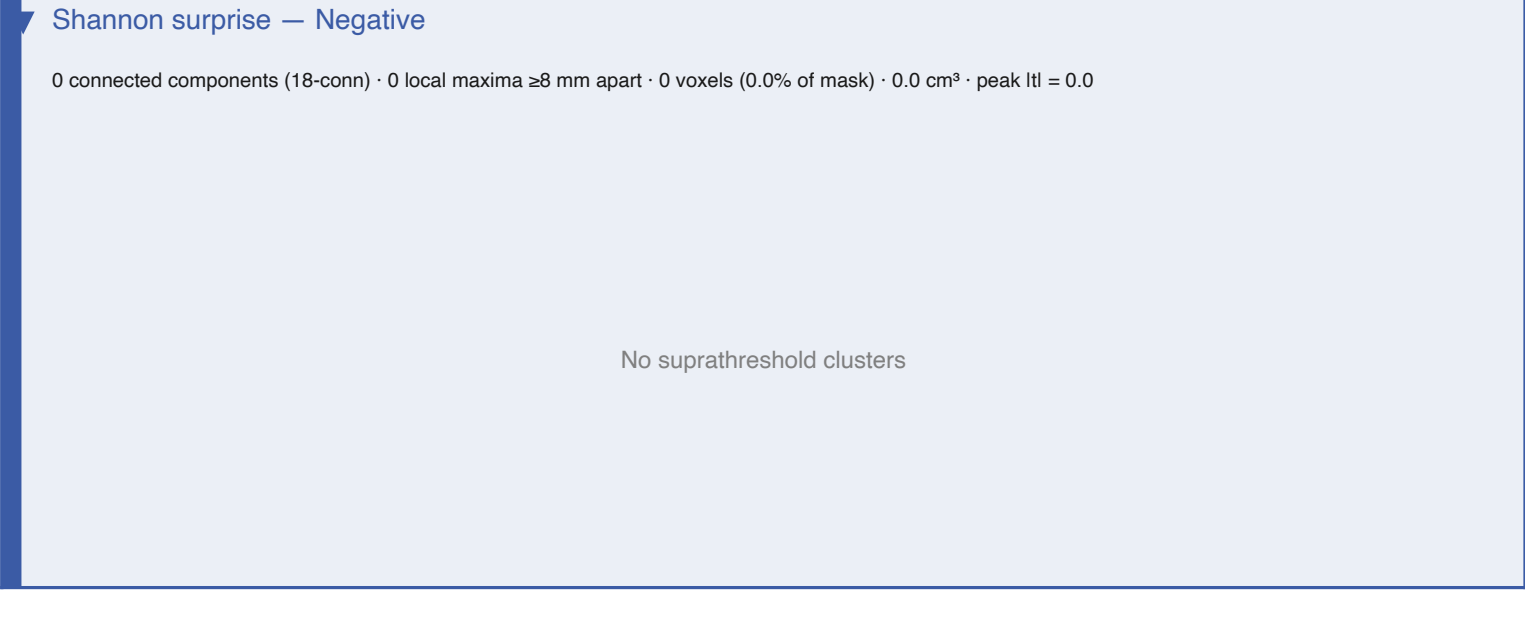
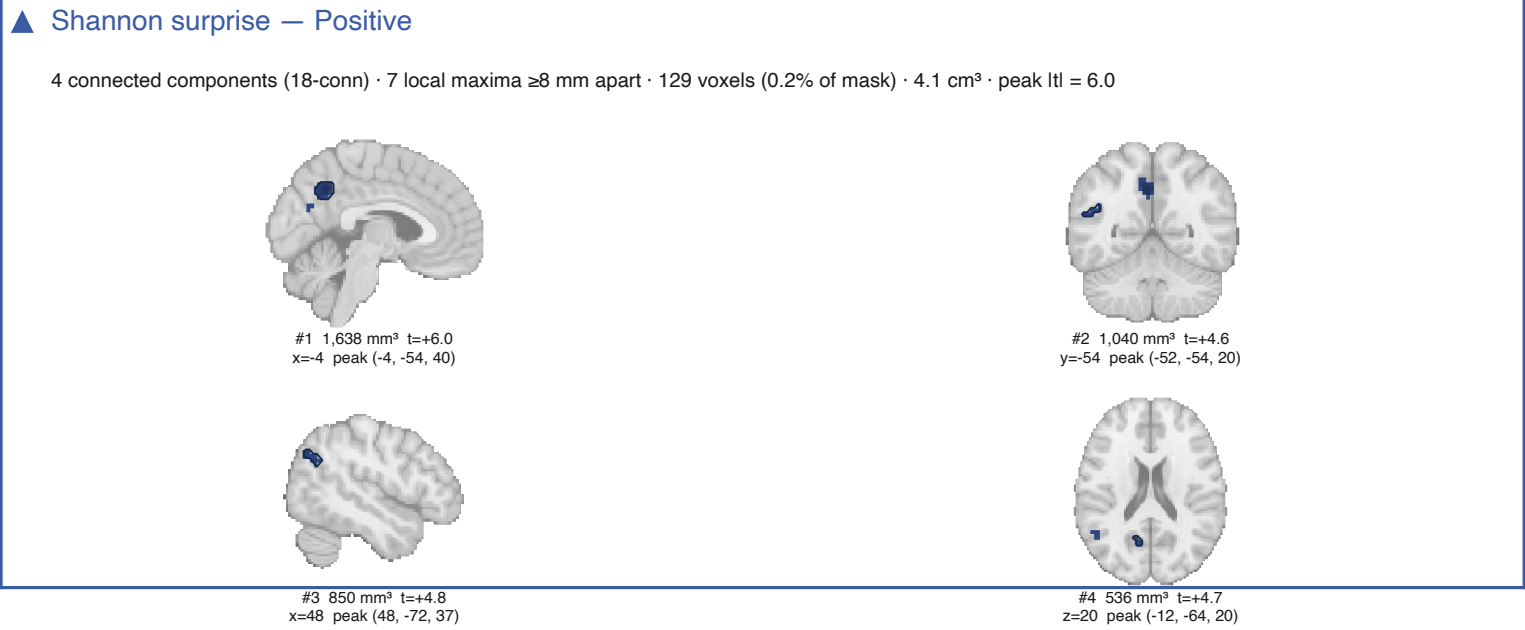
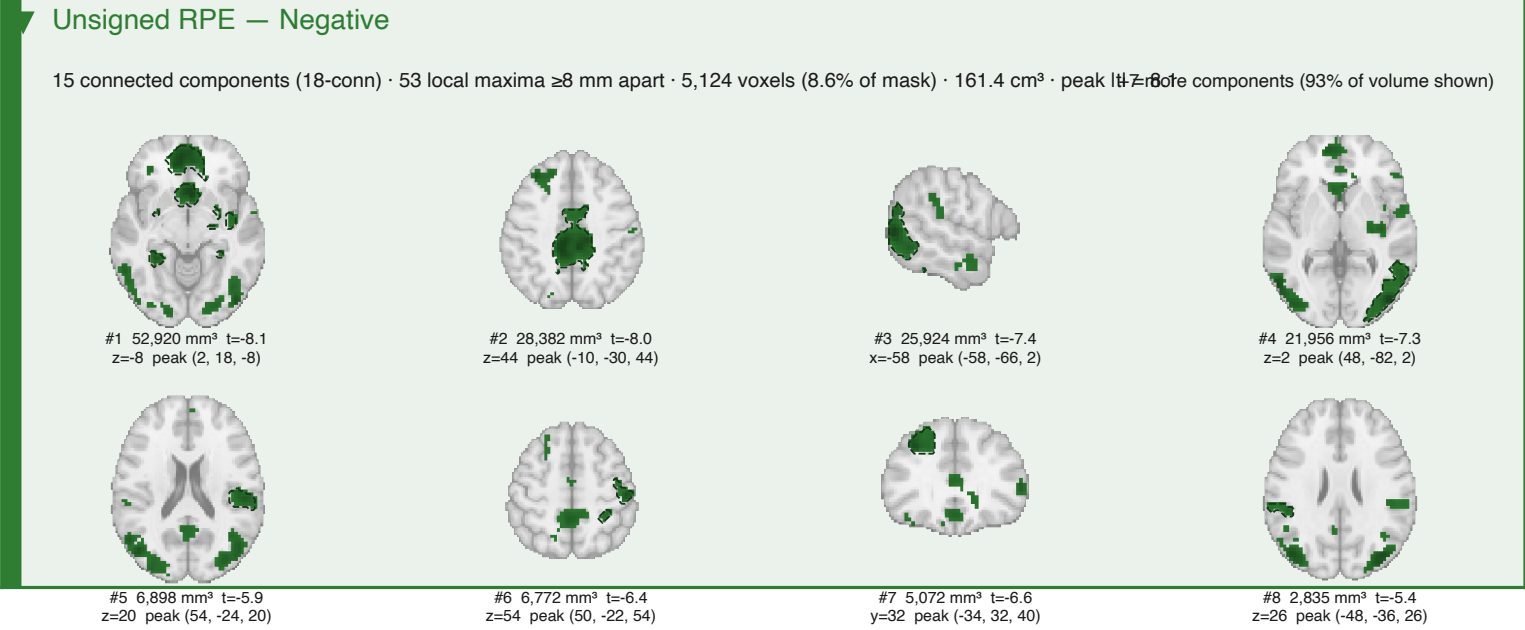
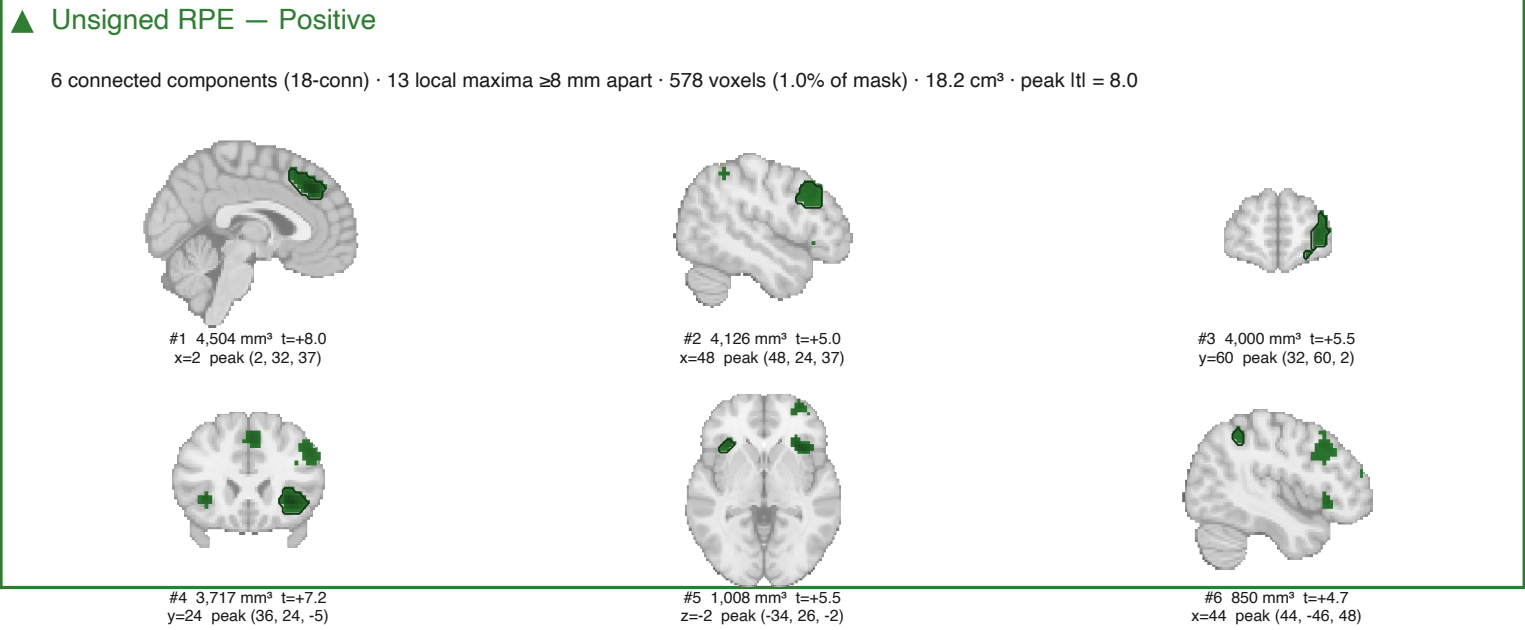
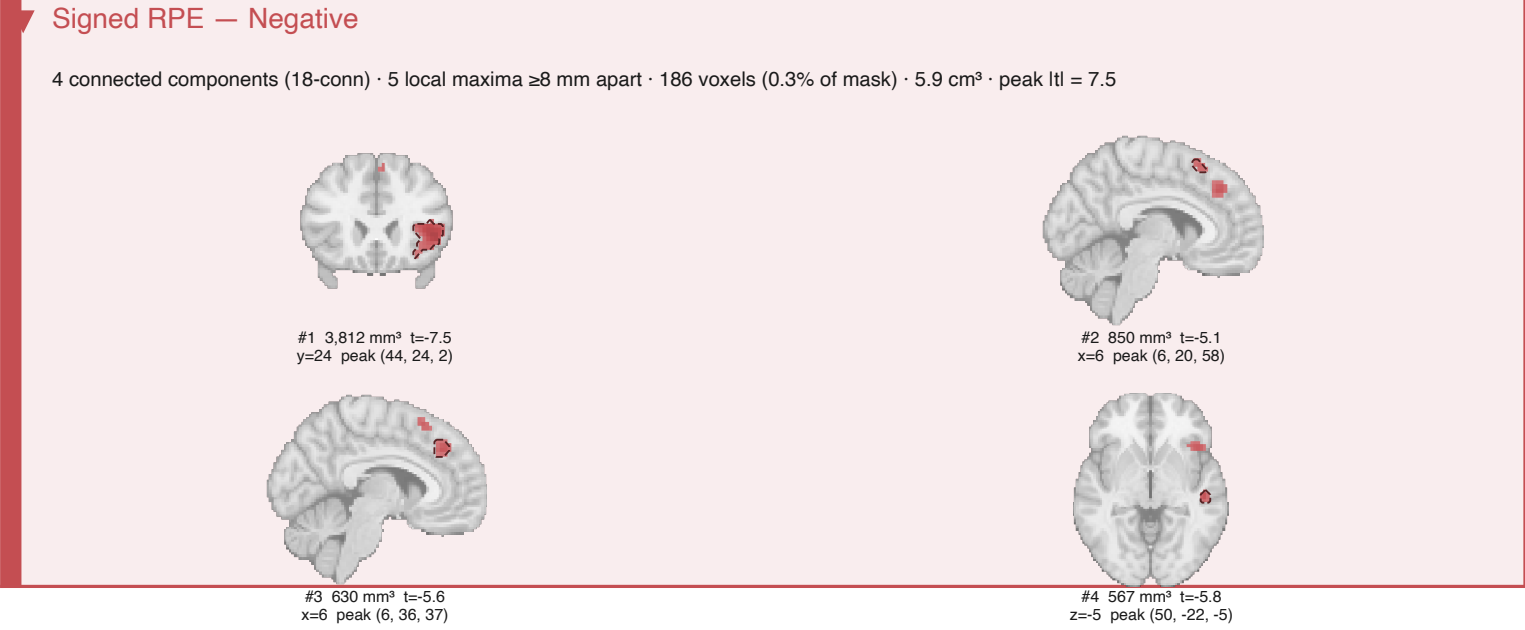
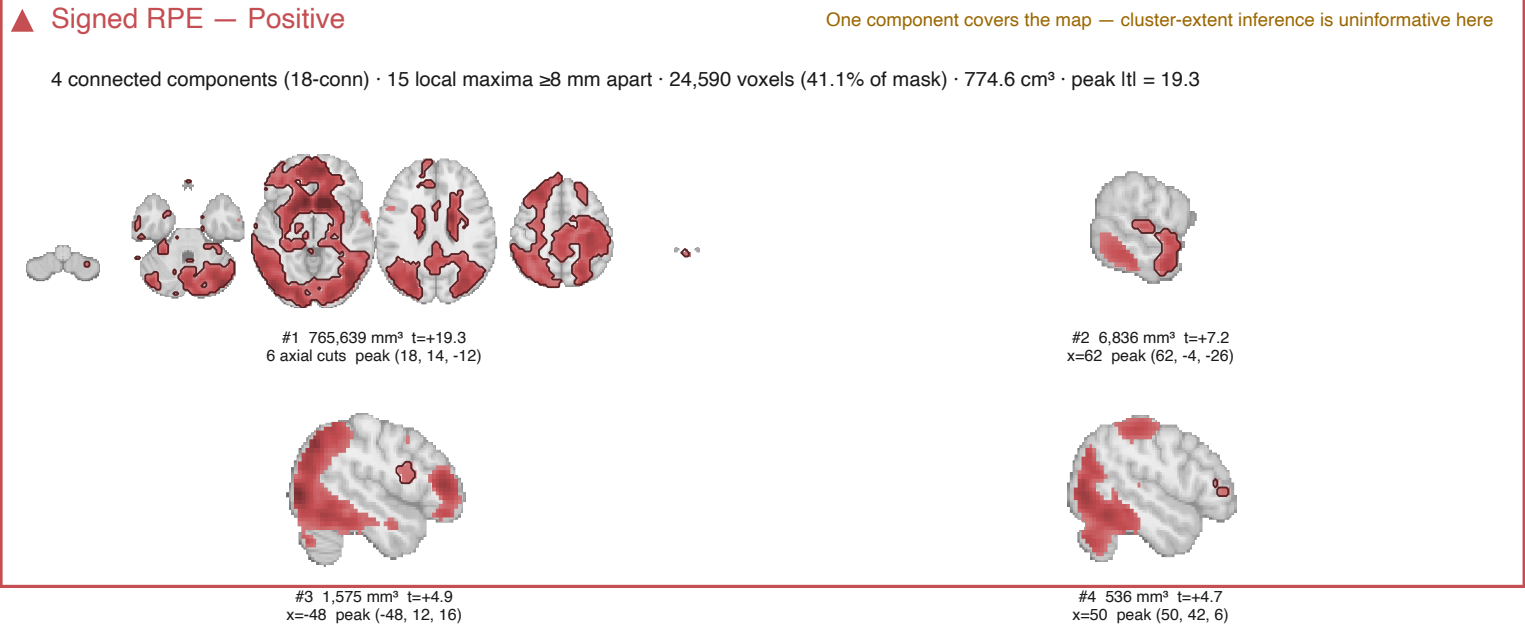
nilearn cluster-extent FWE — model2

Cluster-forming $t > 3.2395$ (one-tailed $p < 0.001$); clusters with two-tailed $p_{\text{FWE}} < .05$, 5000 sign flips, 18-connectivity.



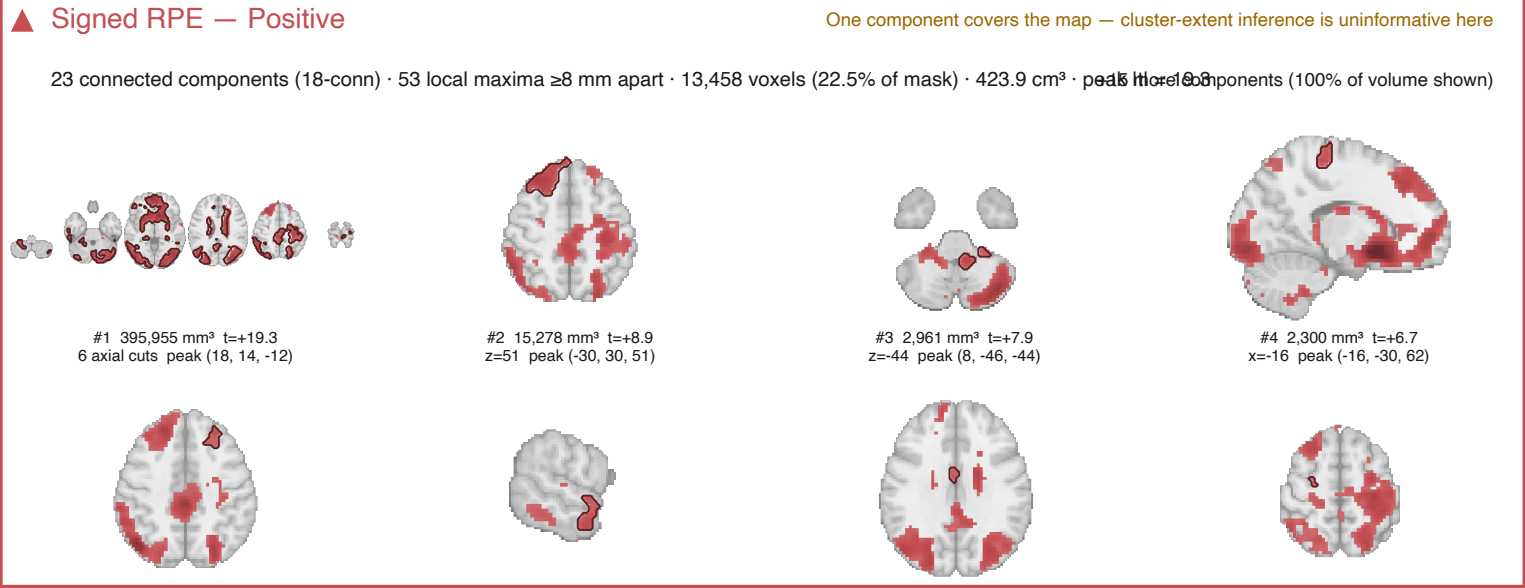
nilearn cluster-extent FWE — model2

Cluster-forming $t > 3.9756$ (one-tailed $p < 0.0001$); clusters with two-tailed $p_{\text{FWE}} < .05$, 5000 sign flips, 18-connectivity.



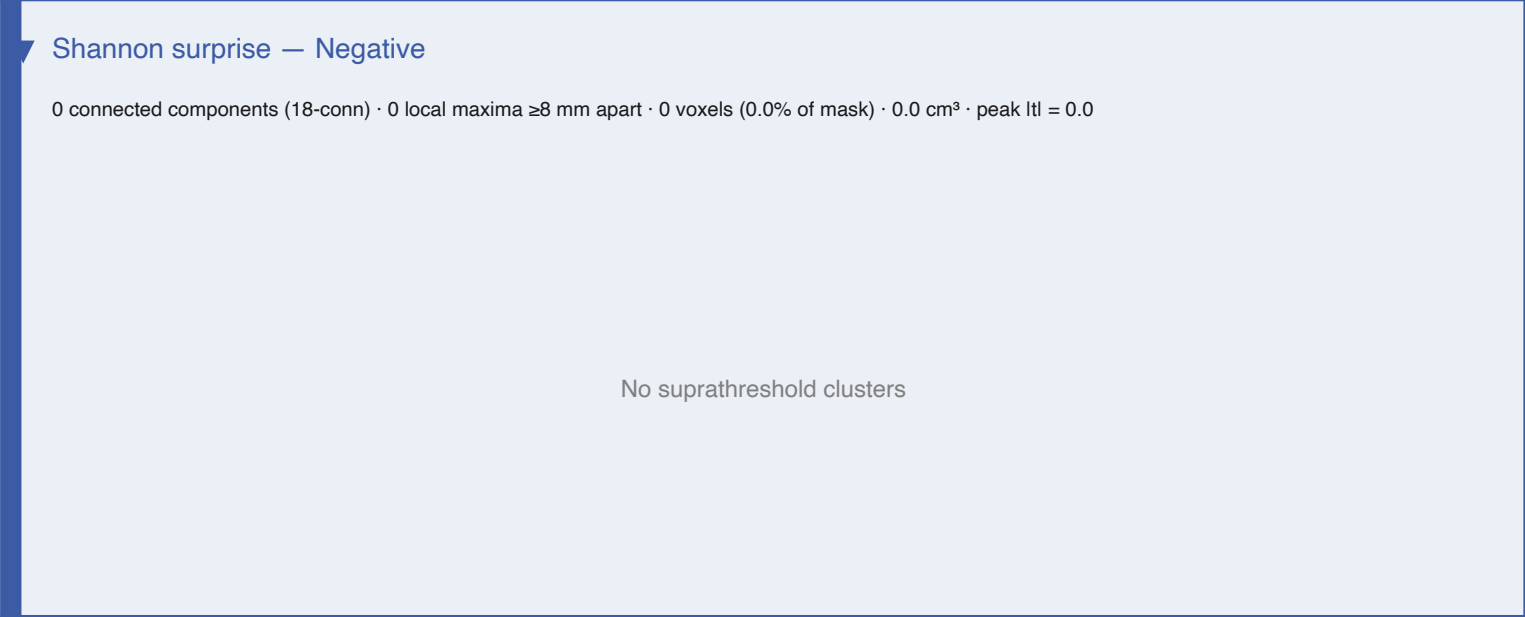
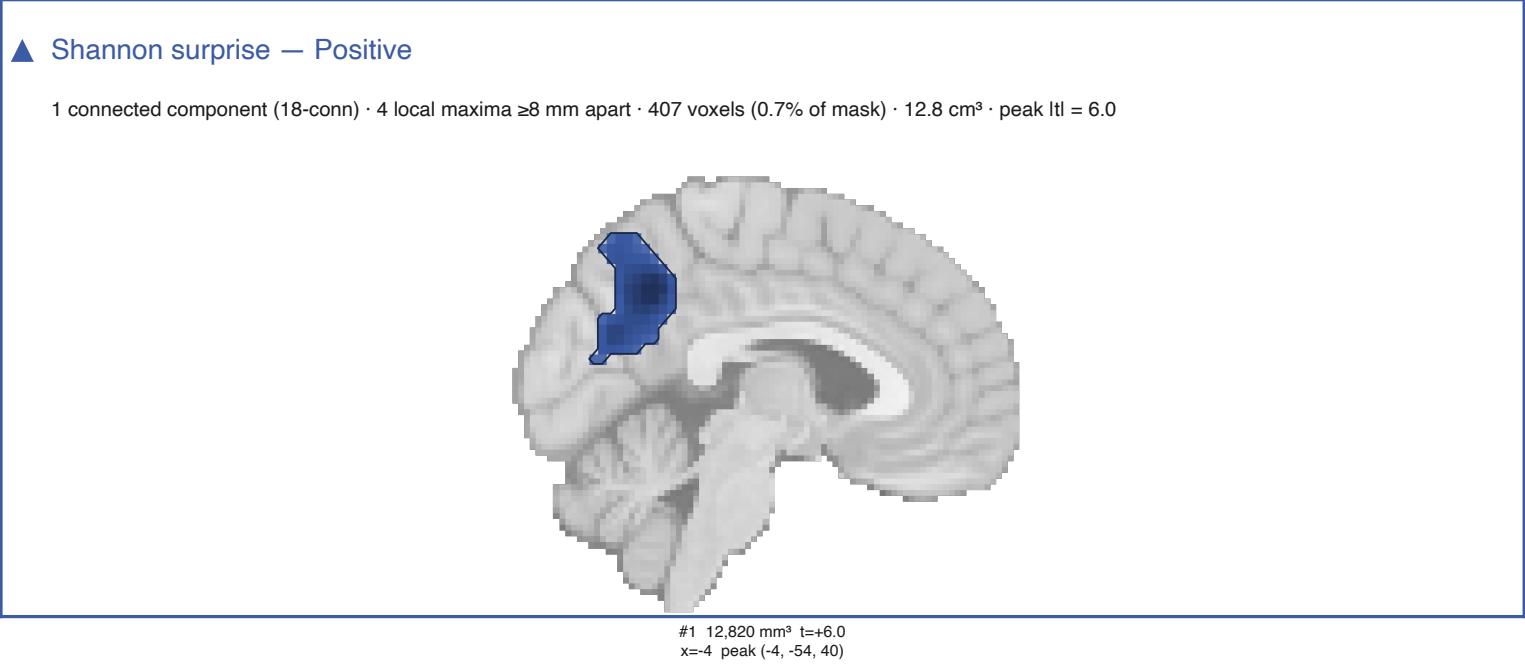
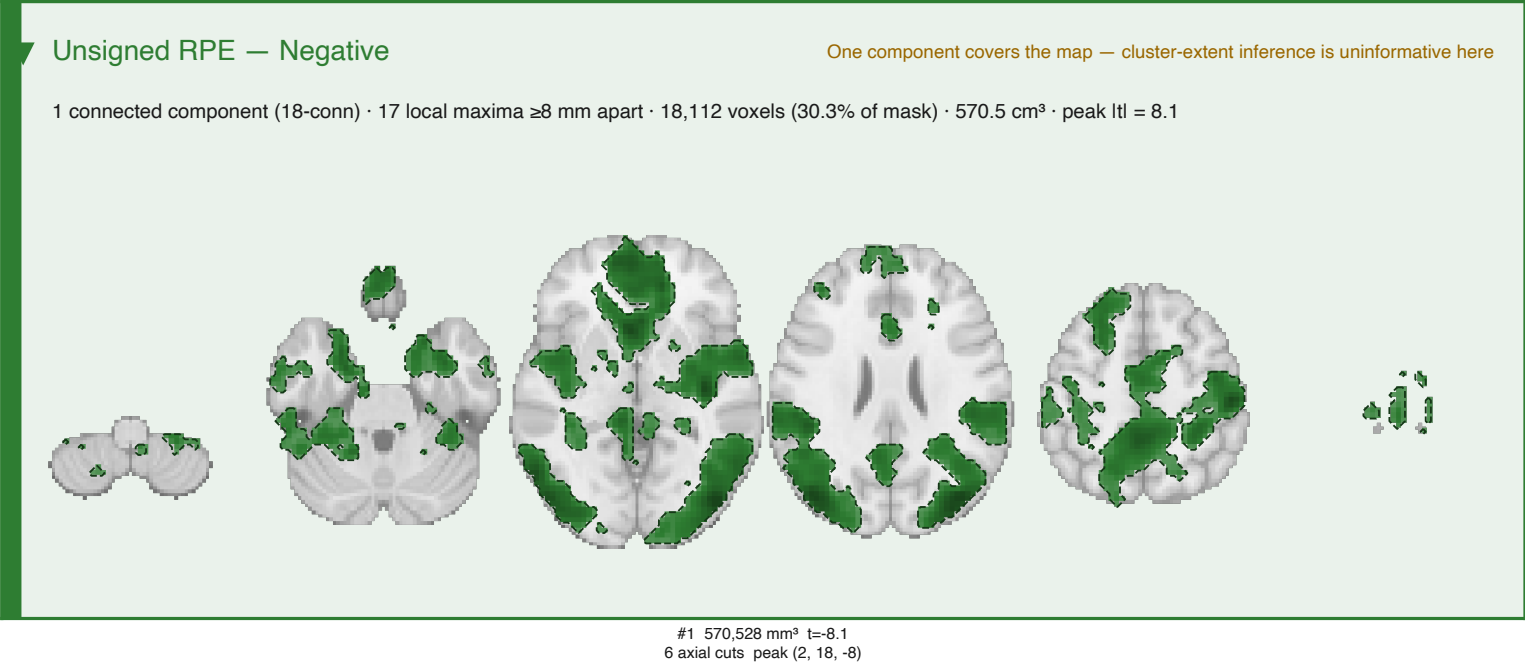
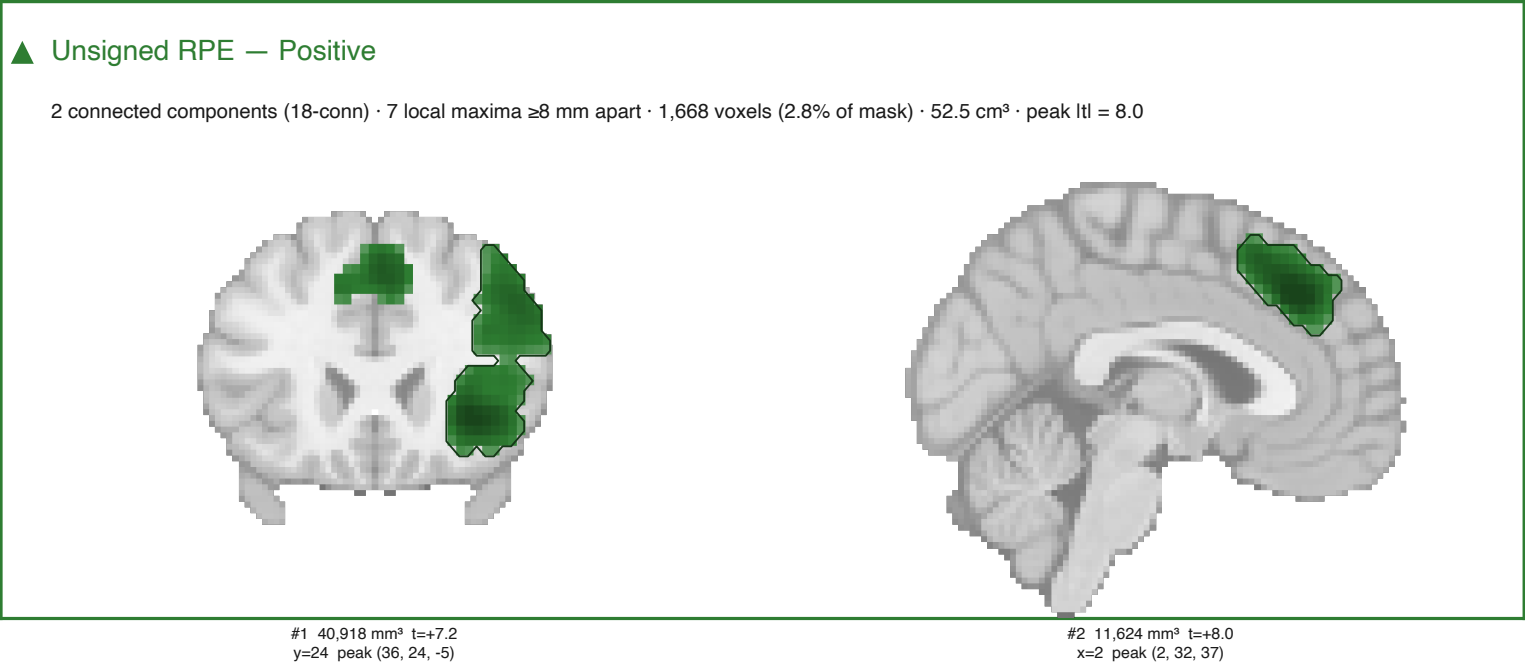
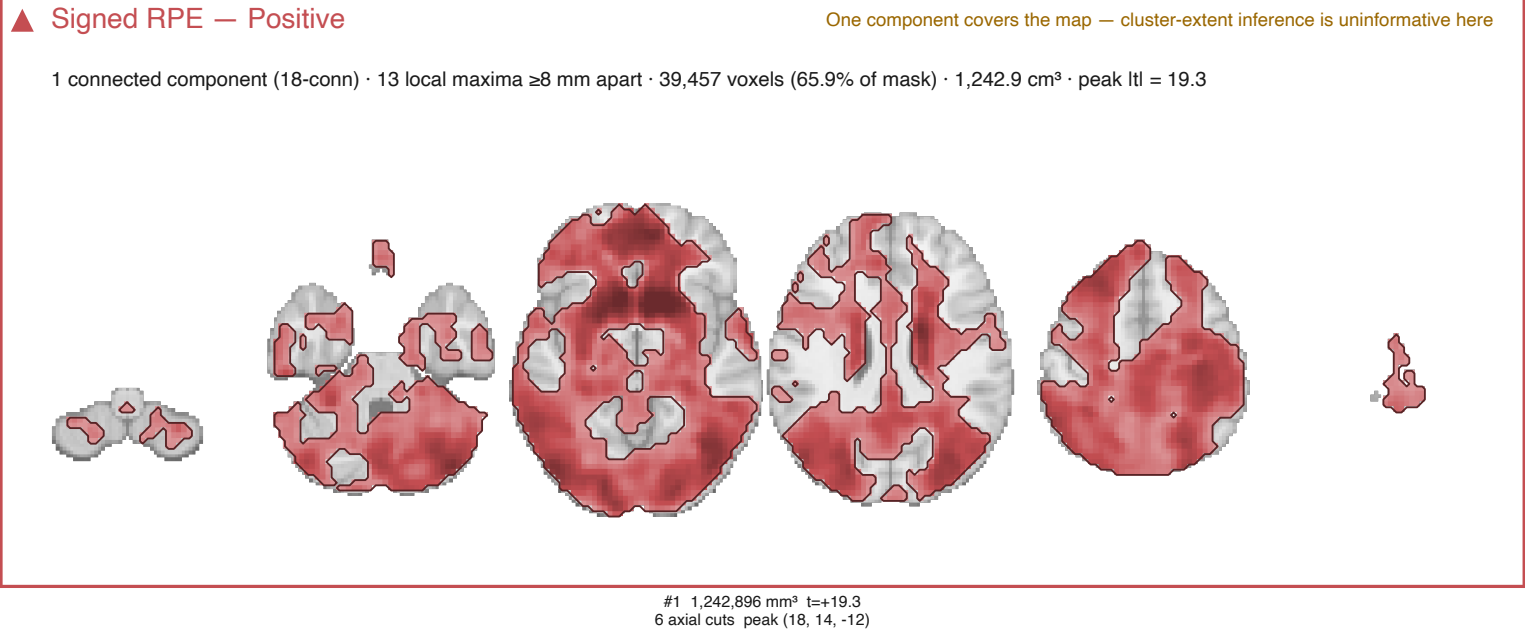
nilearn cluster-extent FWE — model2

Cluster-forming $t > 5.2929$ (one-tailed $p < 1e-06$); clusters with two-tailed $p_FWE < .05$, 5000 sign flips, 18-connectivity.



nilearn cluster-MASS FWE — model2

Cluster-forming $t > 2.3936$ (one-tailed $p < 0.01$); clusters with two-tailed $p_{\text{FWE}} < .05$ on summed excess t rather than extent.



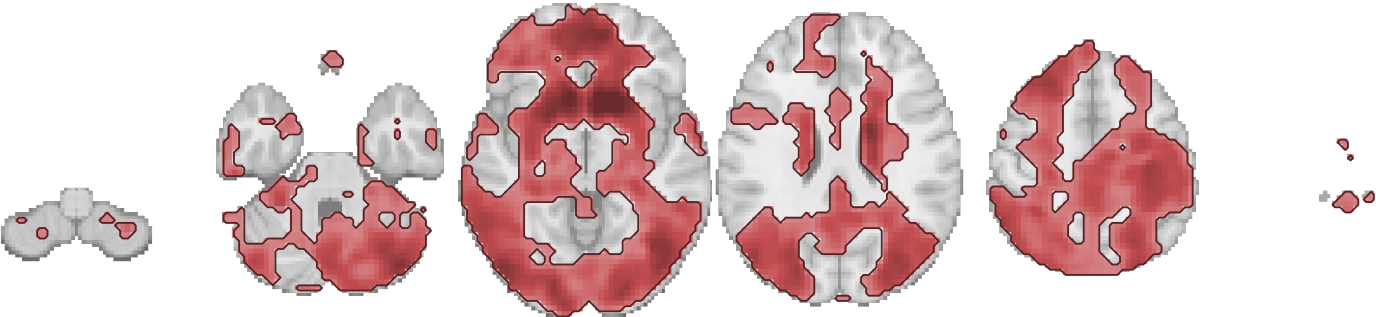
nilearn cluster-MASS FWE — model2

Cluster-forming $t > 3.2395$ (one-tailed $p < 0.001$); clusters with two-tailed $p_{\text{FWE}} < .05$ on summed excess t rather than extent.

▲ Signed RPE — Positive

One component covers the map — cluster-extent inference is uninformative here

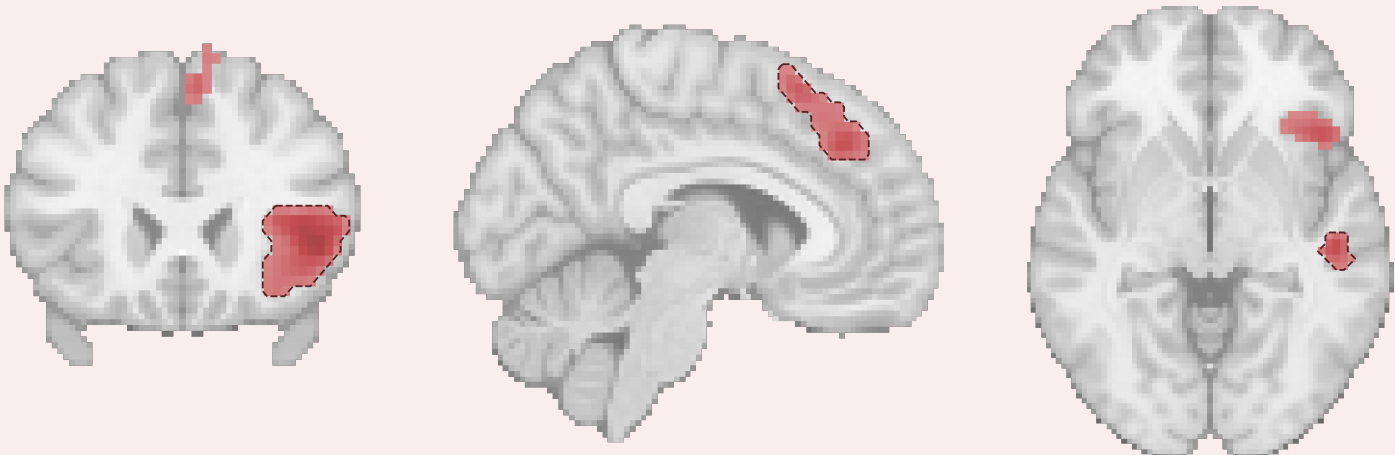
1 connected component (18-conn) · 8 local maxima ≥ 8 mm apart · 31,420 voxels (52.5% of mask) · 989.7 cm³ · peak $l_{tI} = 19.3$



#1 989,730 mm³ $t=+19.3$
6 axial cuts peak (18, 14, -12)

▶ Signed RPE — Negative

3 connected components (18-conn) · 6 local maxima ≥ 8 mm apart · 375 voxels (0.6% of mask) · 11.8 cm³ · peak $l_{tI} = 7.5$



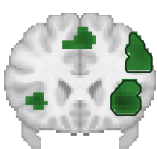
#1 6,363 mm³ $t=-7.5$
y=24 peak (44, 24, 2)

#2 4,032 mm³ $t=-5.6$
x=6 peak (6, 36, 37)

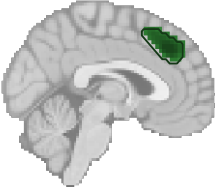
#3 1,418 mm³ $t=-5.8$
z=-5 peak (50, -22, -5)

▲ Unsigned RPE — Positive

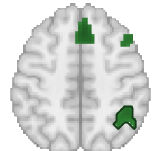
4 connected components (18-conn) · 13 local maxima ≥ 8 mm apart · 1,215 voxels (2.0% of mask) · 38.3 cm³ · peak $l_{tI} = 8.0$



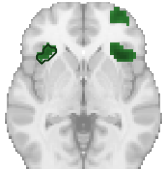
#1 25,546 mm³ $t=+7.2$
y=24 peak (36, 24, -5)



#2 7,402 mm³ $t=+8.0$
x=2 peak (2, 32, 37)



#3 3,150 mm³ $t=+4.7$
z=48 peak (44, -46, 48)



#4 2,174 mm³ $t=+5.5$
z=-2 peak (-34, 26, -2)

▶ Unsigned RPE — Negative

One component covers the map — cluster-extent inference is uninformative here

4 connected components (18-conn) · 28 local maxima ≥ 8 mm apart · 10,011 voxels (16.7% of mask) · 315.3 cm³ · peak $l_{tI} = 8.1$



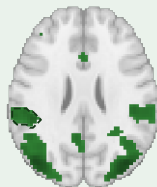
#1 257,292 mm³ $t=-8.1$
6 axial cuts peak (2, 18, -8)



#2 41,769 mm³ $t=-7.3$
x=48 peak (48, -82, 2)



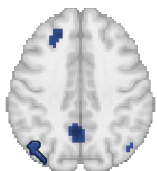
#3 9,576 mm³ $t=-6.6$
y=32 peak (-34, 32, 40)



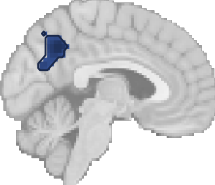
#4 6,710 mm³ $t=-5.4$
z=26 peak (-48, -36, 26)

▲ Shannon surprise — Positive

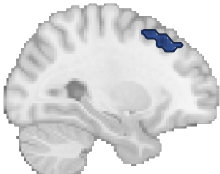
4 connected components (18-conn) · 16 local maxima ≥ 8 mm apart · 510 voxels (0.9% of mask) · 16.1 cm³ · peak $l_{tI} = 6.0$



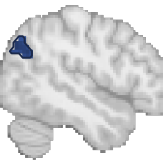
#1 5,576 mm³ $t=+4.6$
z=40 peak (-36, -82, 40)



#2 5,418 mm³ $t=+6.0$
x=-4 peak (-4, -54, 40)



#3 2,835 mm³ $t=+5.0$
x=-28 peak (-28, 24, 48)



#4 2,236 mm³ $t=+4.8$
x=48 peak (48, -72, 37)

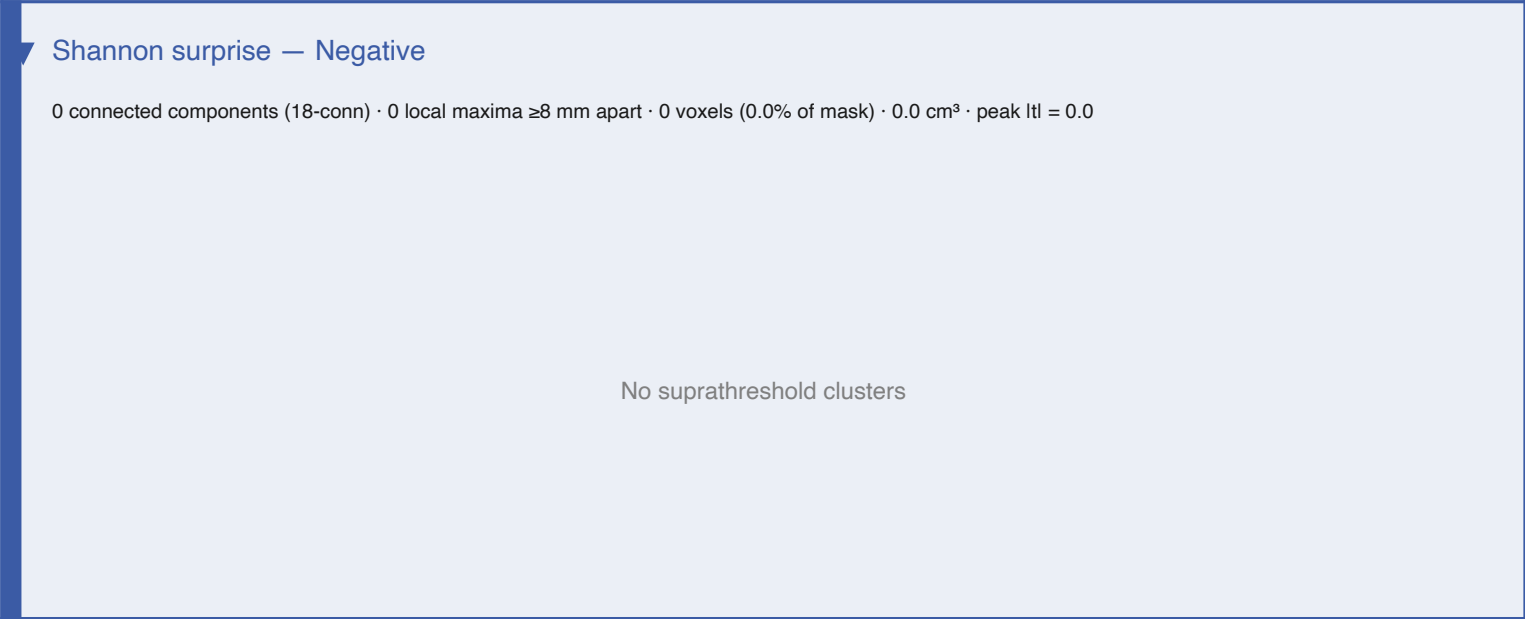
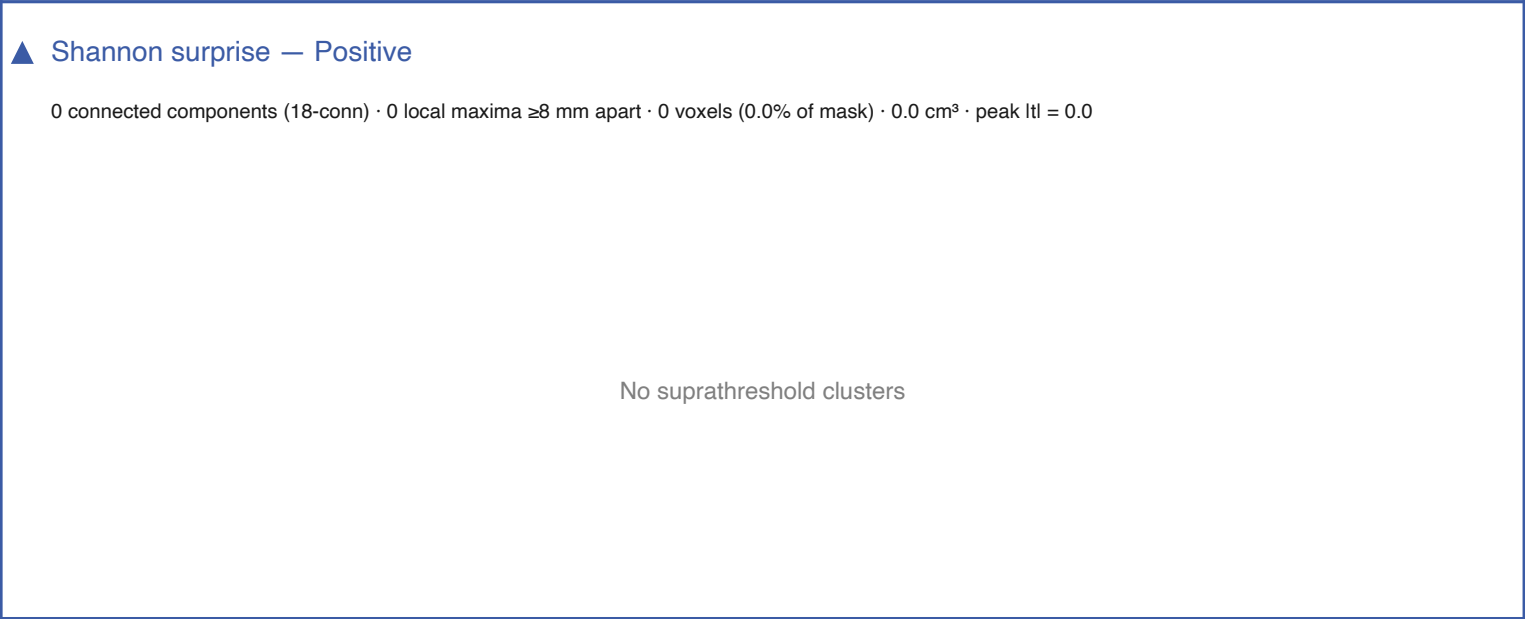
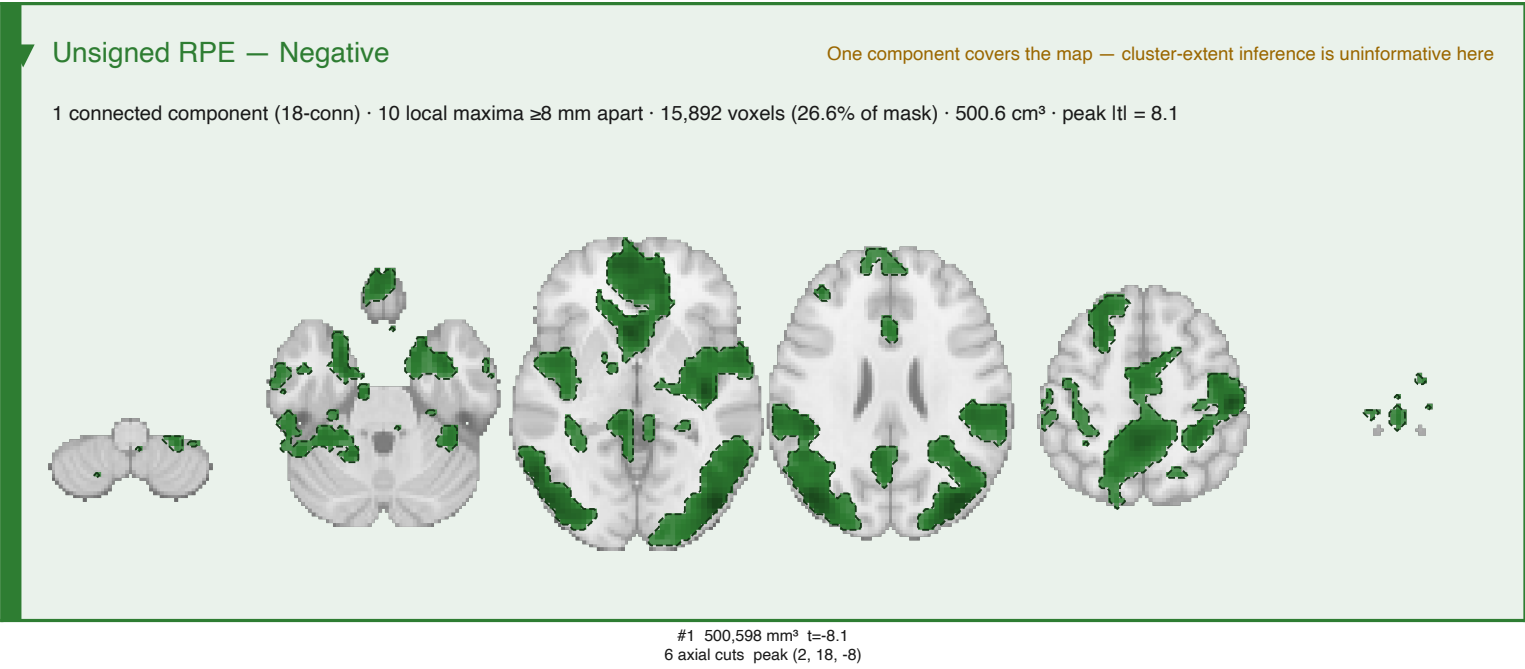
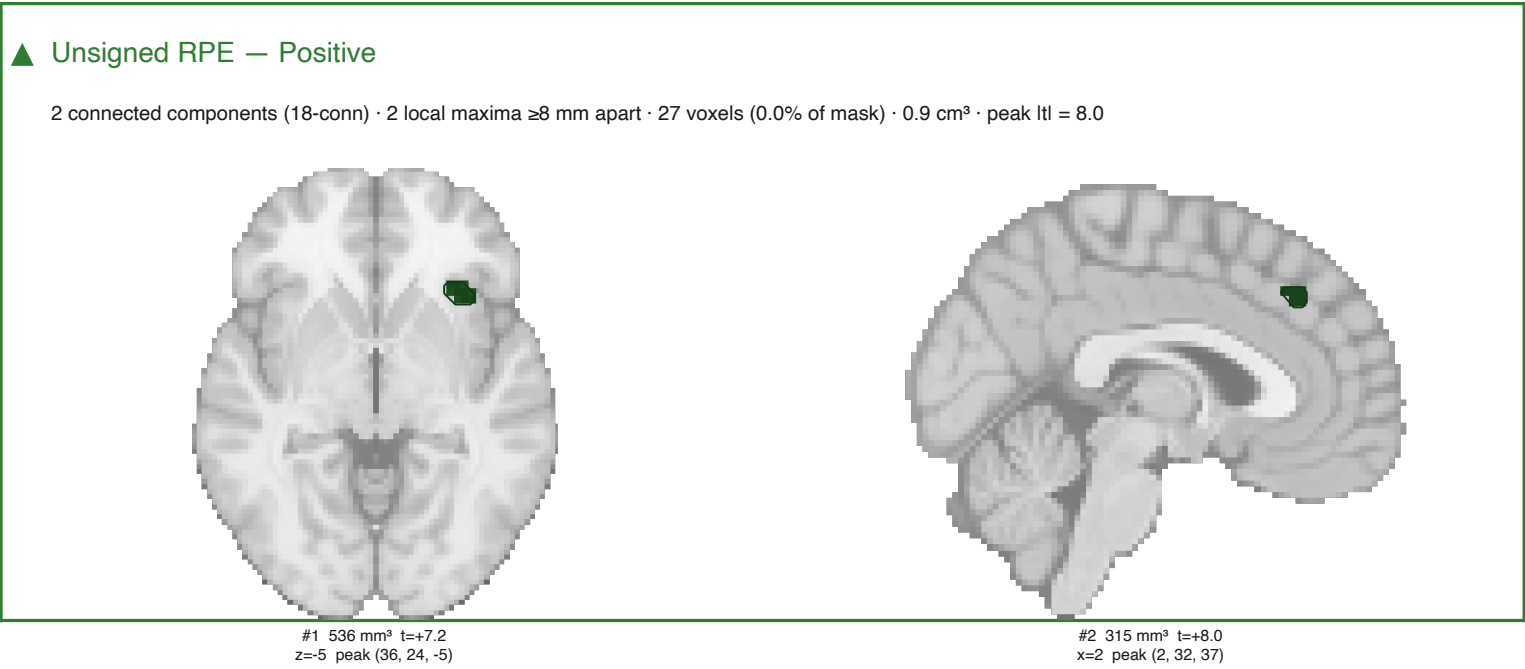
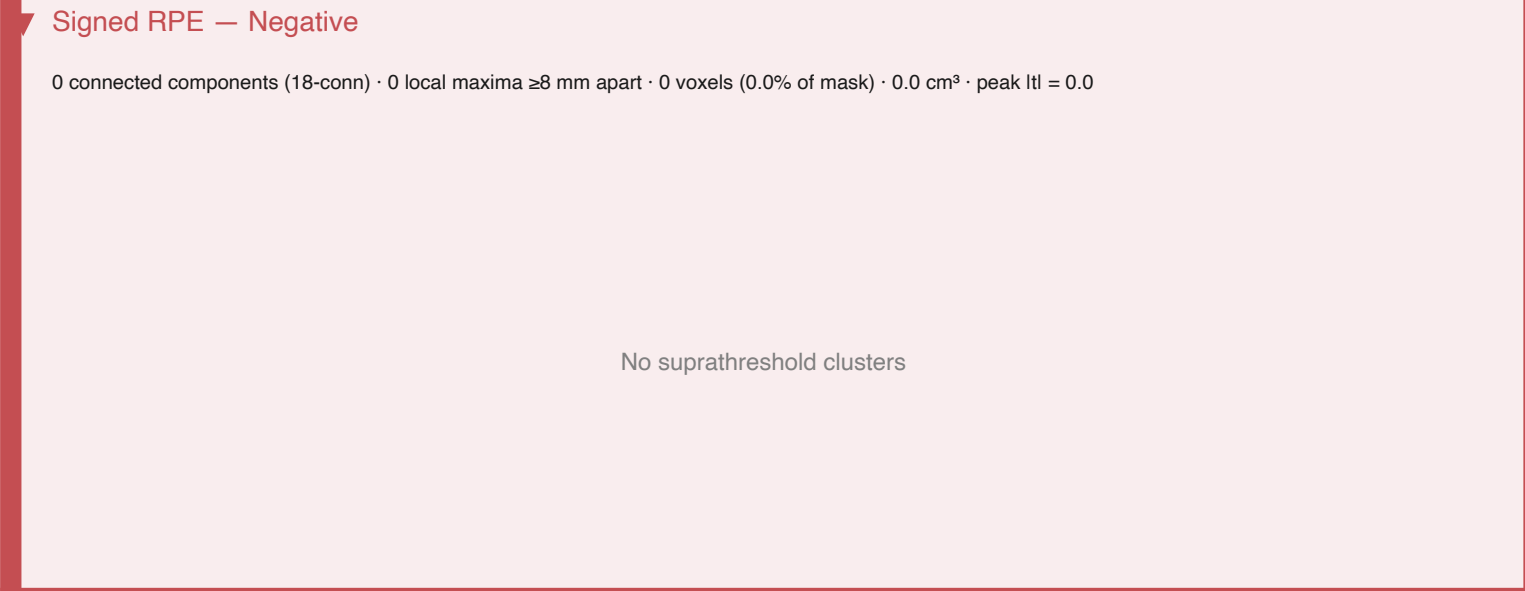
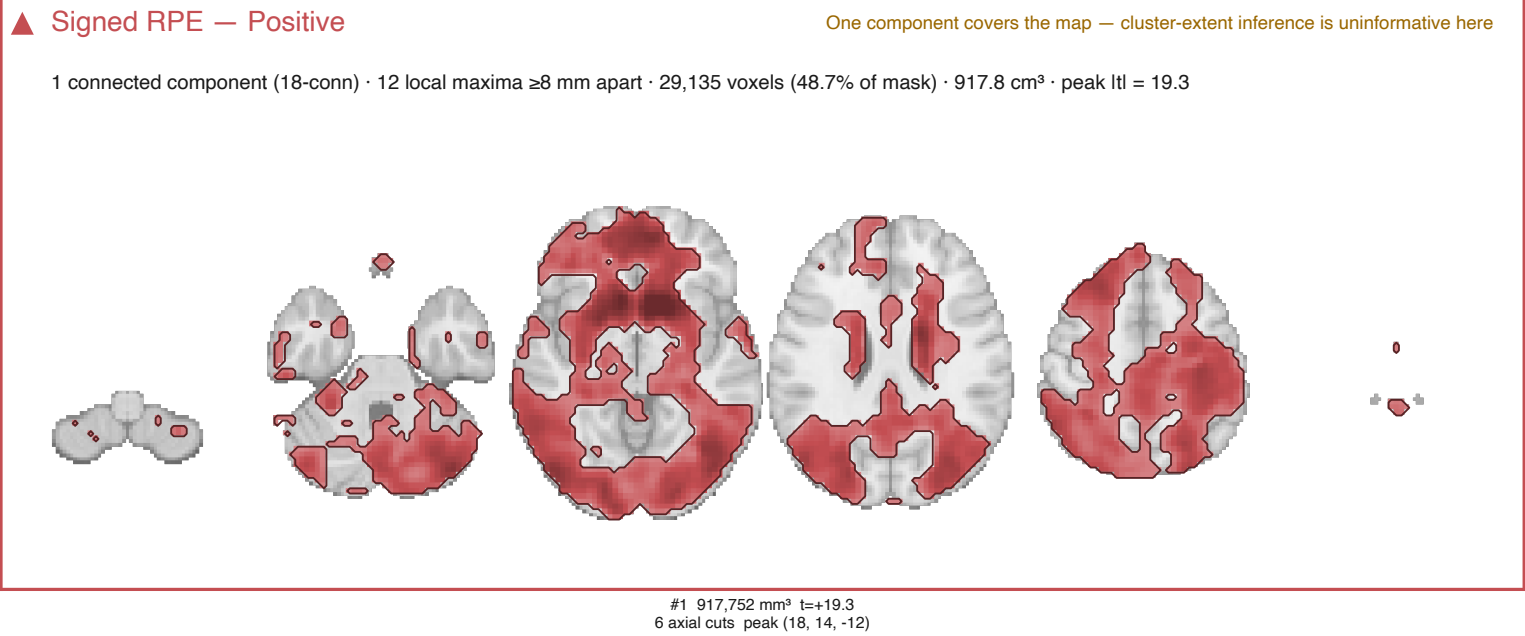
▶ Shannon surprise — Negative

0 connected components (18-conn) · 0 local maxima ≥ 8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak $l_{tI} = 0.0$

No suprathreshold clusters

TFCE — model2

Threshold-free cluster enhancement, two-tailed p_FWE < .05, 5000 sign flips. No cluster-forming threshold is chosen at all, so this page is the one that does not depend on the arbitrary decision the rest of the document sweeps over.



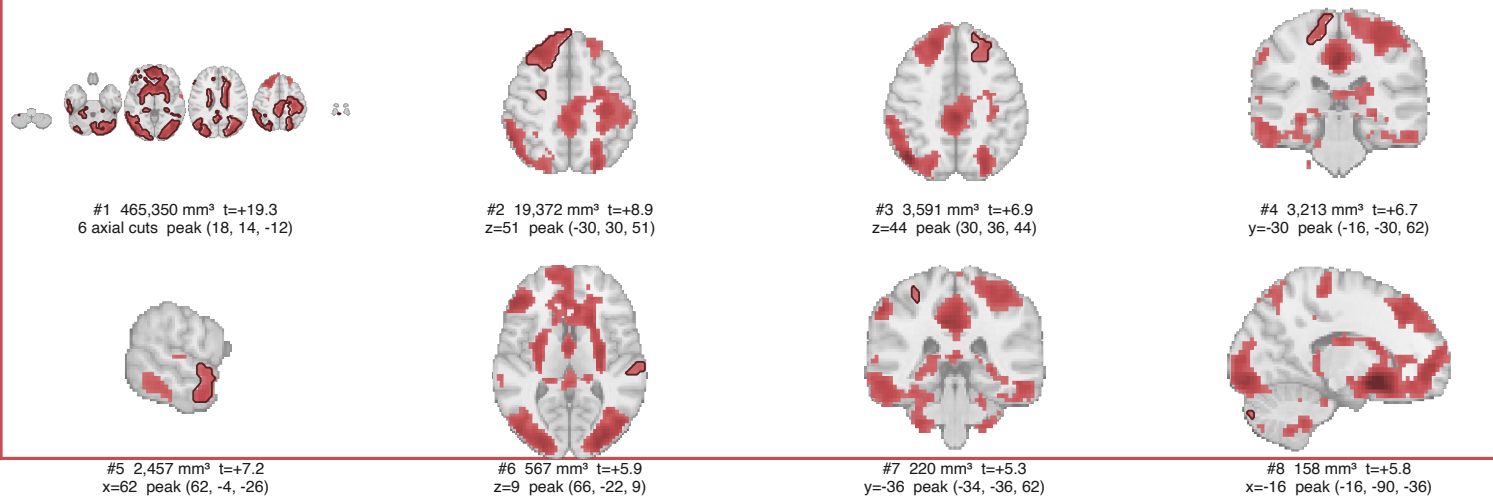
Voxel-wise (max-t) FWE — model2

Voxel-level familywise error, two-tailed $p_{\text{FWE}} < .05$, 5000 sign flips. The strictest and least assumption-laden test here: no spatial criterion of any kind.

▲ Signed RPE — Positive

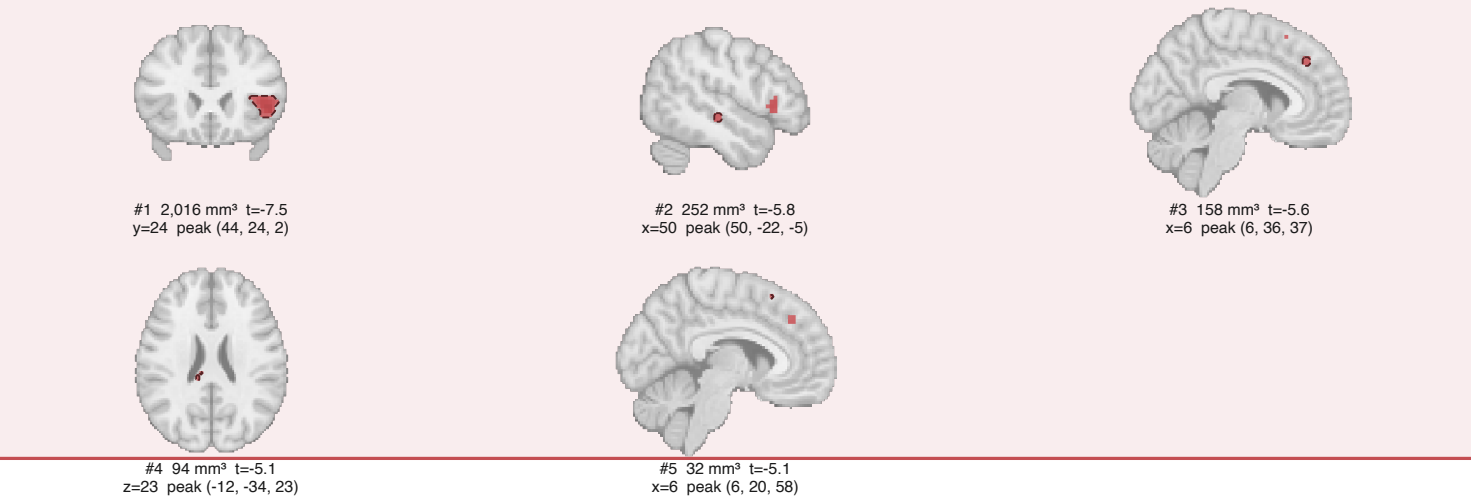
One component covers the map — cluster-extent inference is uninformative here

15 connected components (18-conn) · 39 local maxima ≥ 8 mm apart · 15,726 voxels (26.3% of mask) · 495.4 cm³ · peak 1100168 · 15 connected components (100% of volume shown)



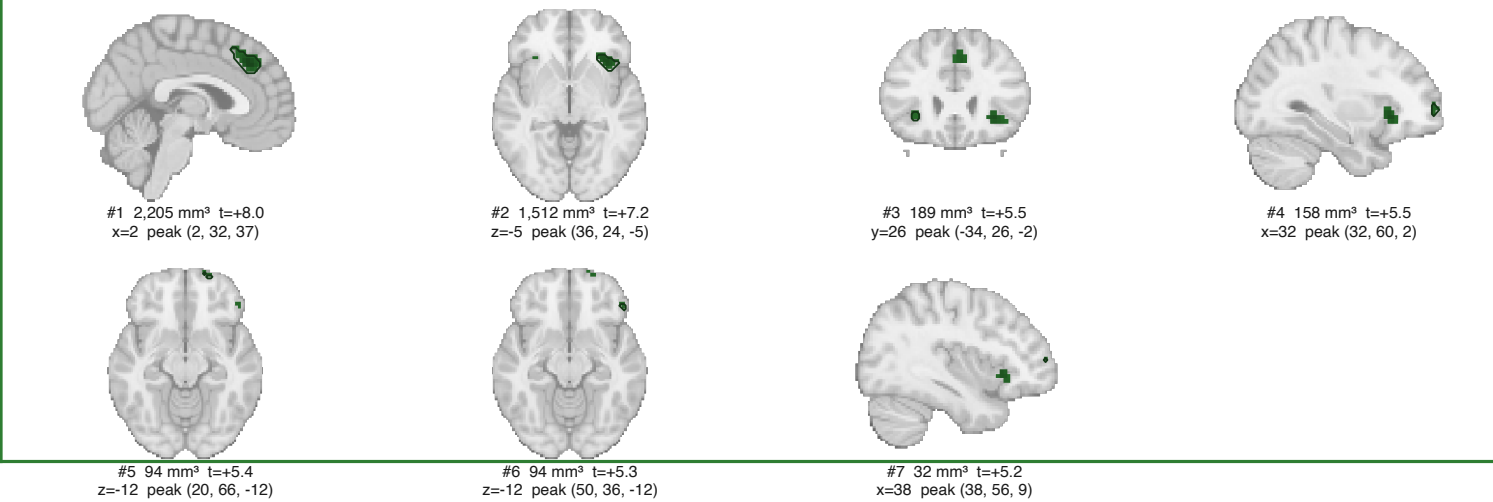
Signed RPE — Negative

5 connected components (18-conn) · 6 local maxima ≥ 8 mm apart · 81 voxels (0.1% of mask) · 2.6 cm³ · peak $l_{tl} = 7.5$



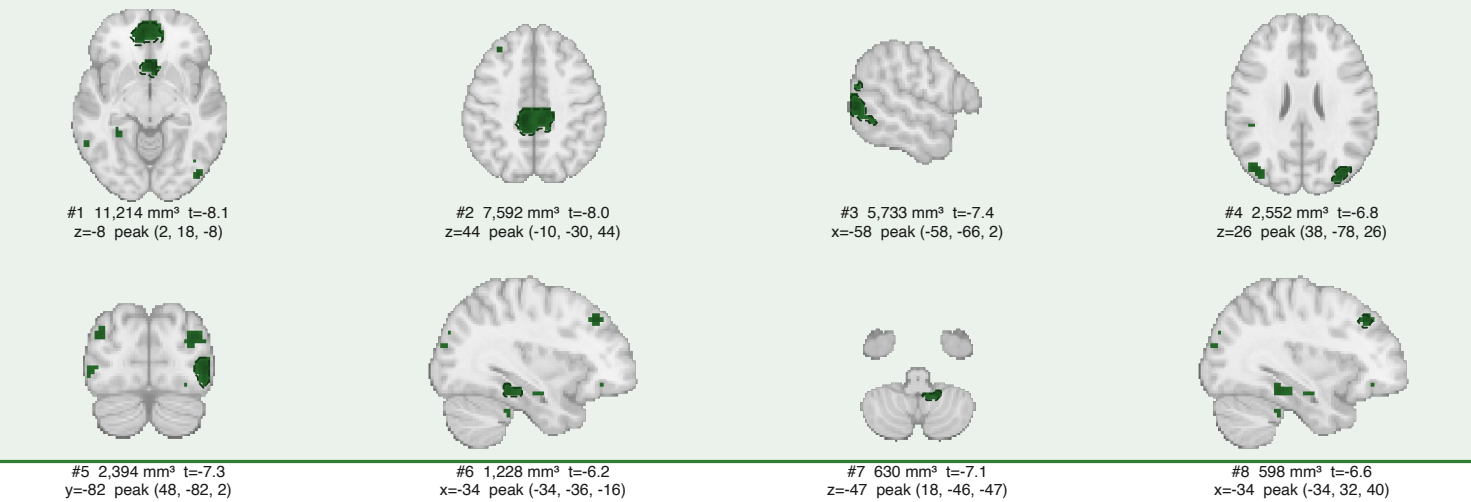
▲ Unsigned RPE — Positive

7 connected components (18-conn) · 8 local maxima ≥ 8 mm apart · 136 voxels (0.2% of mask) · 4.3 cm³ · peak ltl = 8.0



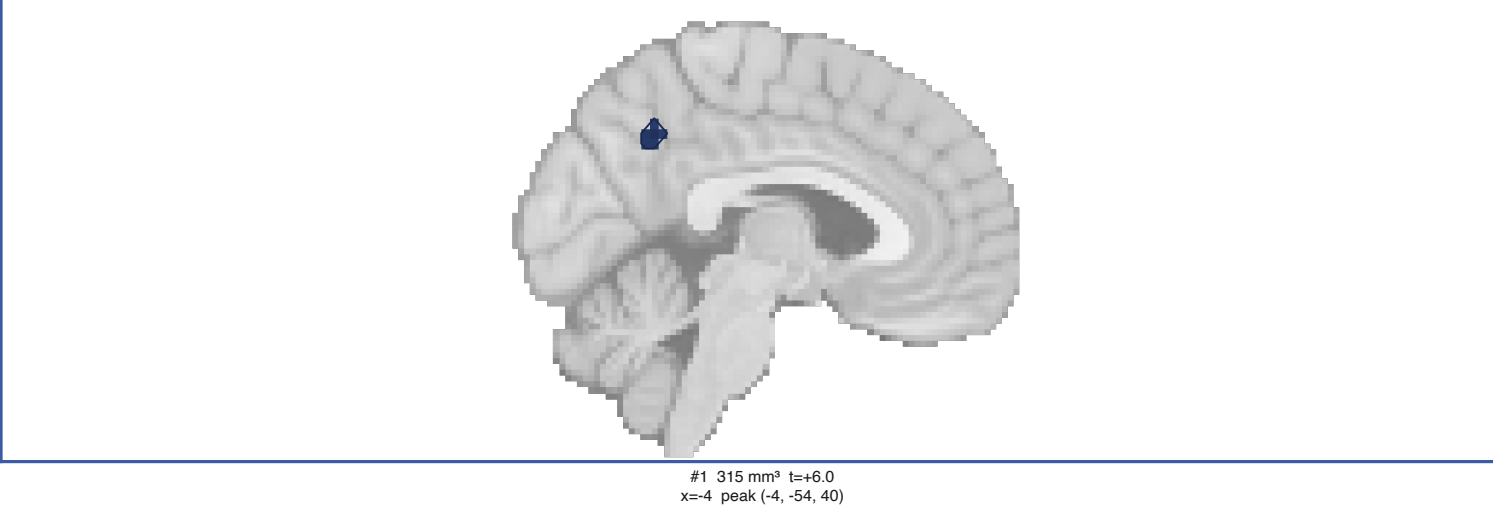
Unsigned RPE — Negative

38 connected components (18-conn) · 59 local maxima ≥ 8 mm apart · 1,198 voxels (2.0% of mask) · 37.7 cm³ · peak 14.308 mm · 308 more components (85% of volume shown)



▲ Shannon surprise — Positive

1 connected component (18-conn) · 1 local maxima ≥ 8 mm apart · 10 voxels (0.0% of mask) · 0.3 cm^3 · peak $\text{I} = 6.0$



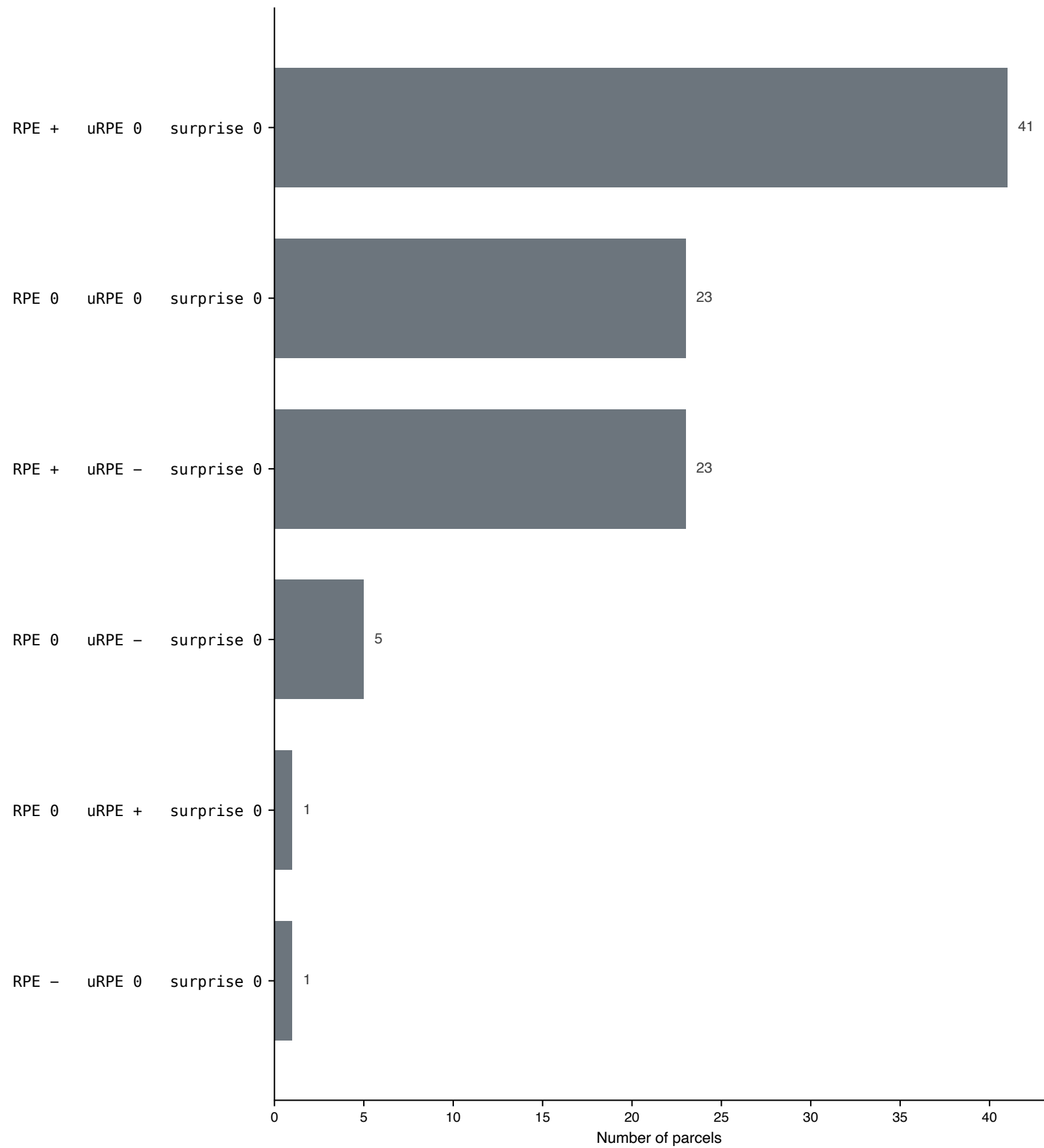
Shannon surprise — Negative

0 connected components (18-conn) · 0 local maxima ≥ 8 mm apart · 0 voxels (0.0% of mask) · 0.0 cm³ · peak ltl = 0.0

No suprathreshold clusters

Sign patterns across all relevant parcels — model2

94 Harvard-Oxford parcels are touched by at least one signal's surviving clusters at cluster-forming $t > 3.24$. For each parcel the sign of each signal is + / - / 0 by a Holm-corrected one-sample t-test on the parcel mean. Order below is RPE, uRPE, surprise.

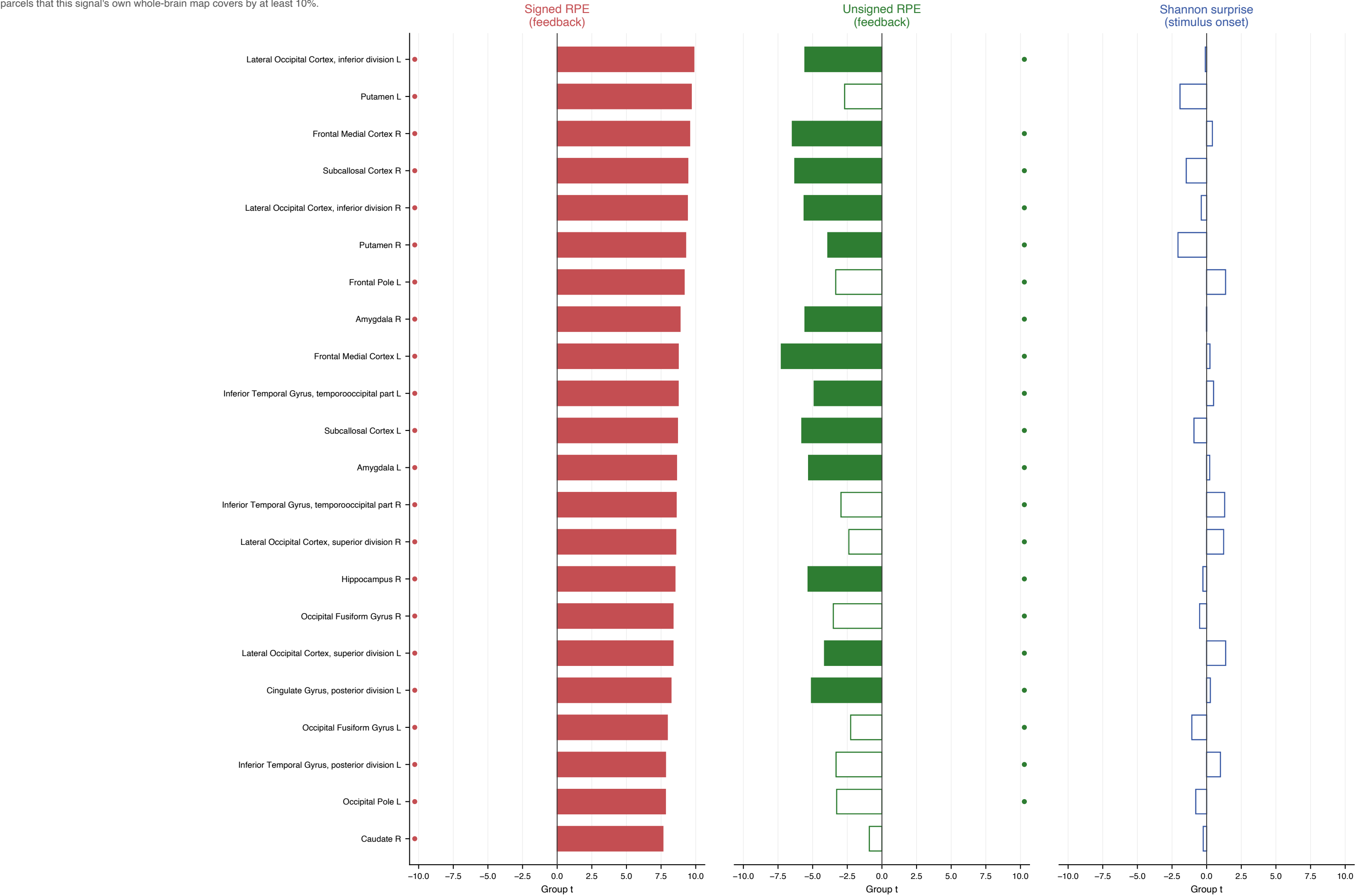


The single most common pattern is + 0 0 (41 of 94 parcels).
RPE is significantly positive in 64 parcels; uRPE is significantly negative in 28; the two co-occur in 23.
Surprise reaches significance in either direction in only 0 parcels.

Read this together with the orthogonalisation caveat: in model7 uRPE is the FIRST feedback modulator and signed RPE the second, so SPM's default serial orthogonalisation gives uRPE every scrap of variance the two share.
A negative uRPE loading therefore cannot be an artefact of the ordering -- the ordering biases uRPE the other way.

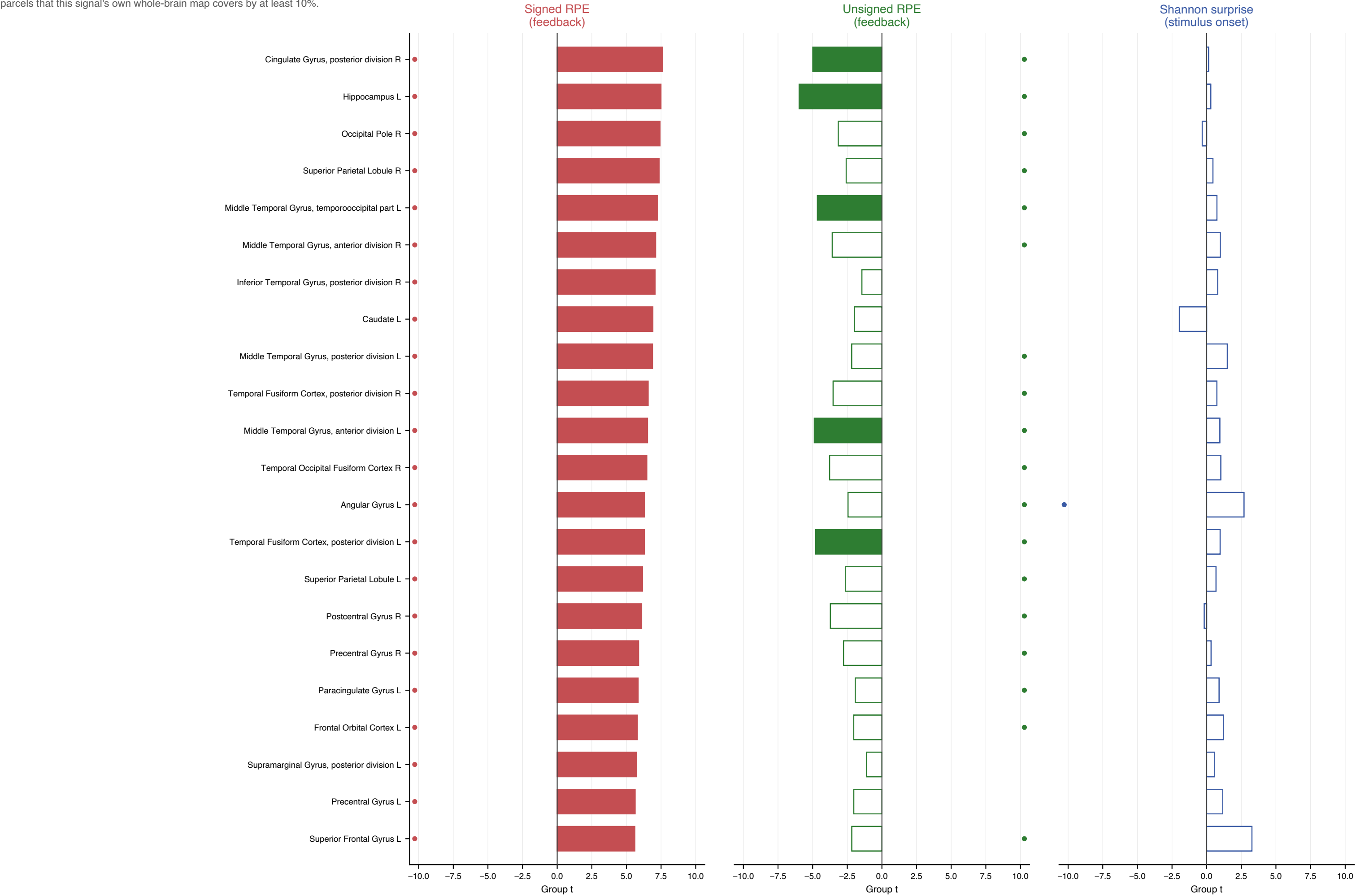
Which regions code which signal, and with which sign — model2 (1/5)

Harvard-Oxford parcels touched by any signal's surviving clusters (cluster-forming $t > 3.24$, $p < .001$). Bars are the group one-sample t of that parcel's mean contrast value; anatomical parcels give no signal a home-field advantage. Filled = Holm-significant across all parcel $\times$ signal tests, open = not. A dot on the far side of the axis marks parcels that this signal's own whole-brain map covers by at least 10%.



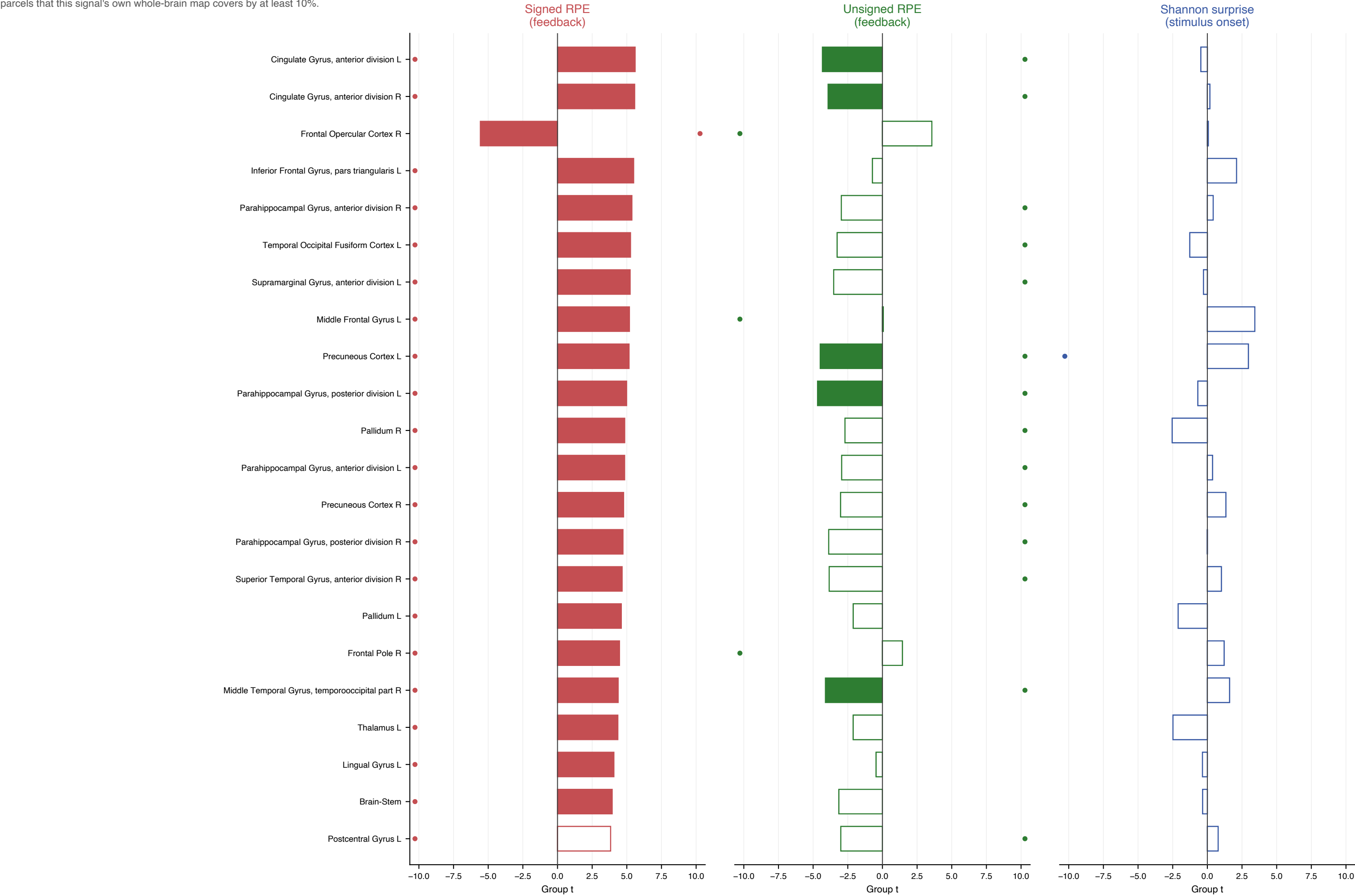
Which regions code which signal, and with which sign — model2 (2/5)

Harvard-Oxford parcels touched by any signal's surviving clusters (cluster-forming $t > 3.24$, $p < .001$). Bars are the group one-sample t of that parcel's mean contrast value; anatomical parcels give no signal a home-field advantage. Filled = Holm-significant across all parcel $\times$ signal tests, open = not. A dot on the far side of the axis marks parcels that this signal's own whole-brain map covers by at least 10%.



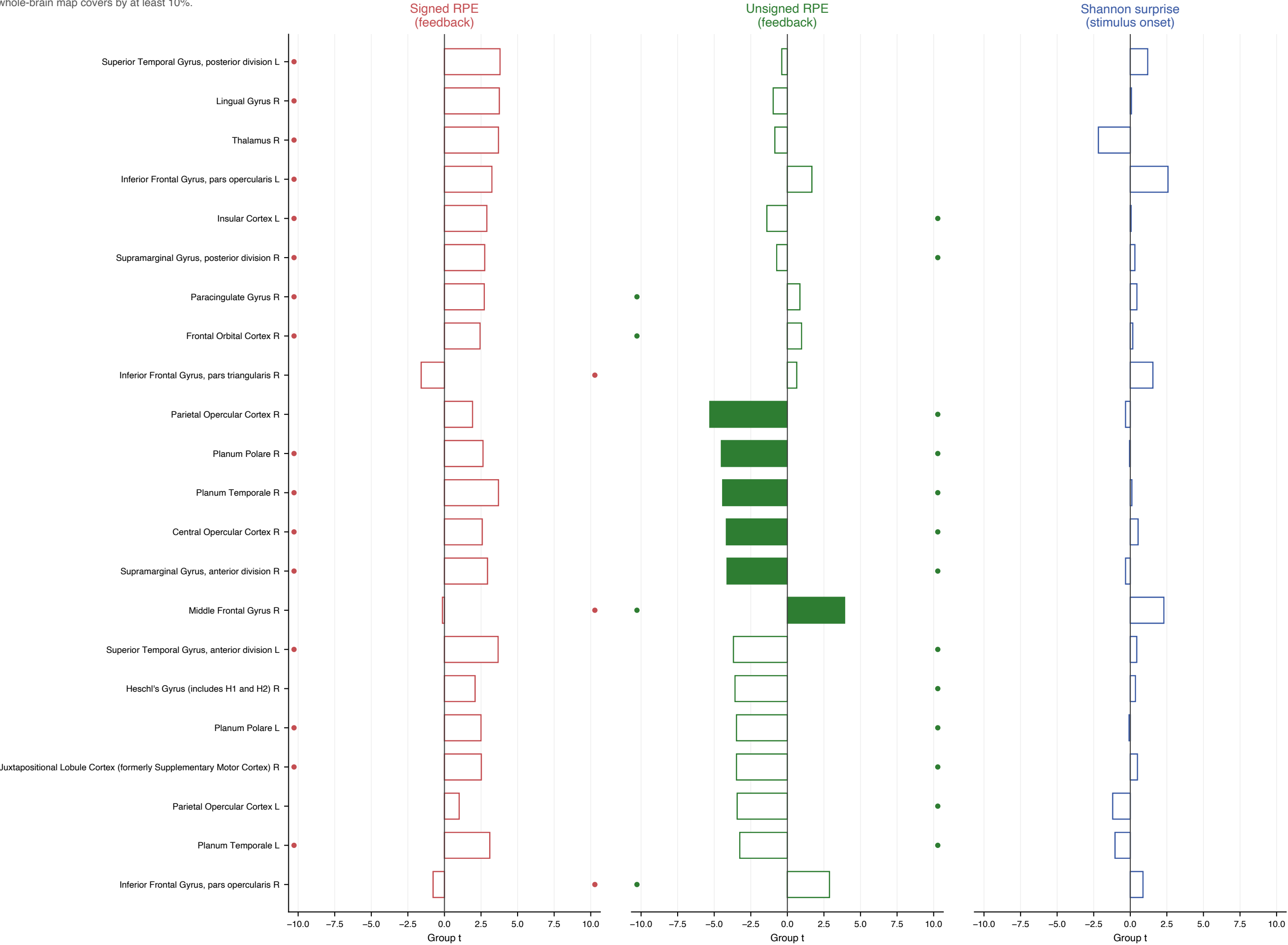
Which regions code which signal, and with which sign — model2 (3/5)

Harvard-Oxford parcels touched by any signal's surviving clusters (cluster-forming $t > 3.24$, $p < .001$). Bars are the group one-sample t of that parcel's mean contrast value; anatomical parcels give no signal a home-field advantage. Filled = Holm-significant across all parcel $\times$ signal tests, open = not. A dot on the far side of the axis marks parcels that this signal's own whole-brain map covers by at least 10%.



Which regions code which signal, and with which sign — model2 (4/5)

Harvard-Oxford parcels touched by any signal's surviving clusters (cluster-forming $t > 3.24$, $p < .001$). Bars are the group one-sample t of that parcel's mean contrast value; anatomical parcels give no signal a home-field advantage. Filled = Holm-significant across all parcel $\times$ signal tests, open = not. A dot on the far side of the axis marks parcels that this signal's own whole-brain map covers by at least 10%.



Which regions code which signal, and with which sign — model2 (5/5)

Harvard-Oxford parcels touched by any signal's surviving clusters (cluster-forming $t > 3.24$, $p < .001$). Bars are the group one-sample t of that parcel's mean contrast value; anatomical parcels give no signal a home-field advantage. Filled = Holm-significant across all parcel $\times$ signal tests, open = not. A dot on the far side of the axis marks parcels that this signal's own whole-brain map covers by at least 10%.

